

Training Multimodal Perception Models

Antonio Rueda-Toicen
November 2025



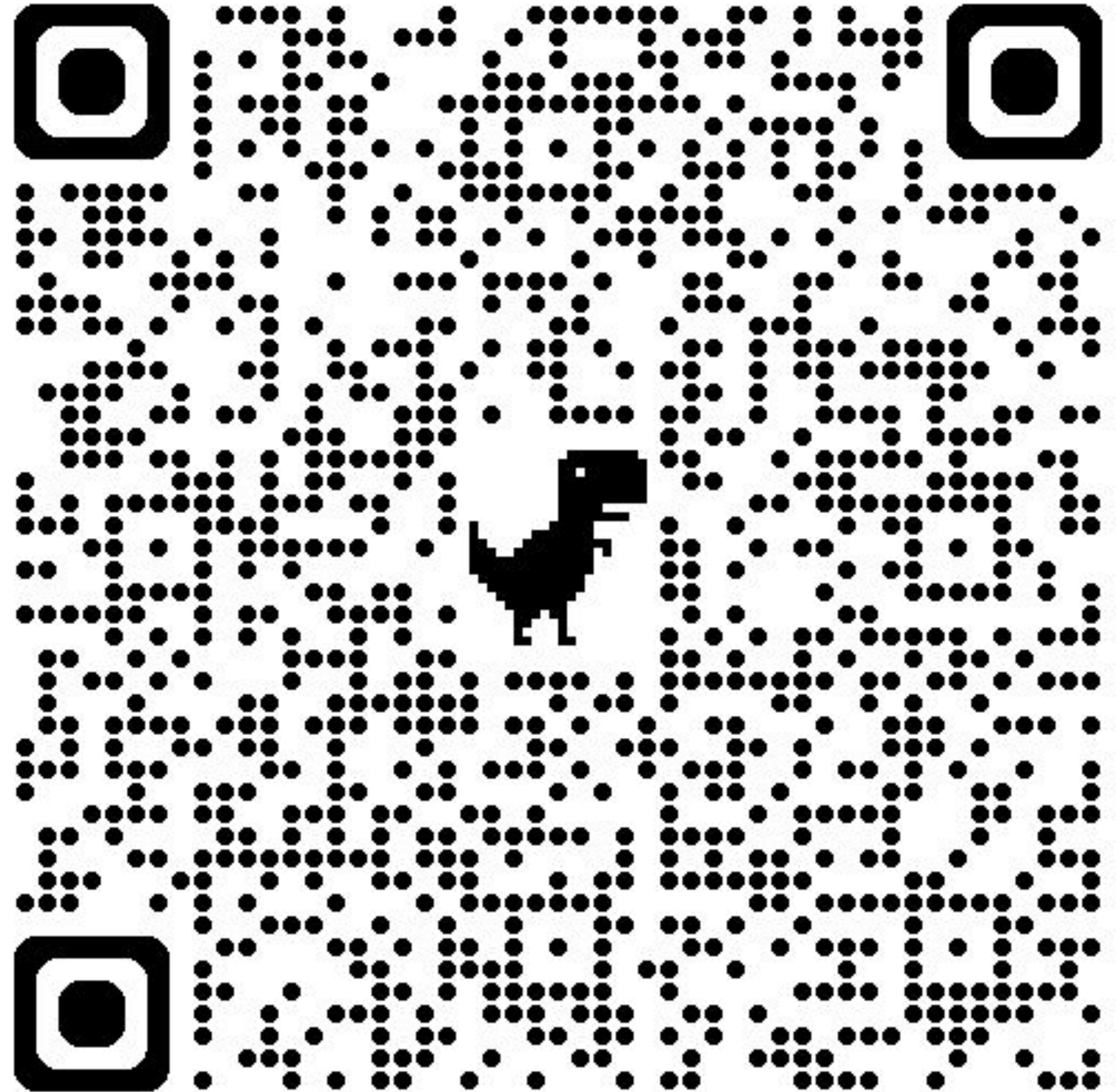
About me

- Machine Learning Consultant at [Kineto.ai](#)
- AI Researcher at [Hasso Plattner Institute](#)
- Instructor at [NVIDIA's Deep Learning Institute](#) and Berlin's [Data Science Retreat](#)
- Organizer of the [Berlin Computer Vision Group](#)



[LinkedIn](#)

Slides



Slack channel

Join **#multimodality-workshop**

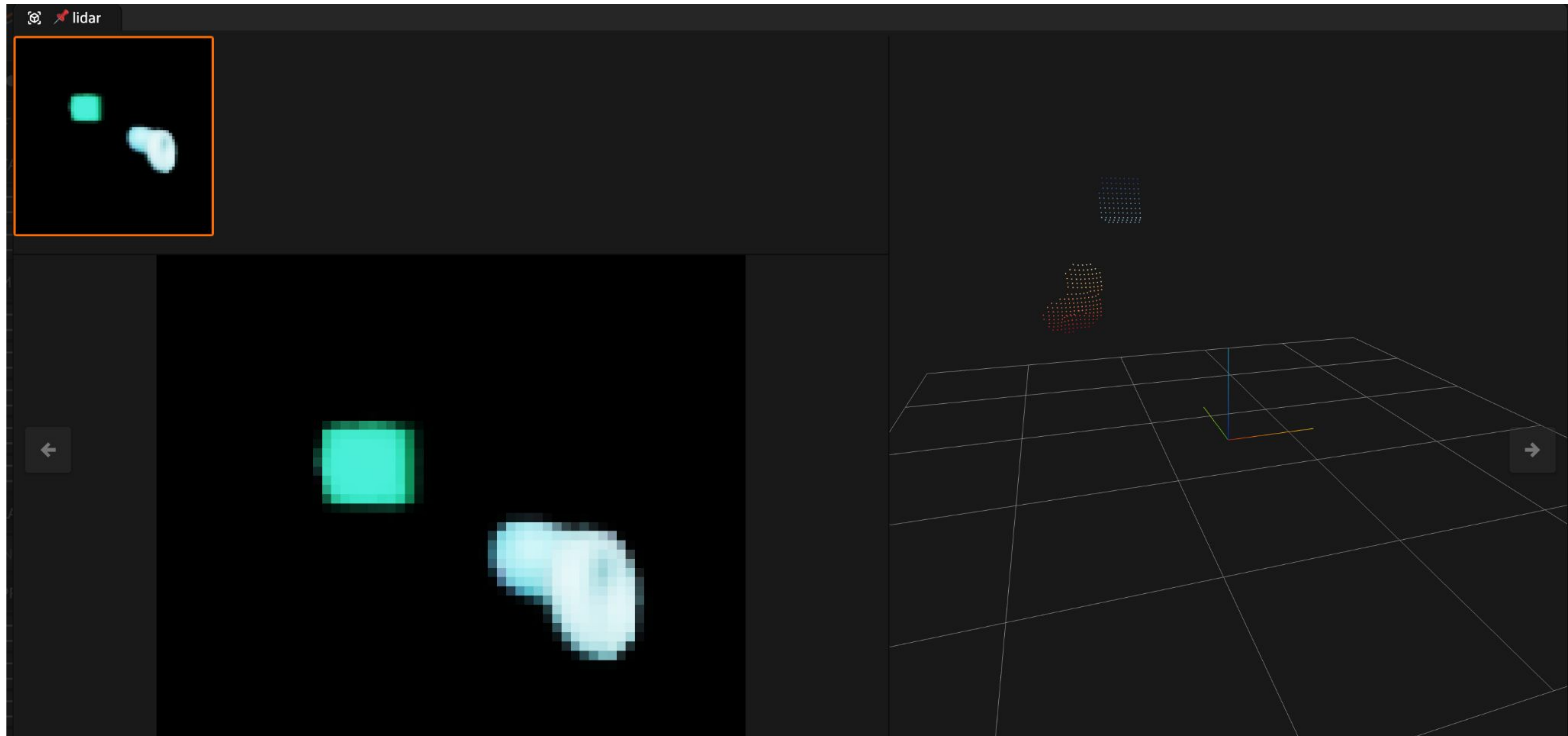
**Wifi password: fatamorgana
(AI Campus Guests)**



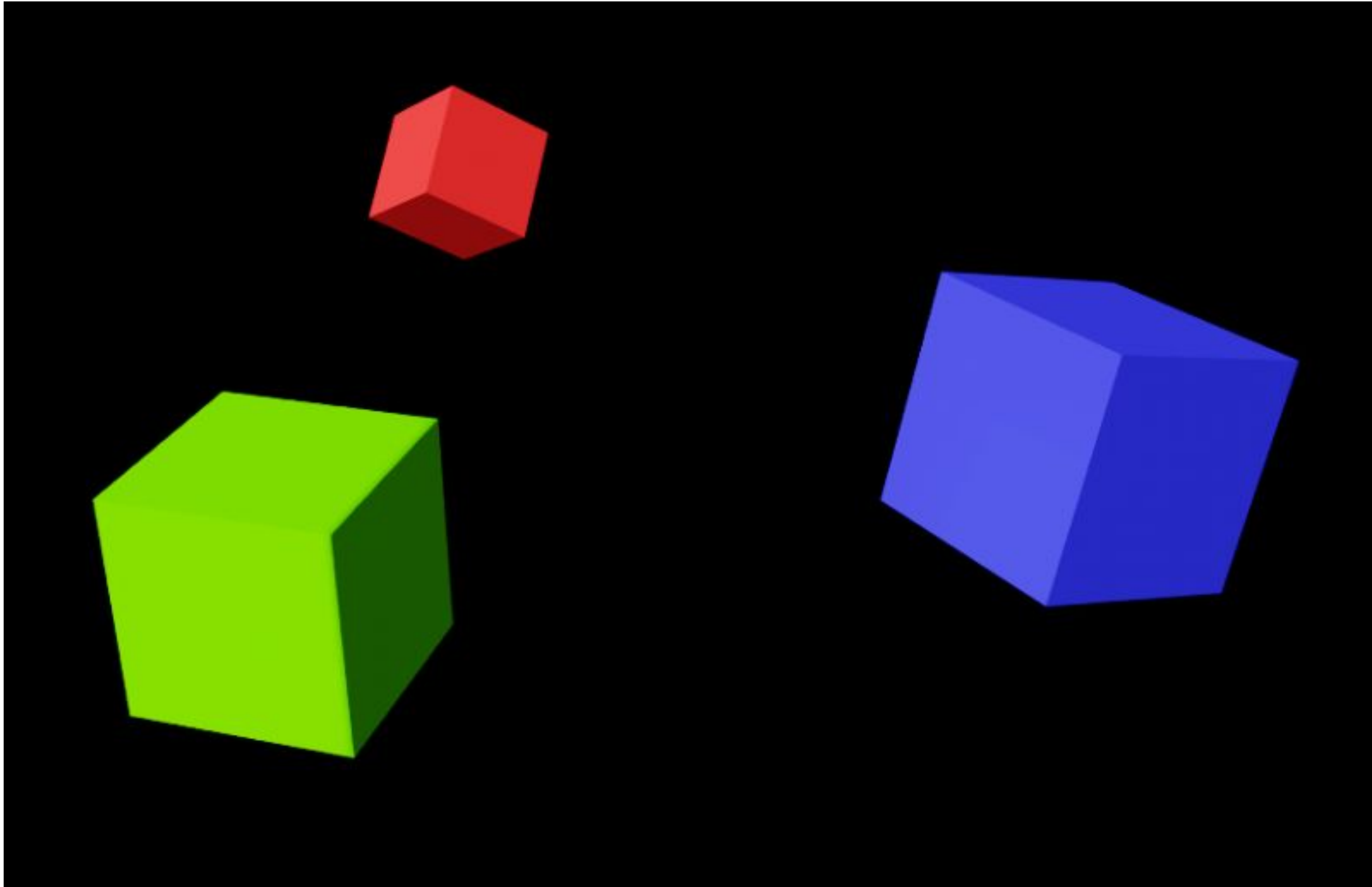
Colab Notebooks

- Early and Late Fusion - LiDAR and RGB
 - Visualization with FiftyOne (**try this notebook first**)
- Intermediate Fusion - Cubes Example
- * Visualization with FiftyOne

Task: Finding the center



Next Experiment



Agenda

- Part 1: Multimodal Data
- Part 2: Intermediate Fusion and CLIP
- Part 3: Cross-modal Projection and OCR Pipelines
- Part 4: VSS and GraphRAG
- Part 5: Wrap-up & Assessment

“

Hypothesis:

Any data source that can be turned into a matrix is useful for neural network training

”

Prerequisites

- Basic familiarity with convolutional neural networks (CNNs)
- Basic familiarity with PyTorch





Multimodality

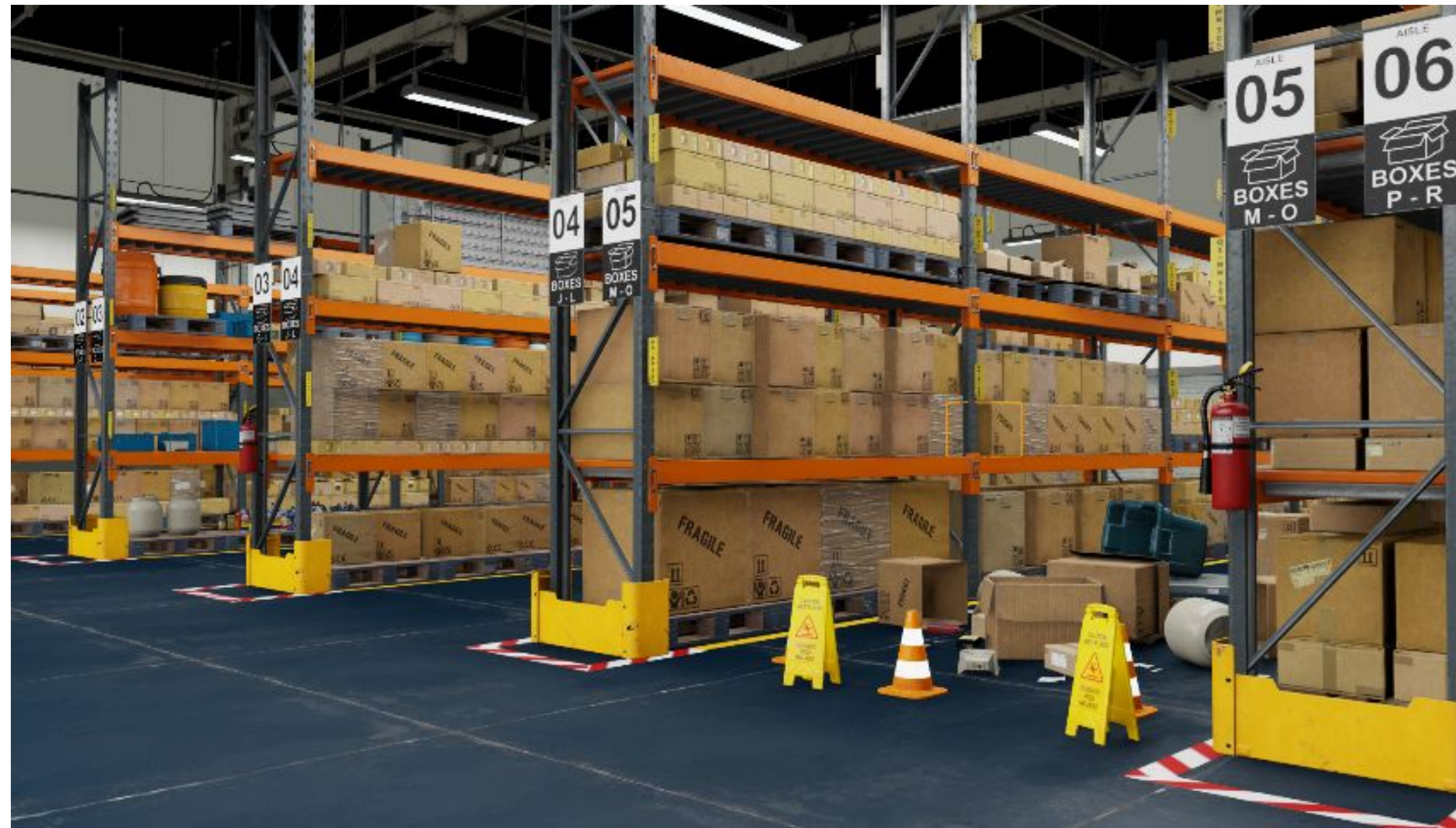
The Five Senses

How Humans Perceive

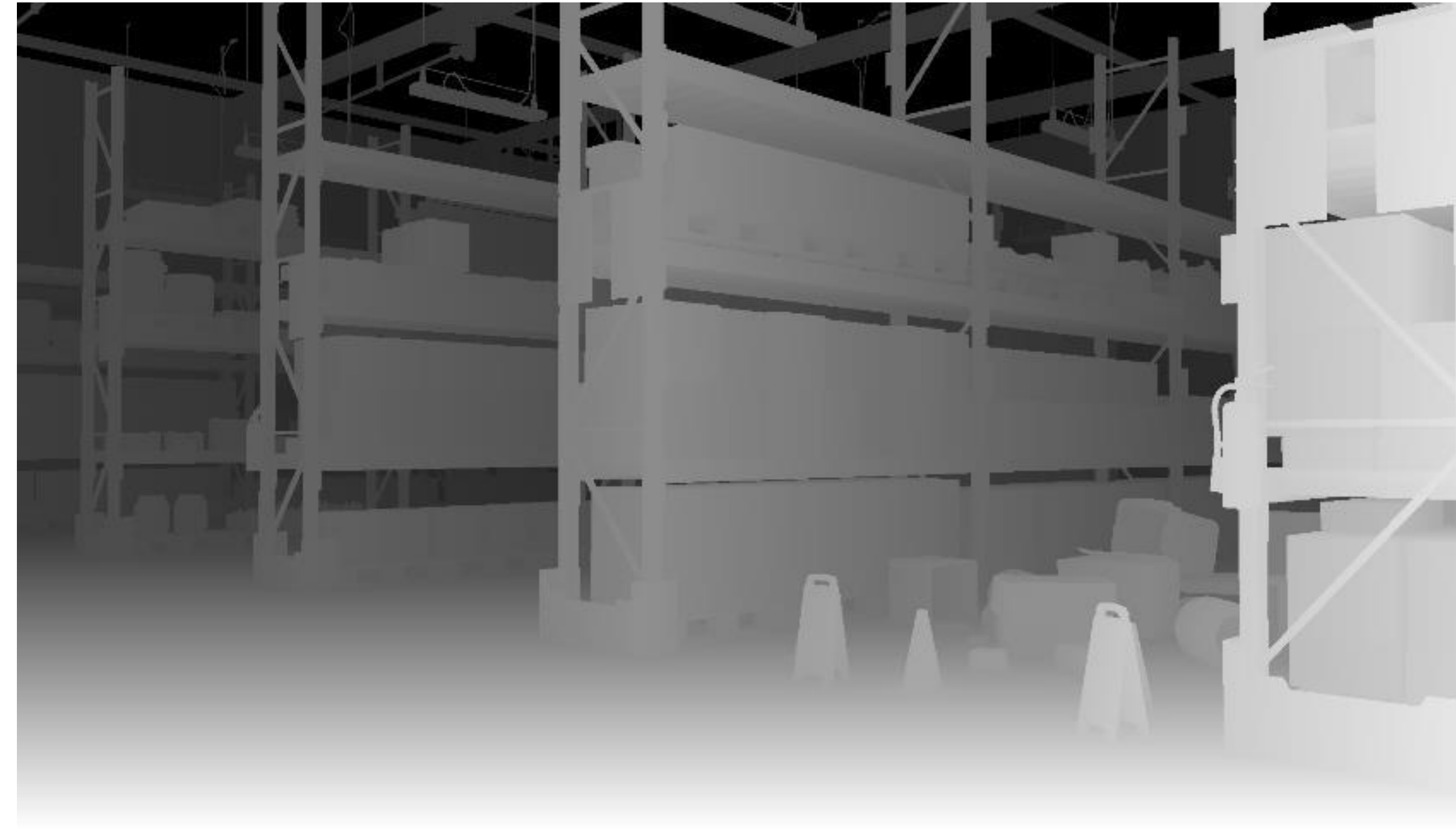


Sight

How A Computer Might See



RGB



Depth



Fisheye

Smell and Taste

How A Computer Might Smell and Taste

pH



Nutrition Facts

Nutrition Facts	
1 serving per container	
Serving size	140 g
Amount per serving	
Calories	100
% Daily Value *	
Total Fat 0.5g	1%
Saturated Fat 0g	0%
Trans Fat 0g	
Cholesterol 0mg	0%
Sodium 0mg	0%
Total Carbohydrate 24g	9%
Dietary Fiber 6g	22%
Total Sugars 15g	
Includes 0g Added Sugars	0%
Protein <1g	
Vitamin D 0mcg	0%
Calcium 0mg	0%
Iron 0.3mg	2%
Potassium 120mg	2%
* The % Daily Value (DV) tells you how much a nutrient in a serving of food contributes to a daily diet. 2,000 calories a day is used for general nutrition advice.	

Molecular Weight



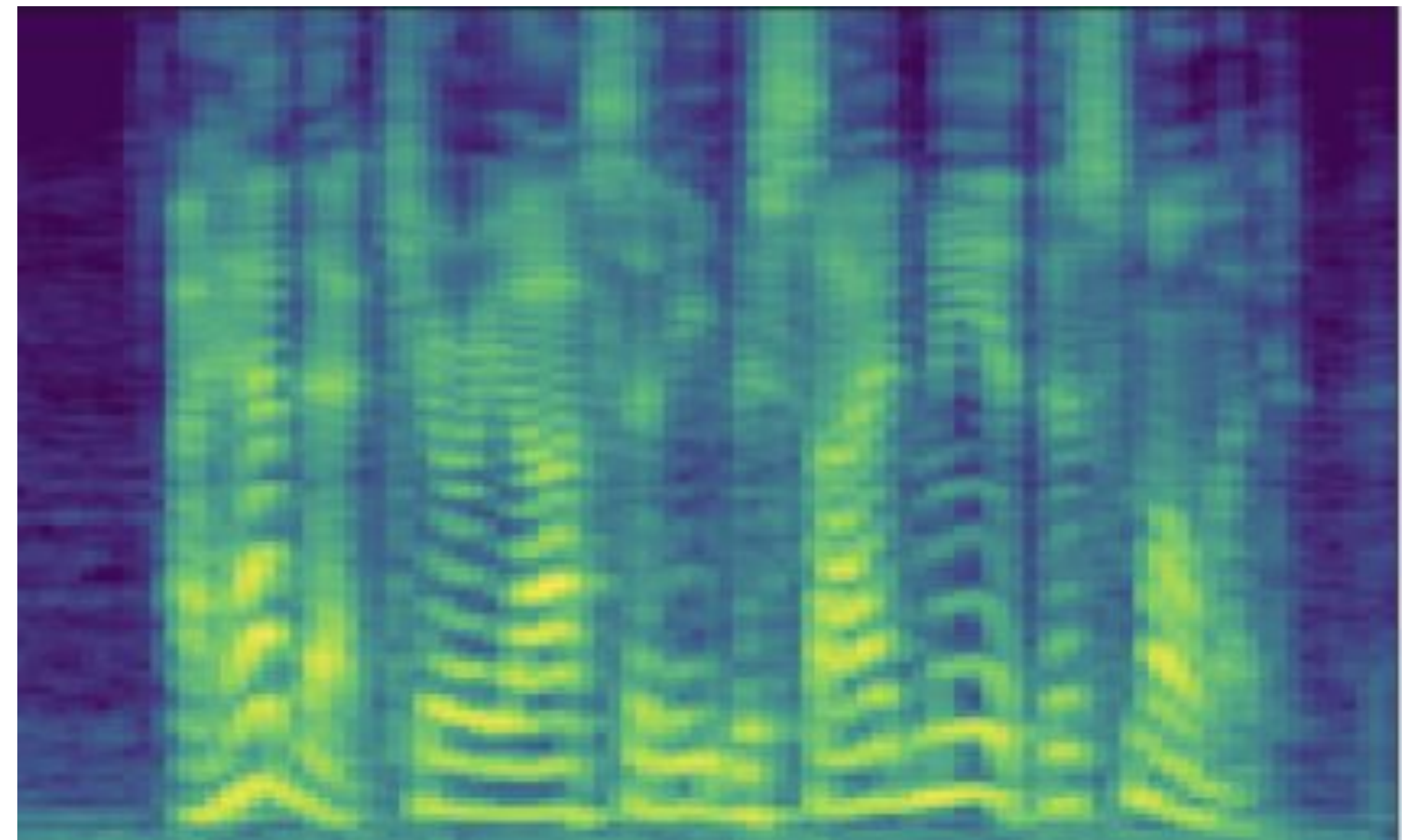
Hearing

How A Computer Might Hear

Sheet Music



Spectrogram



Touch

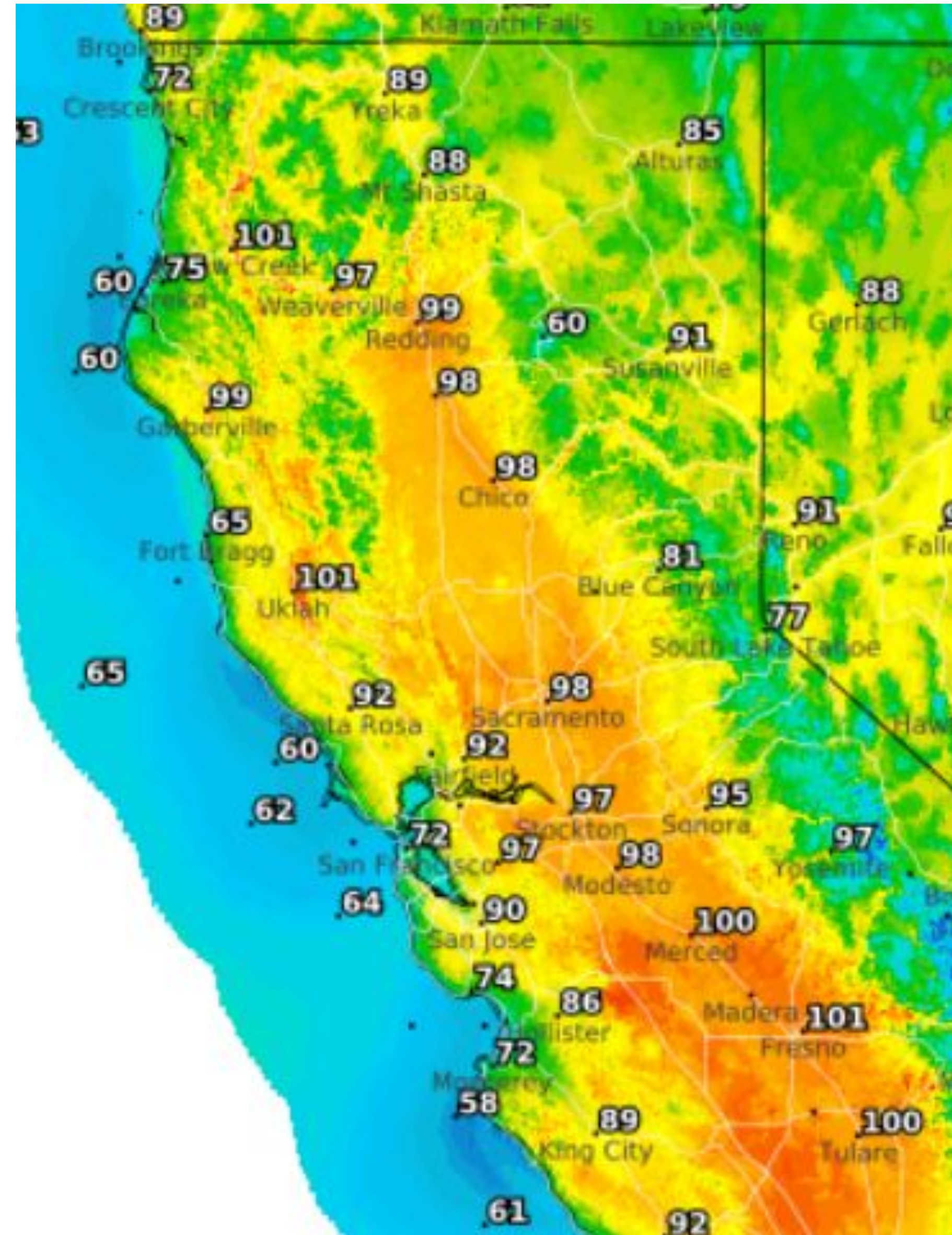
How A Computer Might Feel

Tensile Strength



By Christian V.

Weather Maps



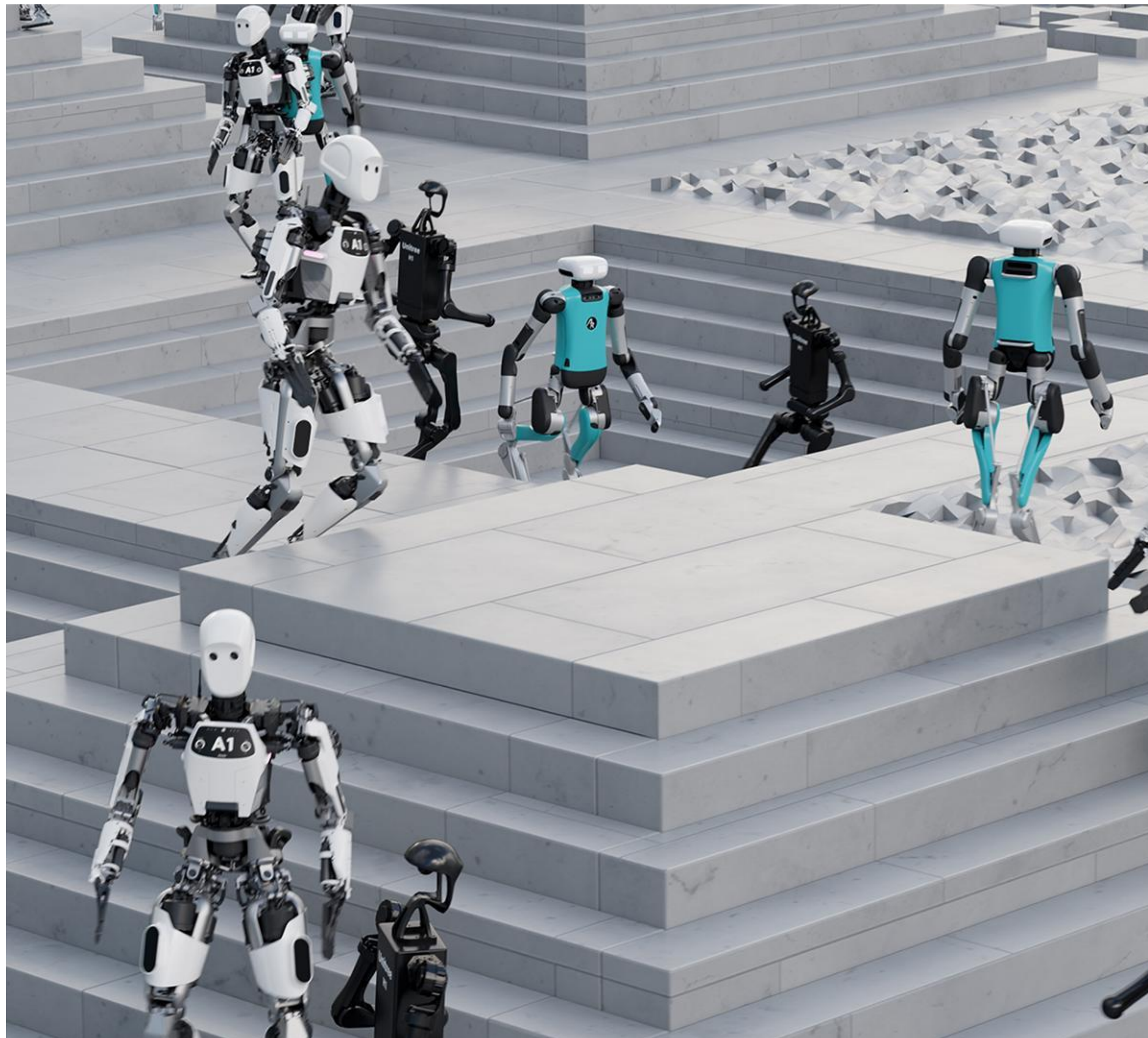
Weight



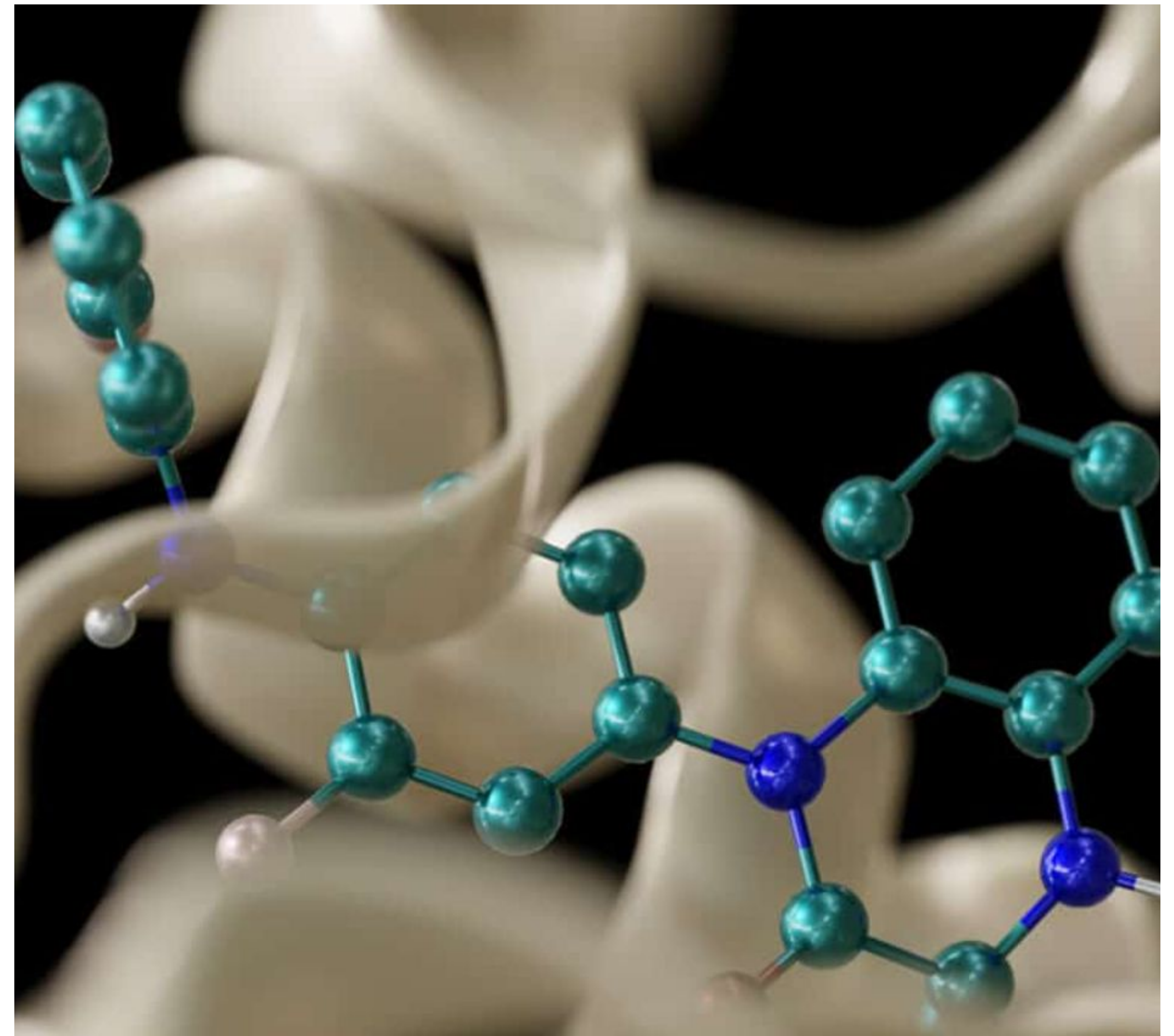
Multimodal Applications

Common Multimodal Use Cases

Robotics

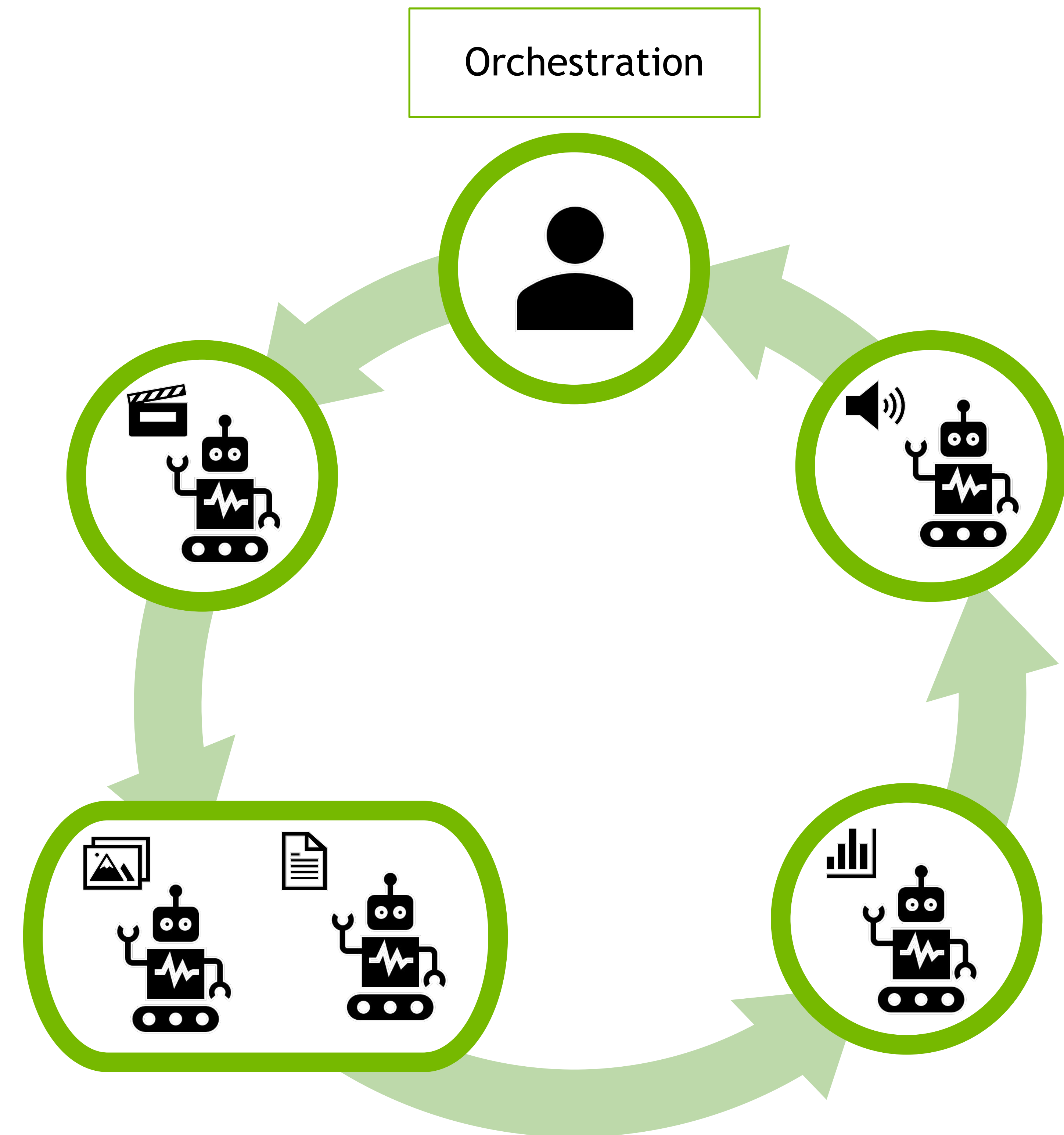
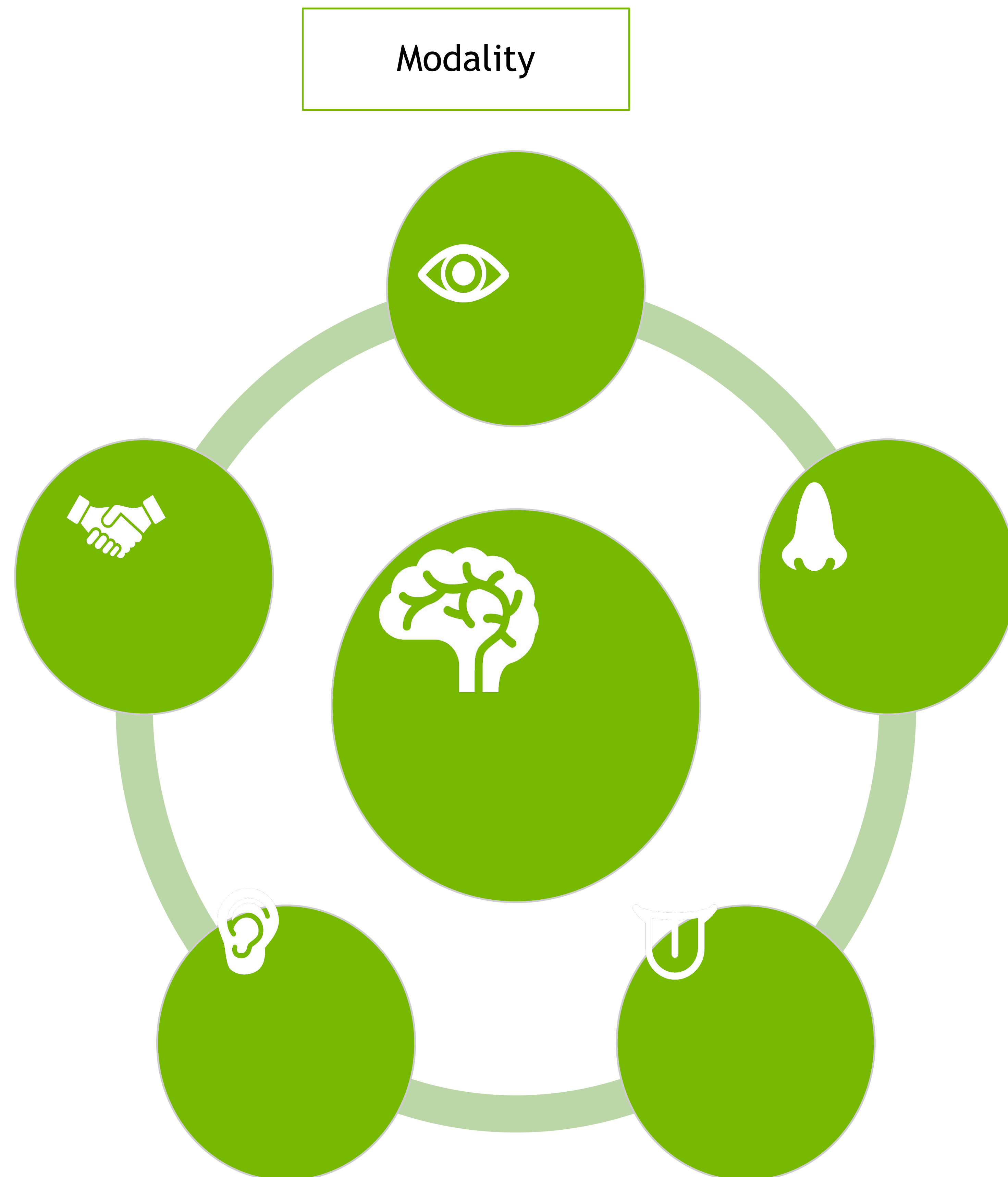


Biology



Multimodal vs Orchestration

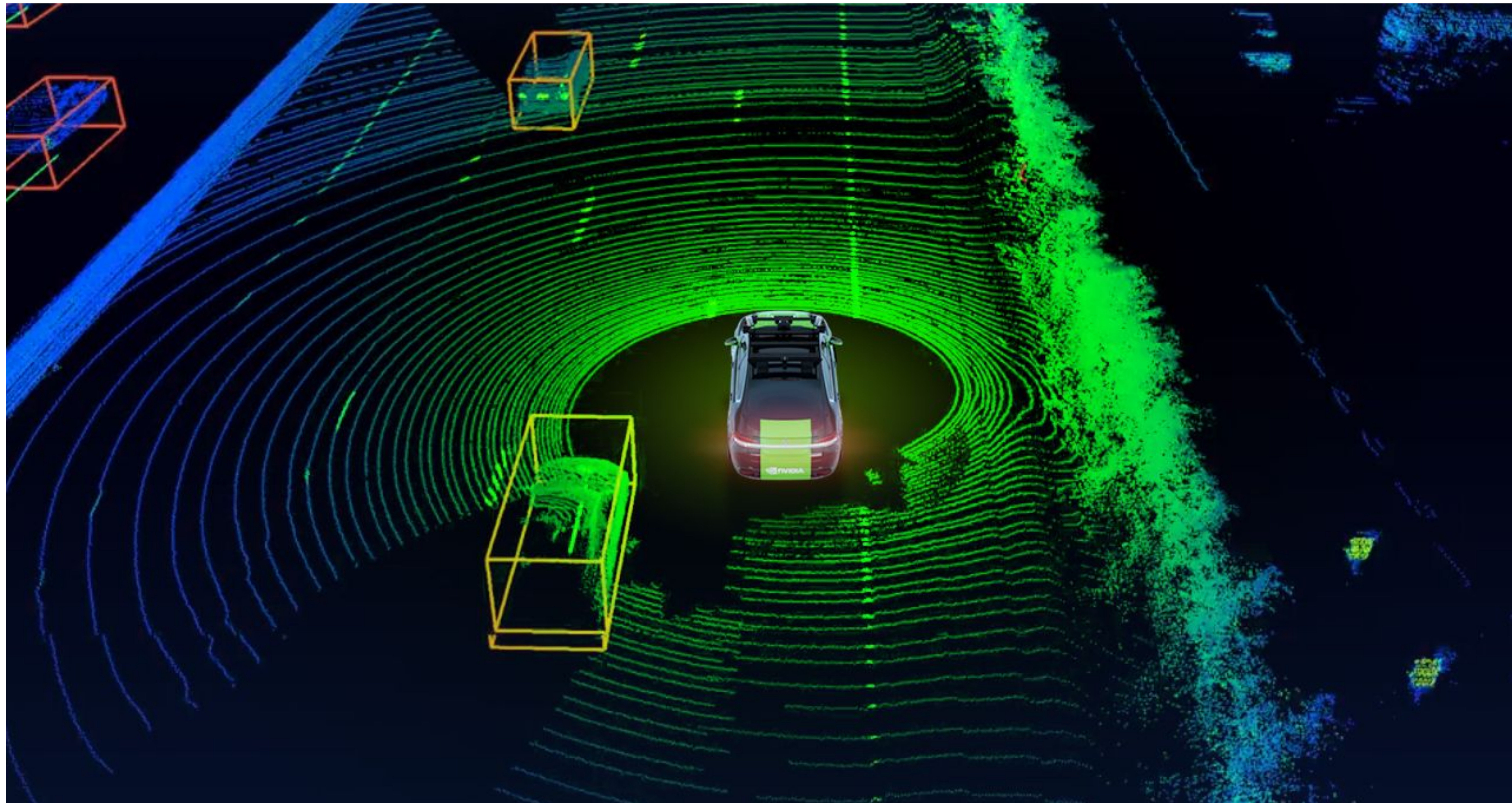
Common Multimodal Use Cases





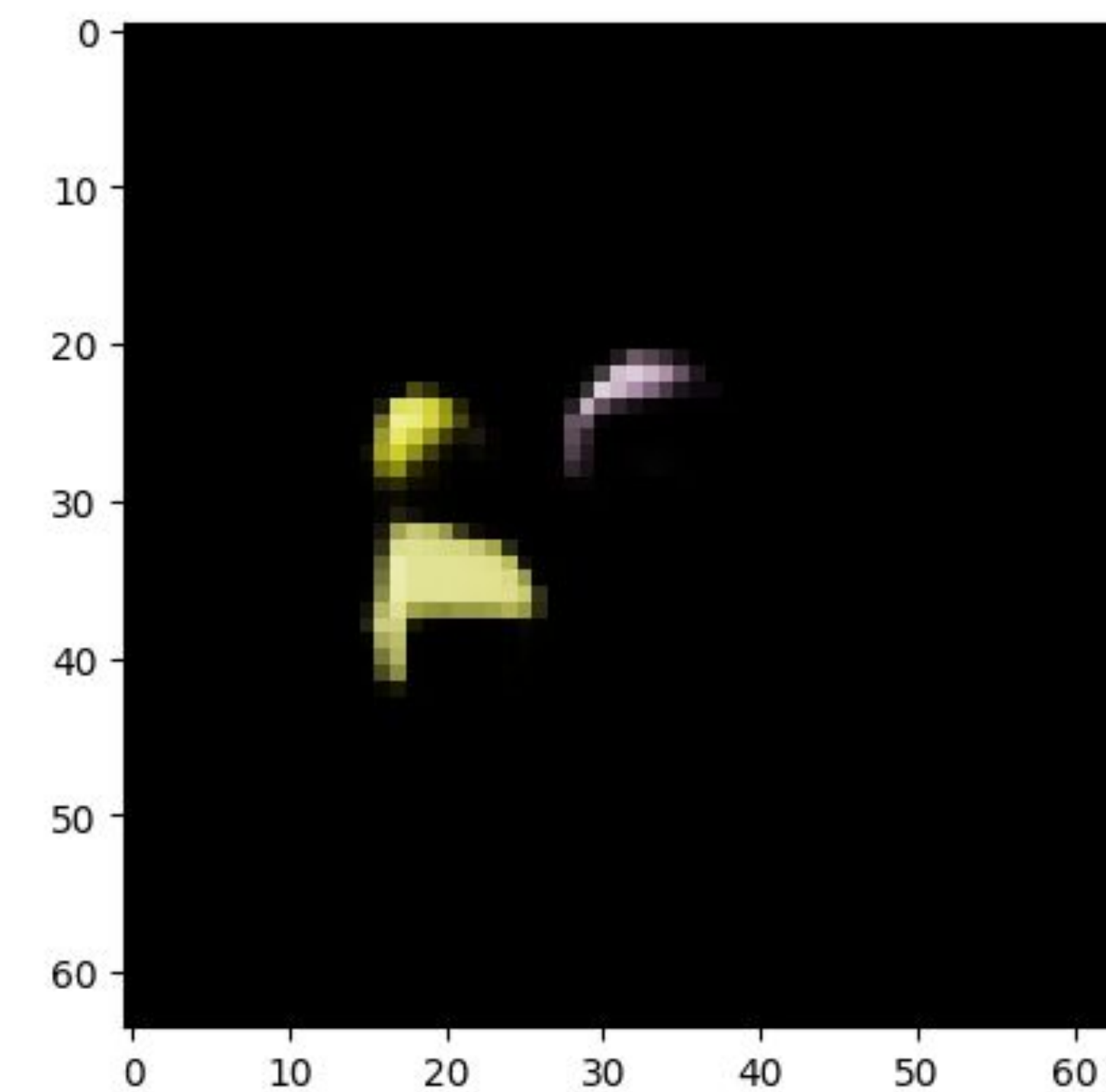
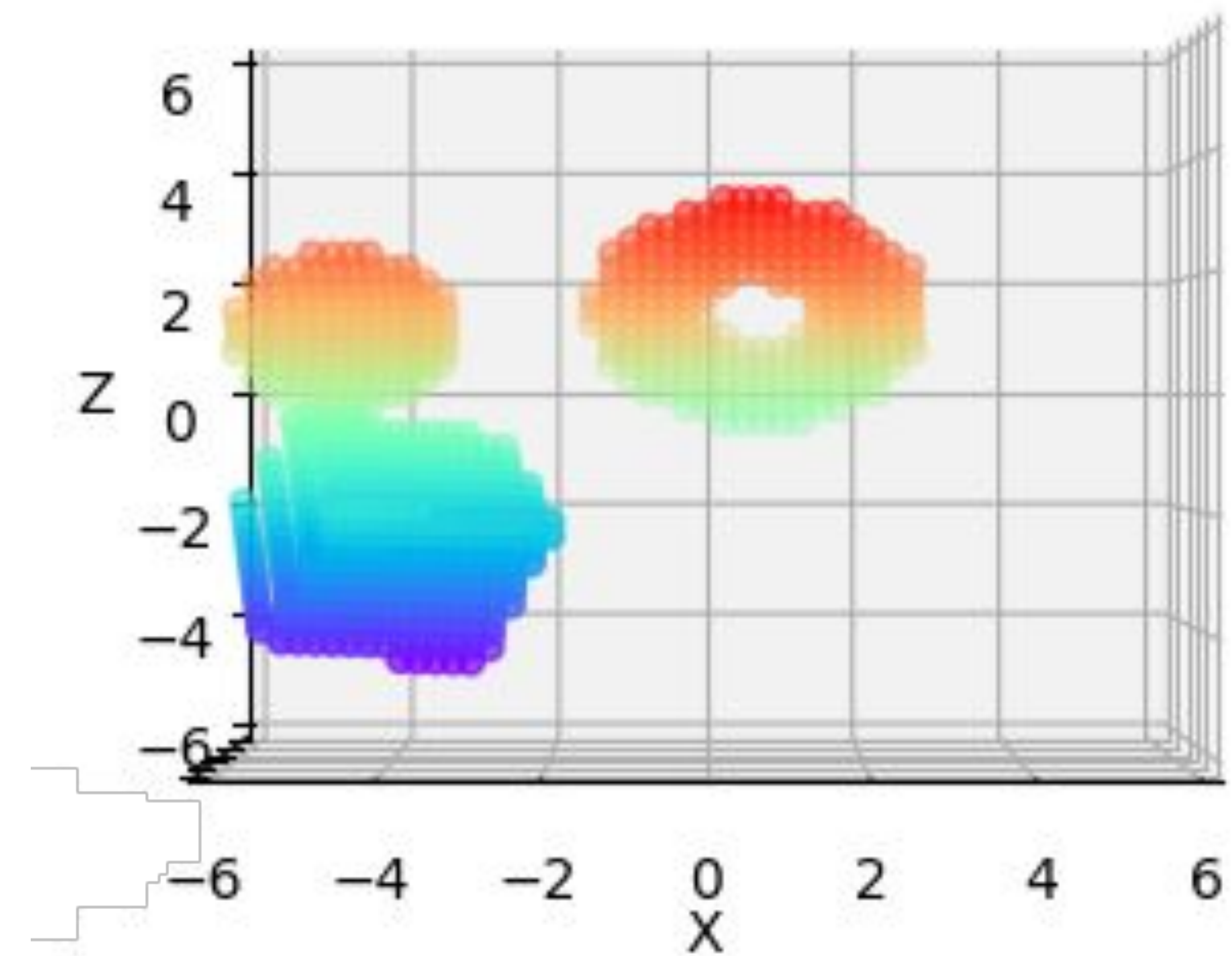
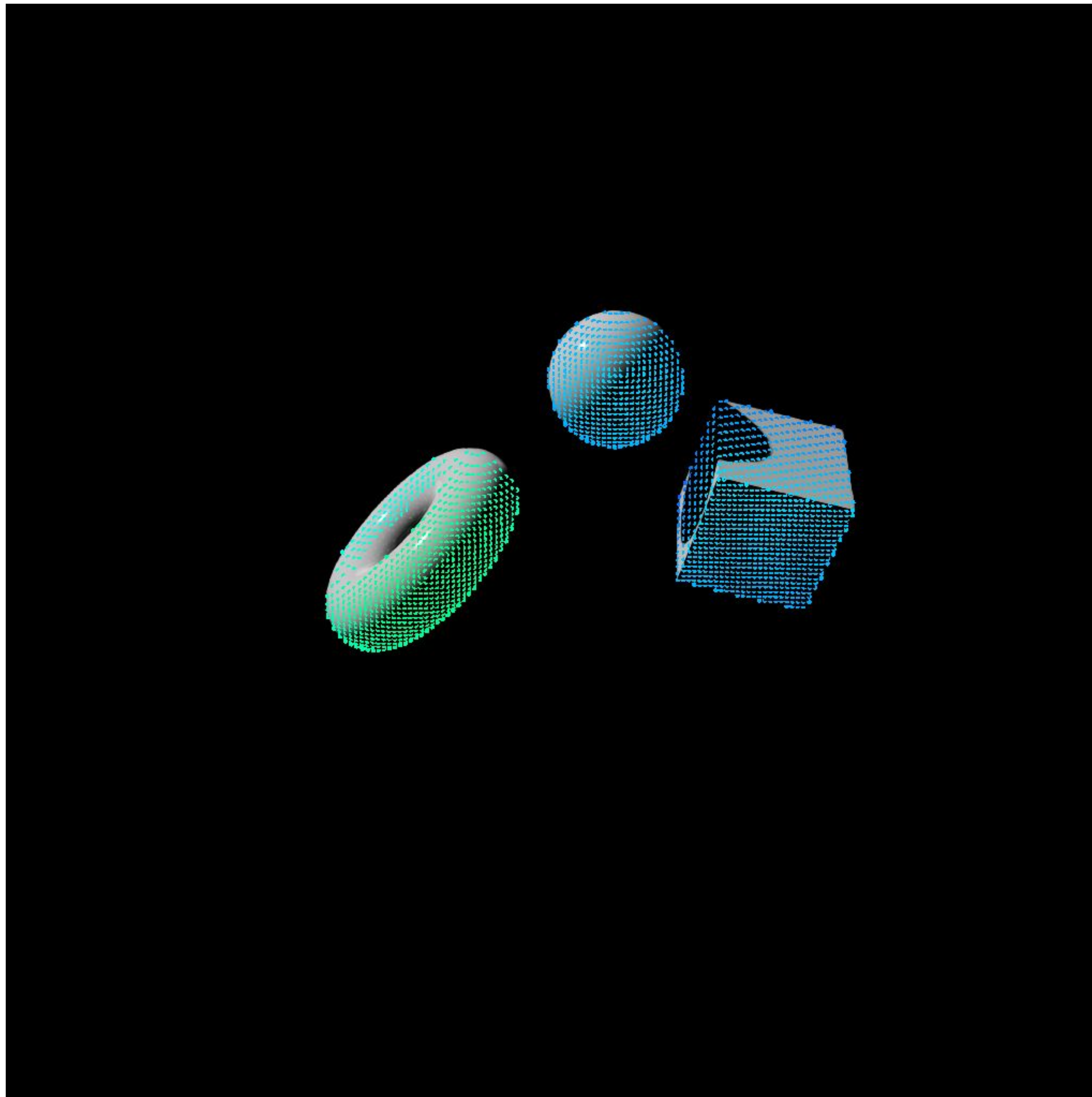
Lab 1a – Early and Late Fusion

Using LiDAR Data



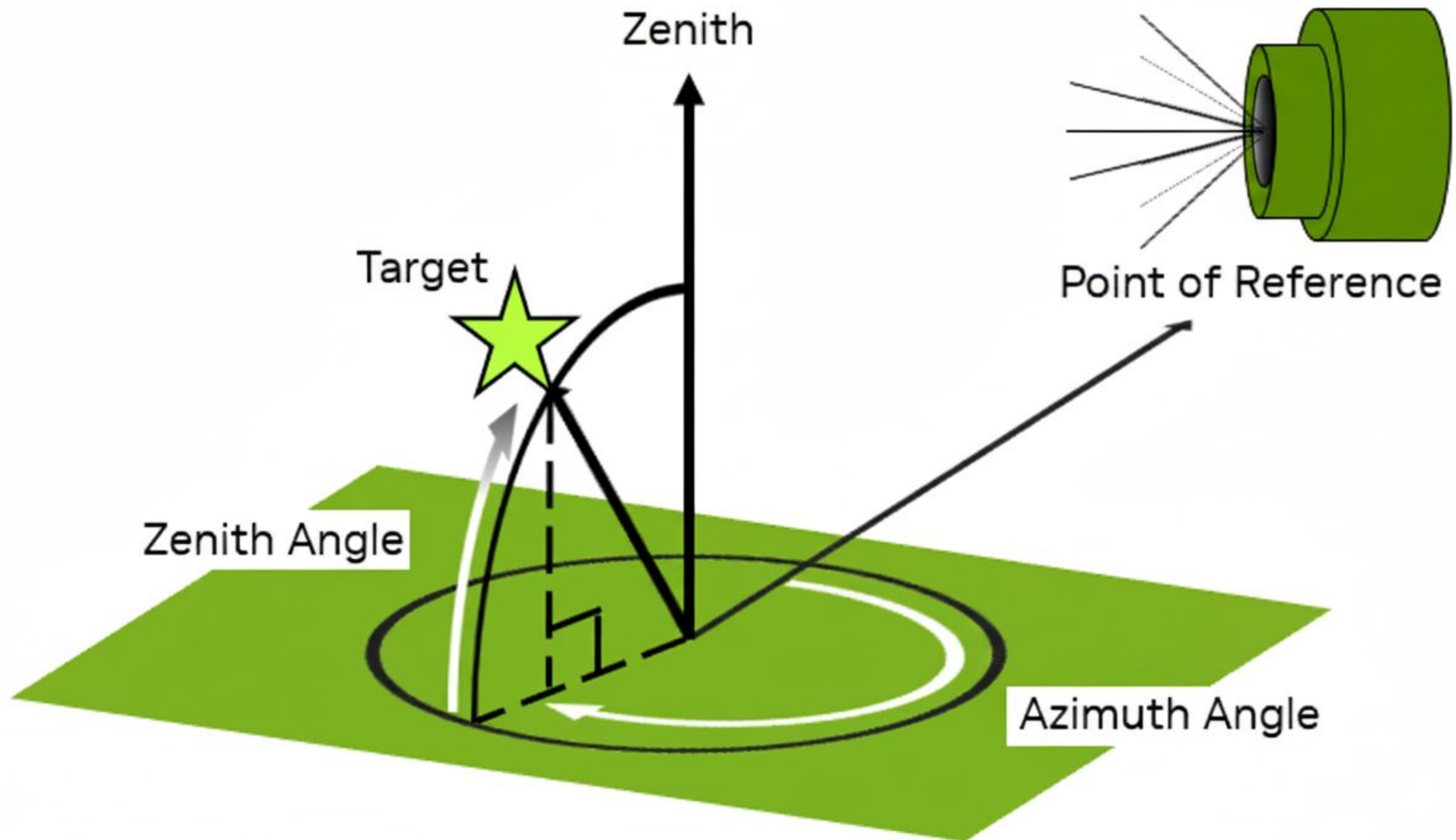
LiDAR vs Camera

Common Multimodal Use Cases

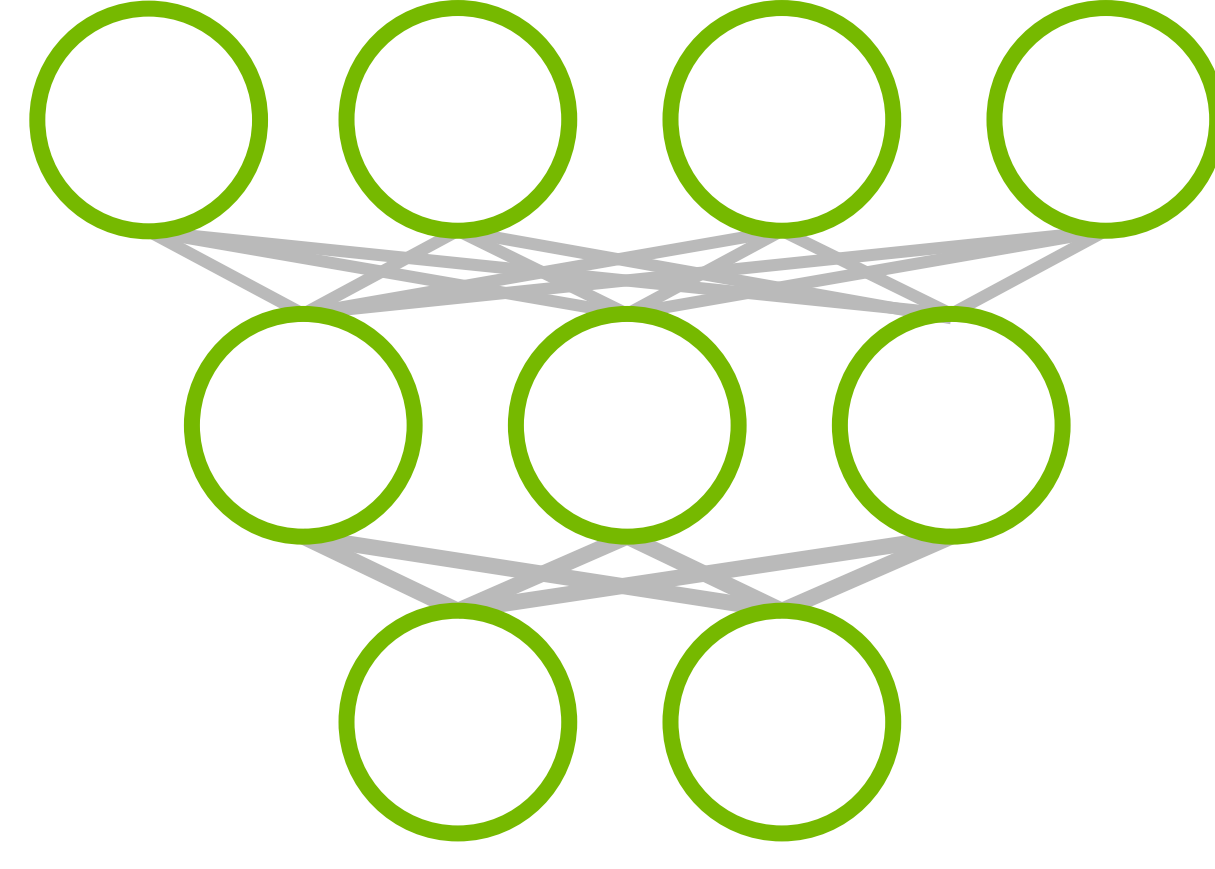
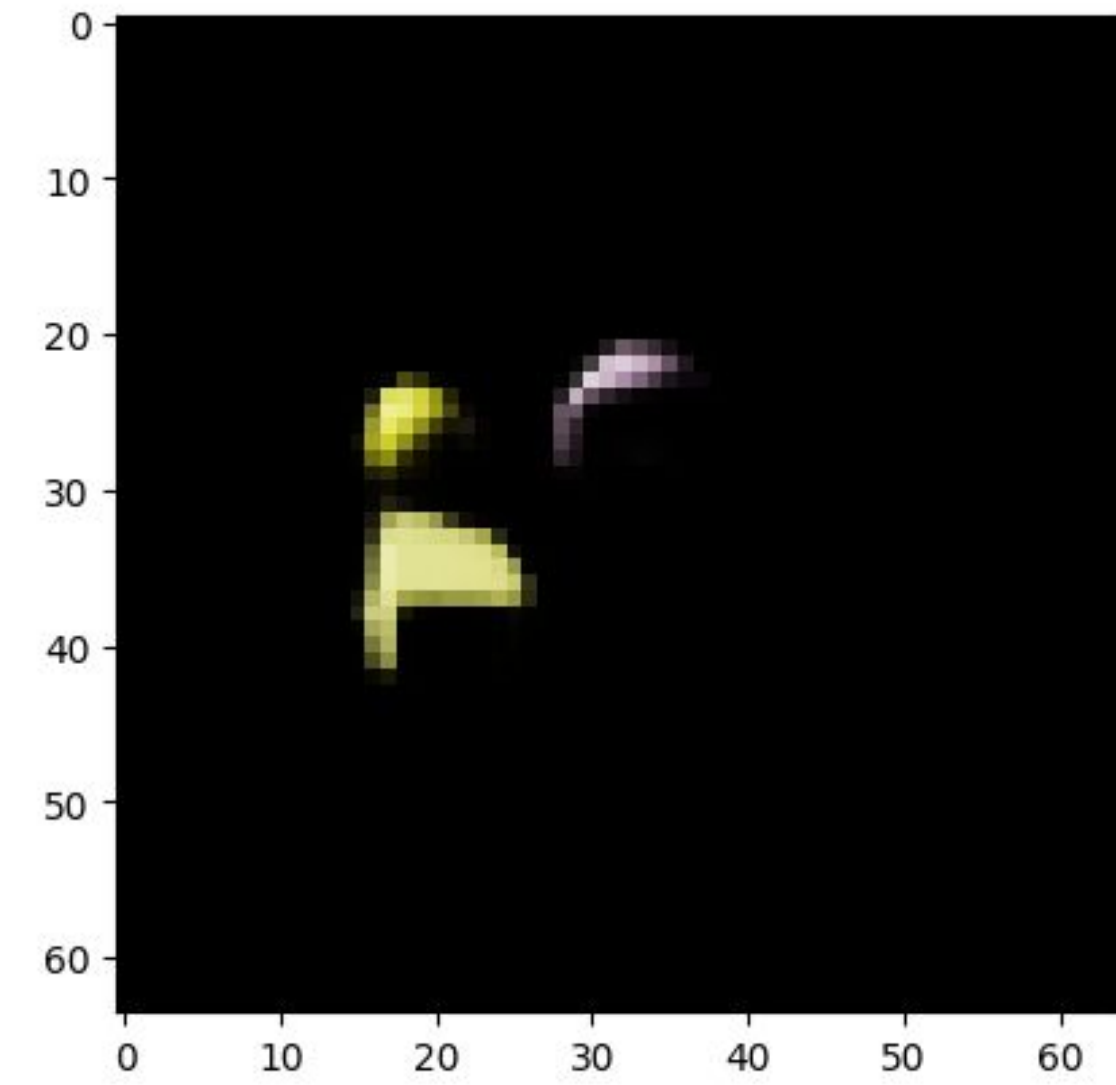
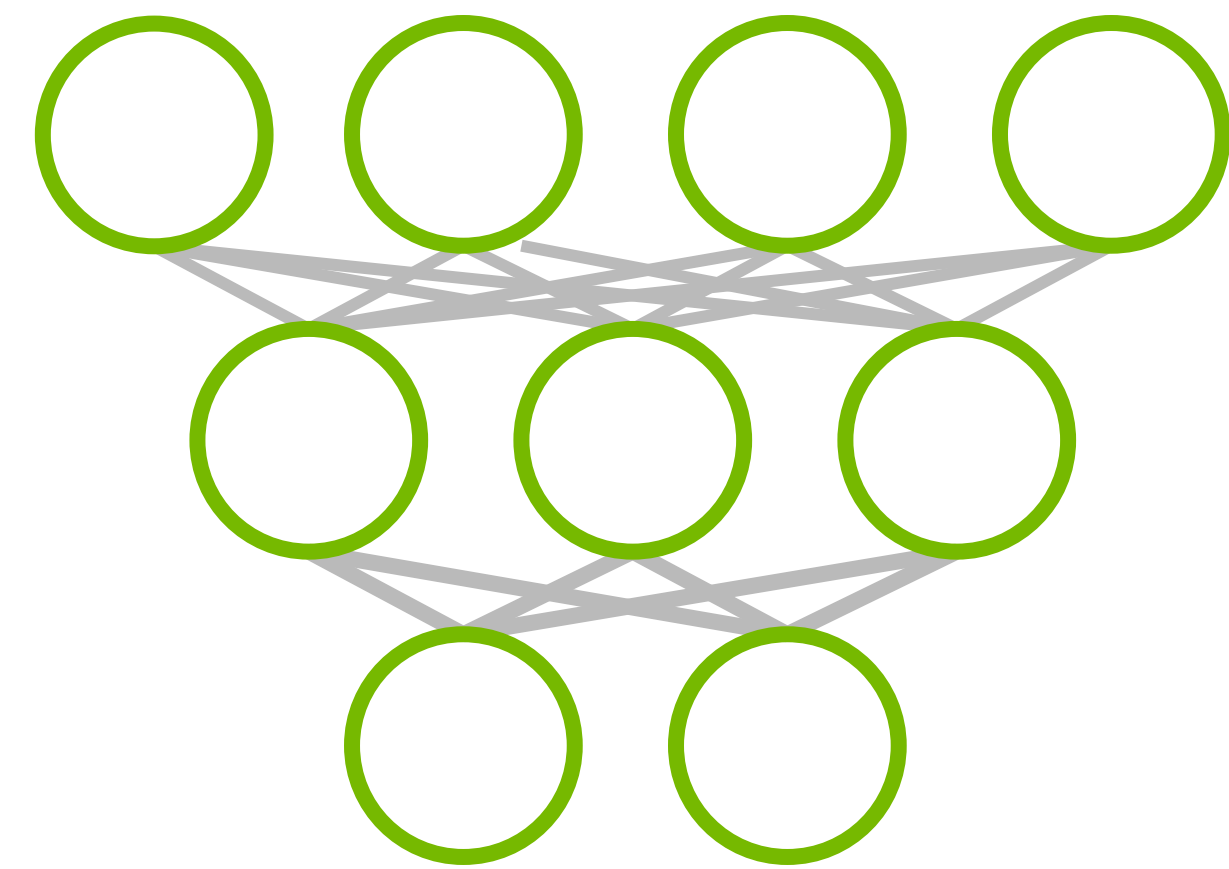
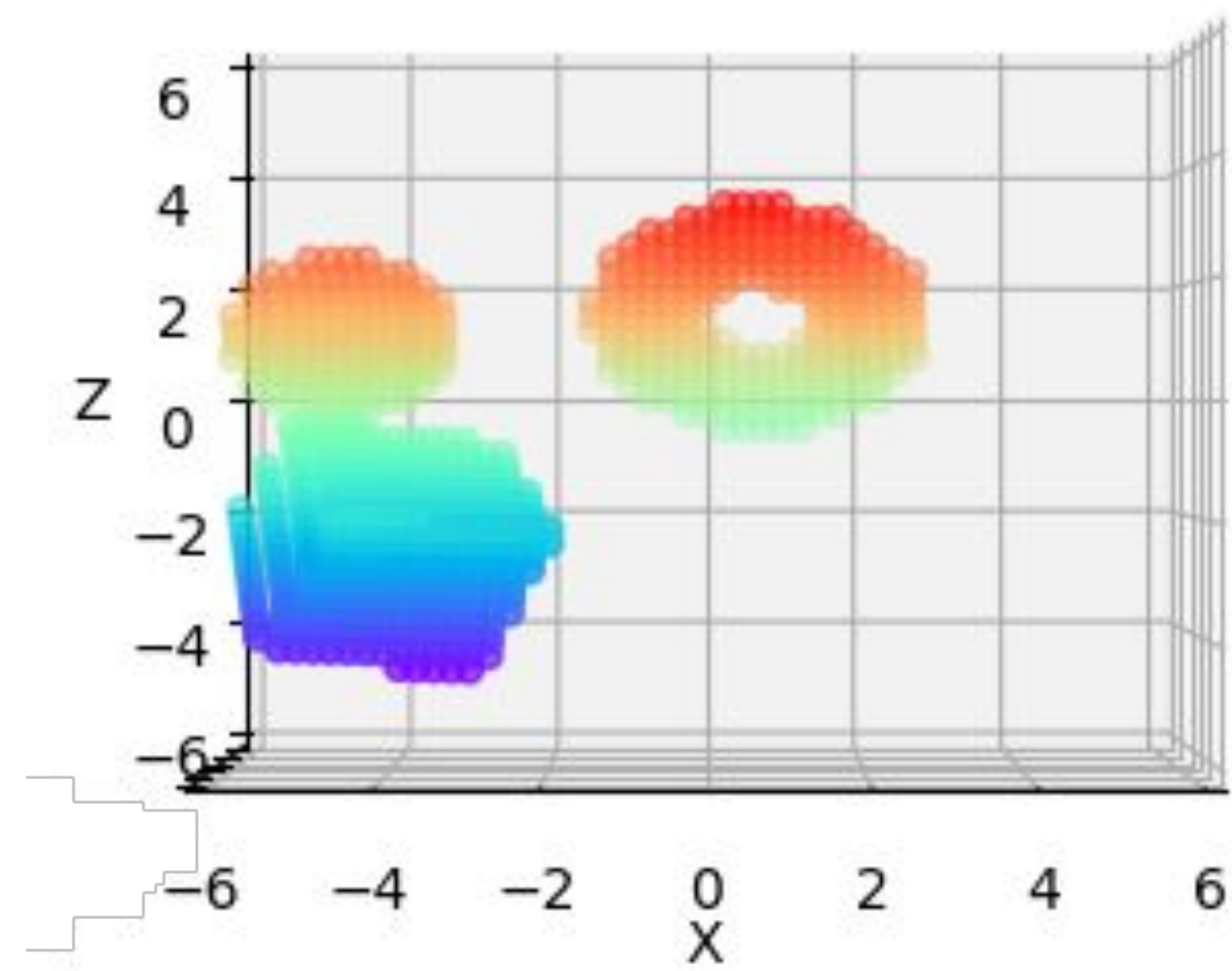


Azimuth and Zenith

Finding Stars

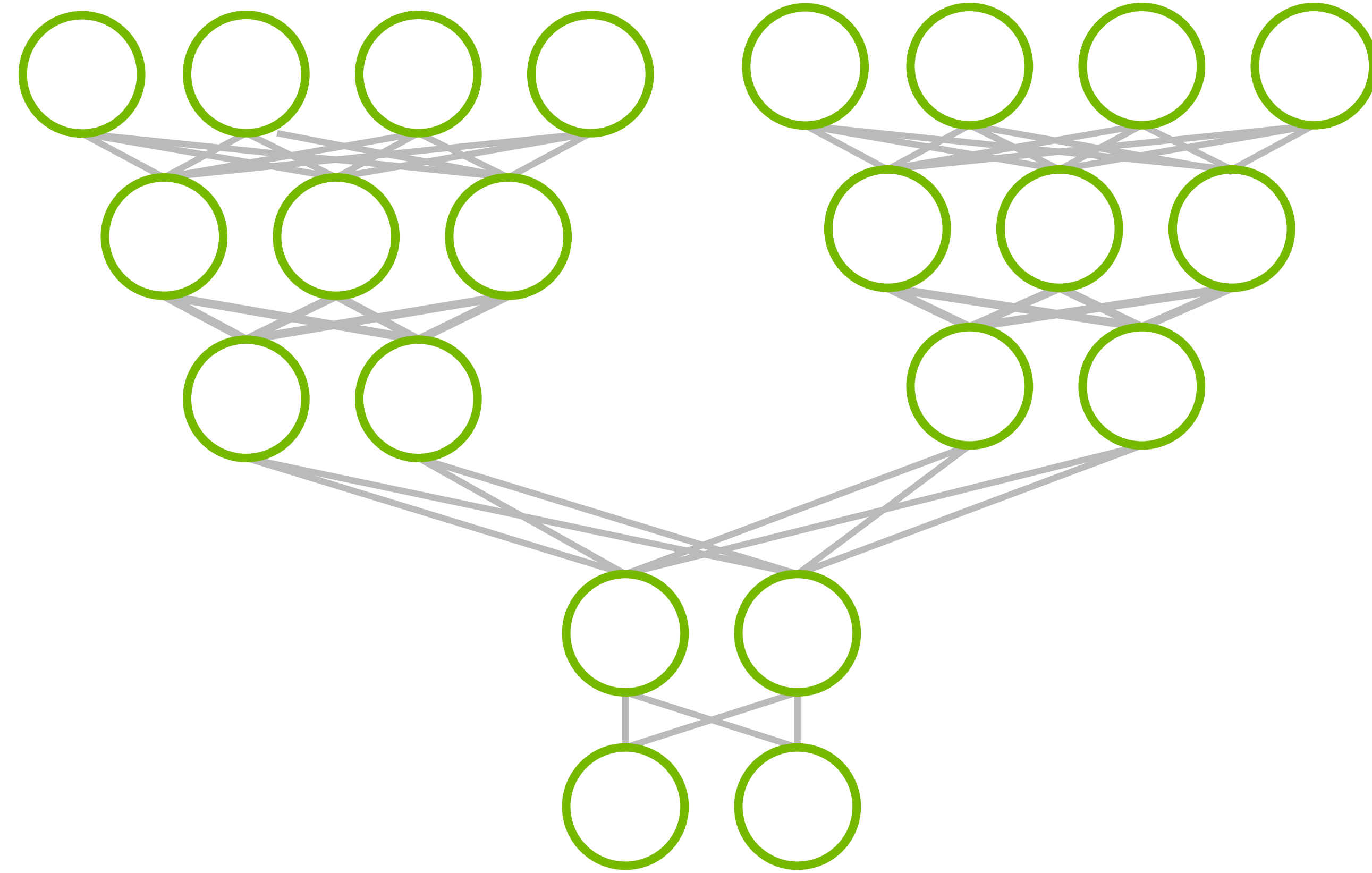
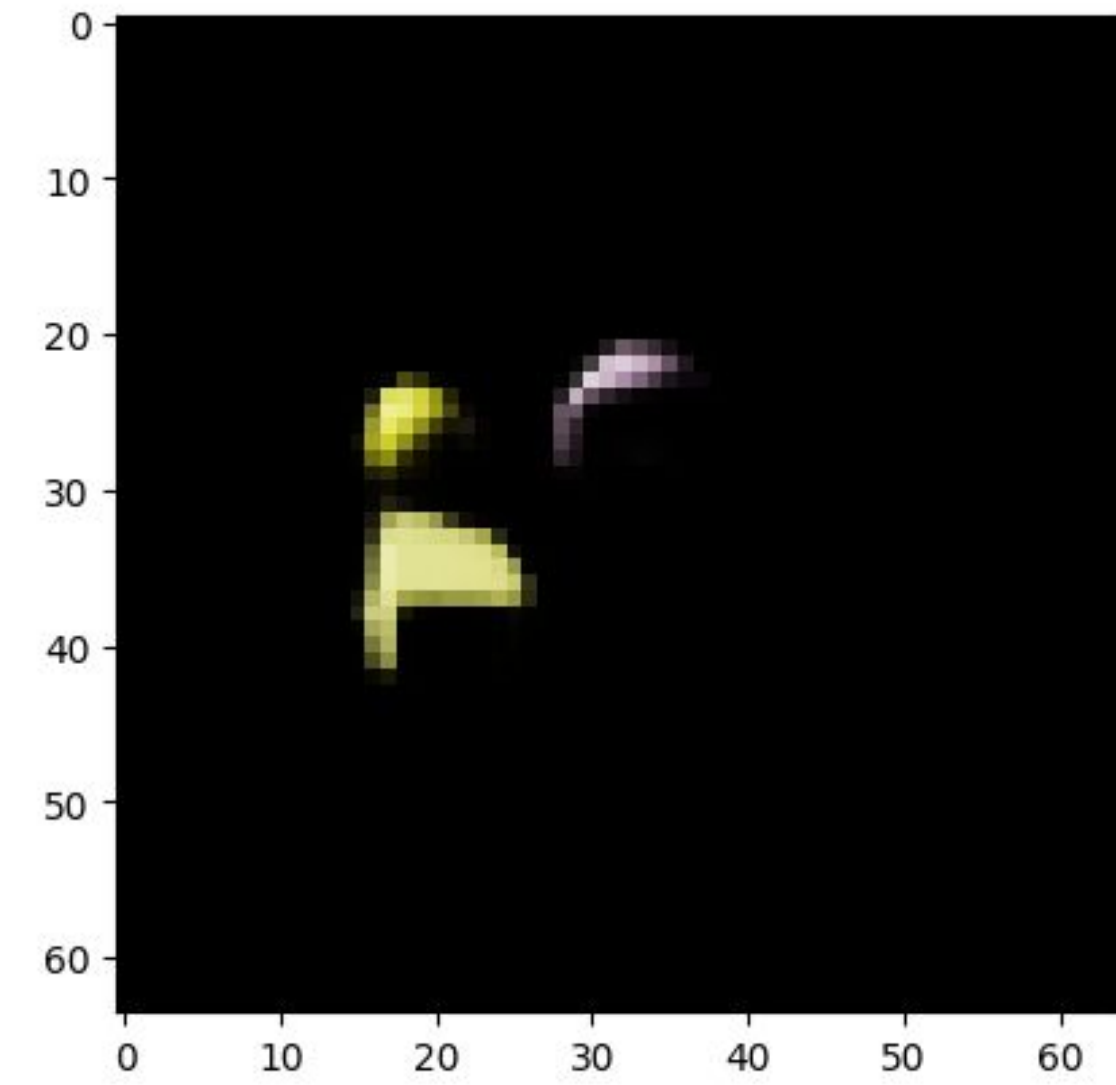
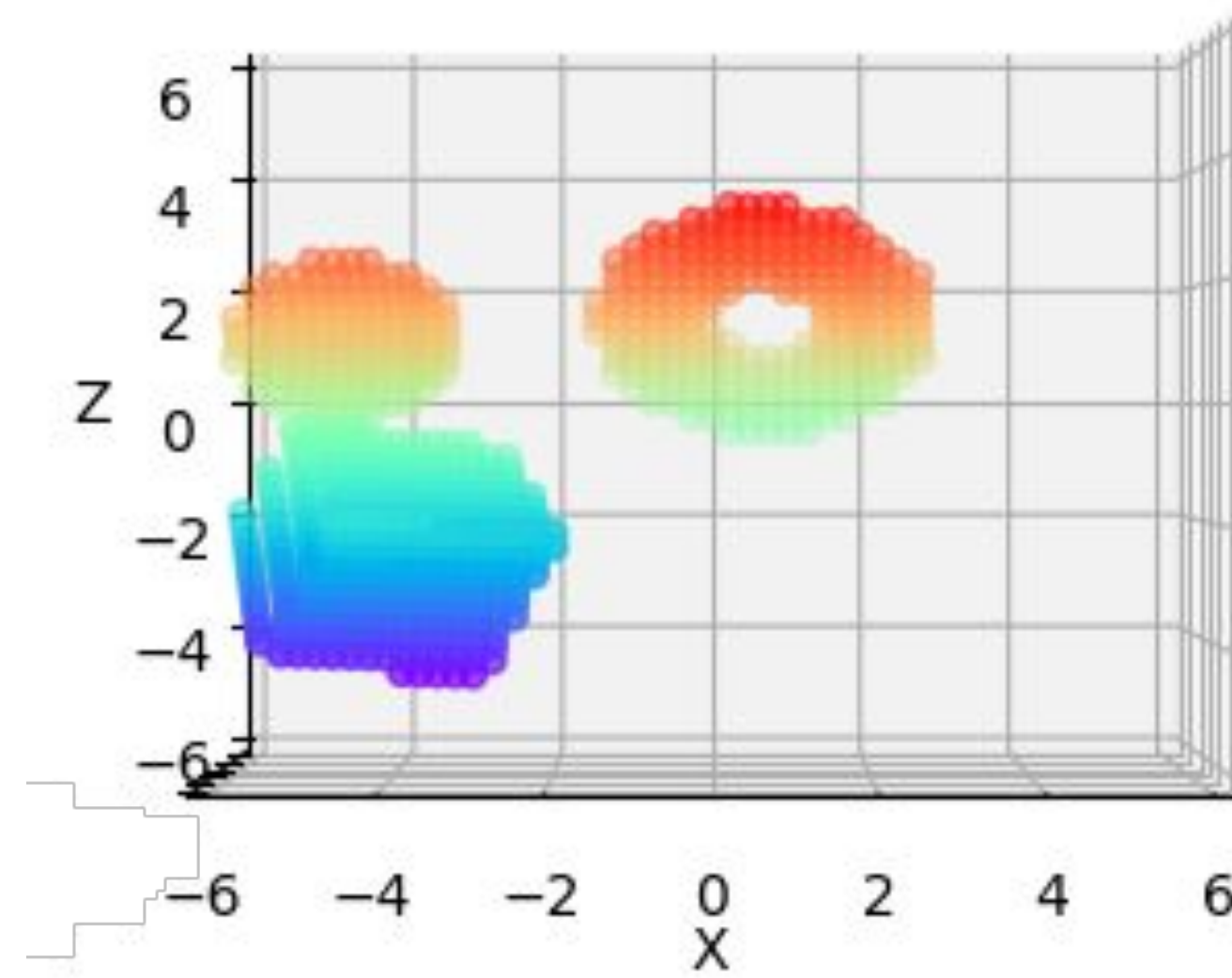


Fusion Techniques



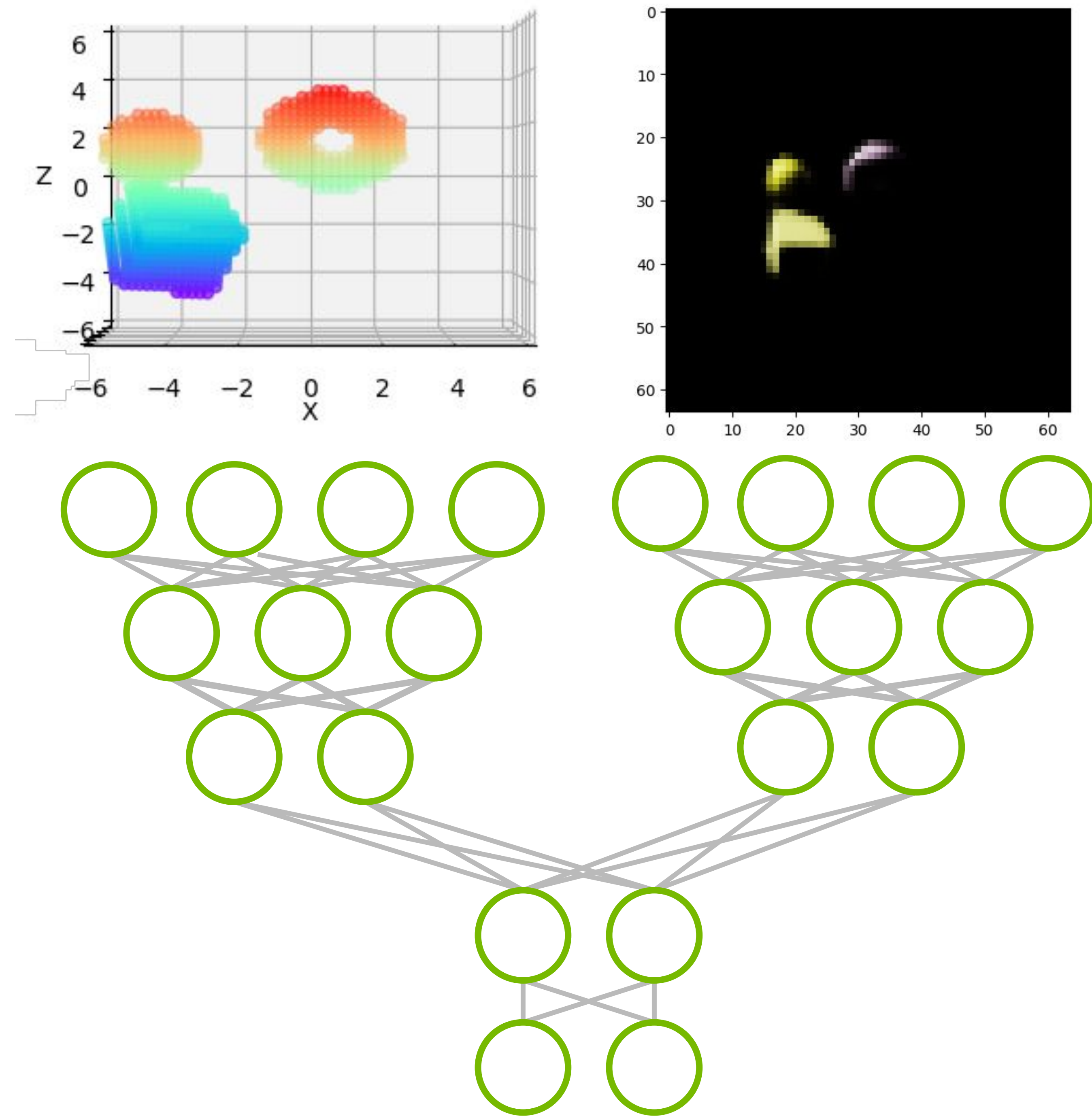
Fusion Techniques

Late

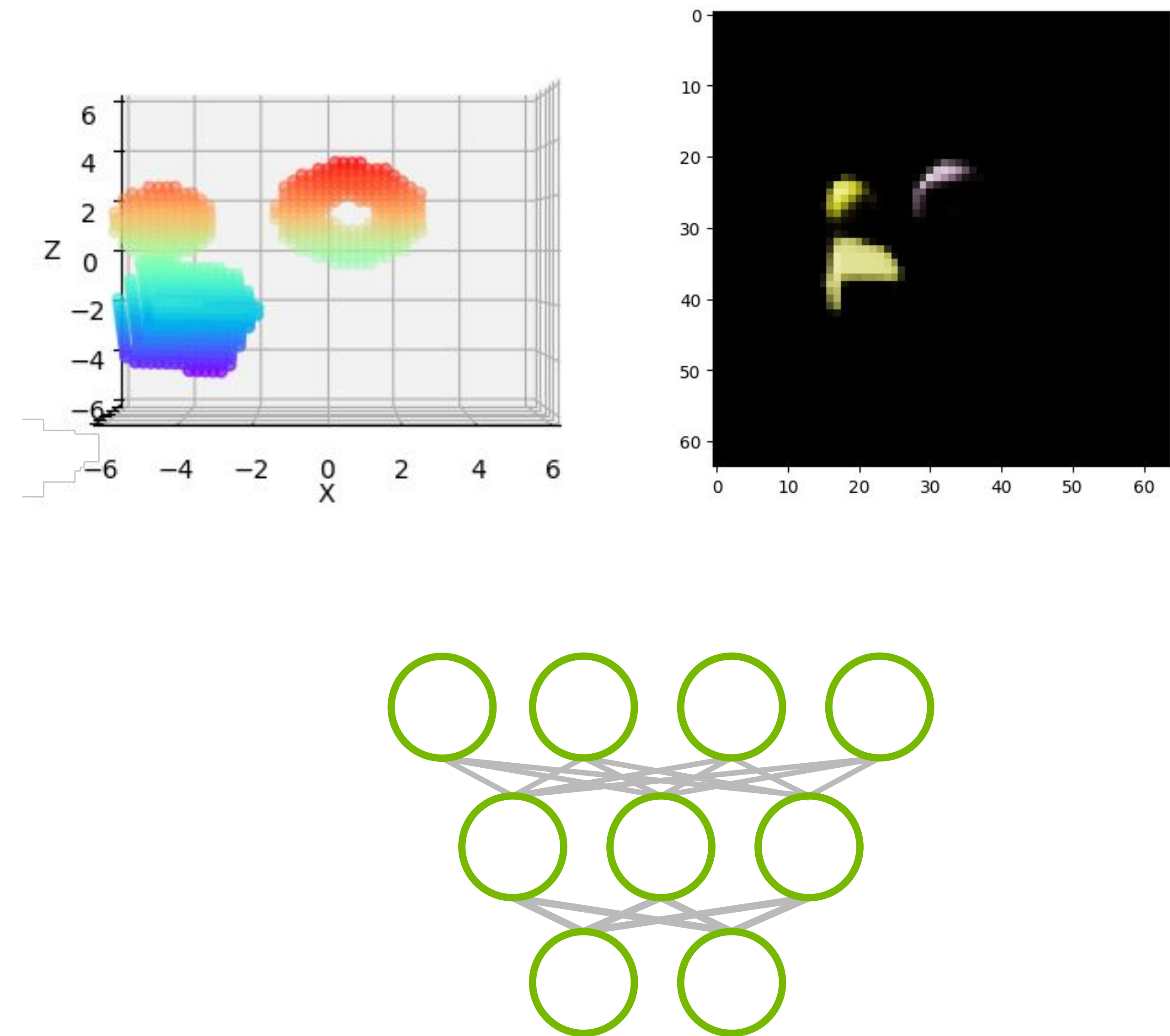


Fusion Techniques

Late



Early

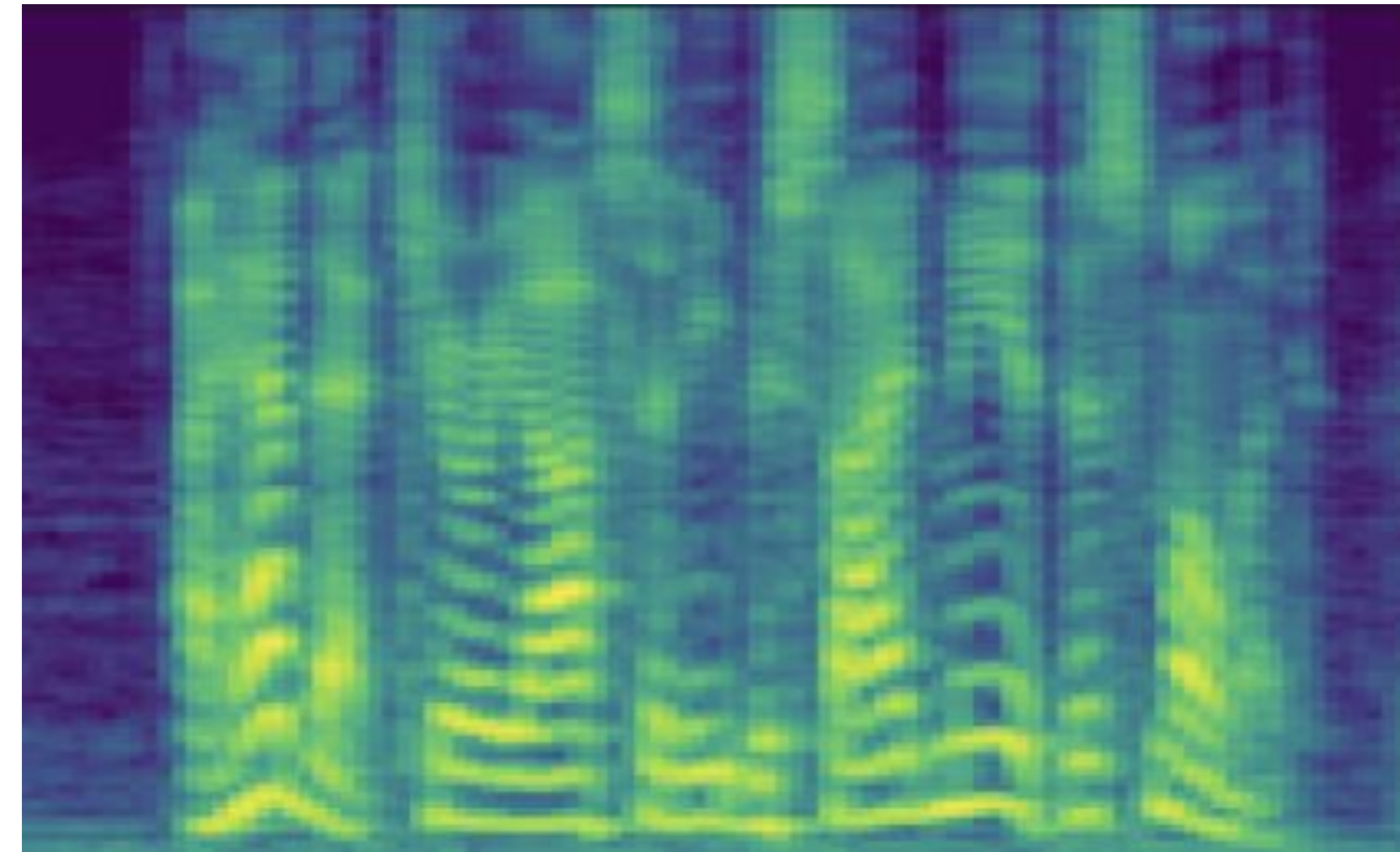
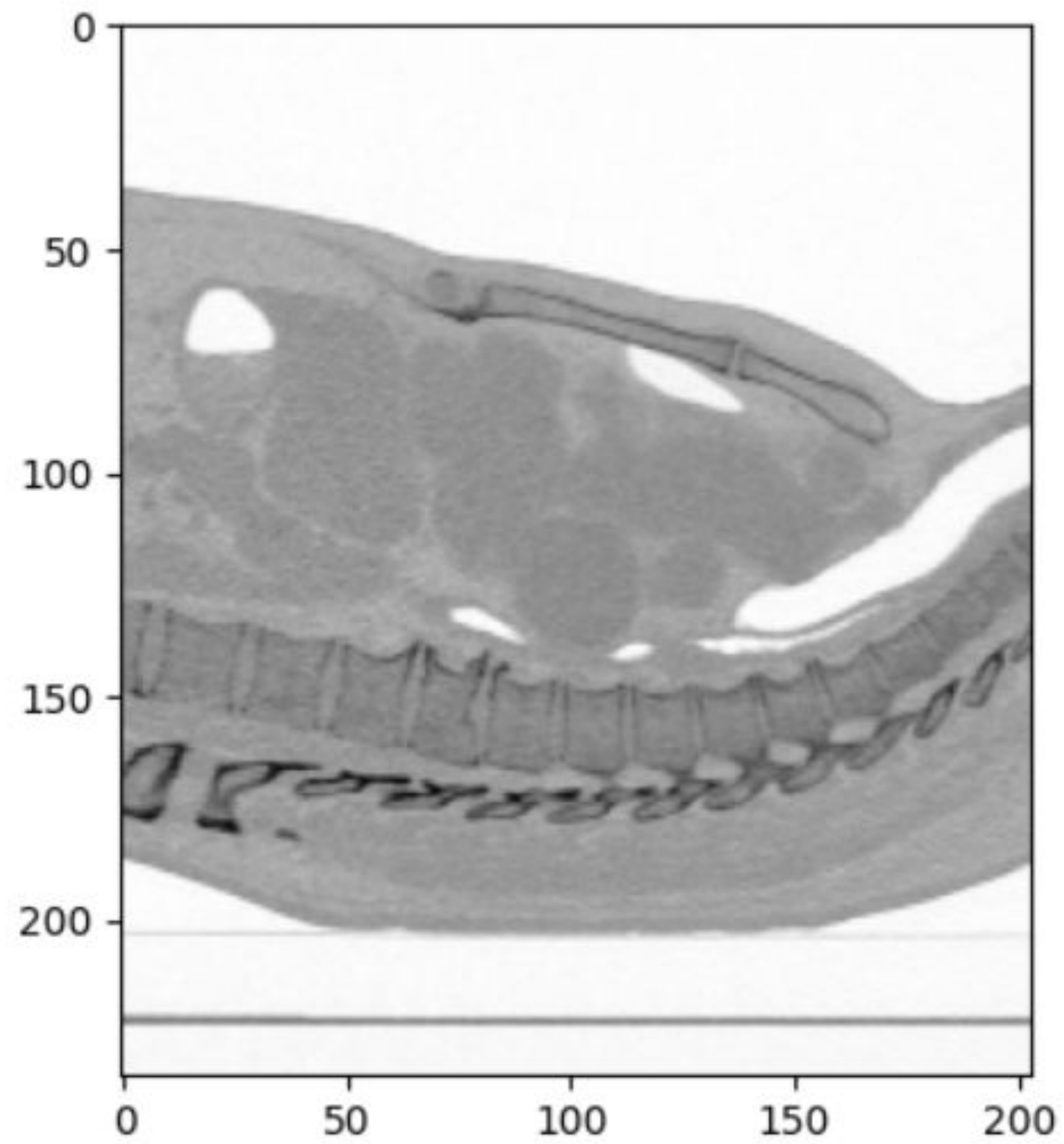


The background features a series of parallel diagonal lines in shades of light green and yellow, creating a sense of depth and movement. Overlaid on these are several overlapping, rounded rectangular shapes in various shades of green, from light to dark, which appear to be layered or stacked.

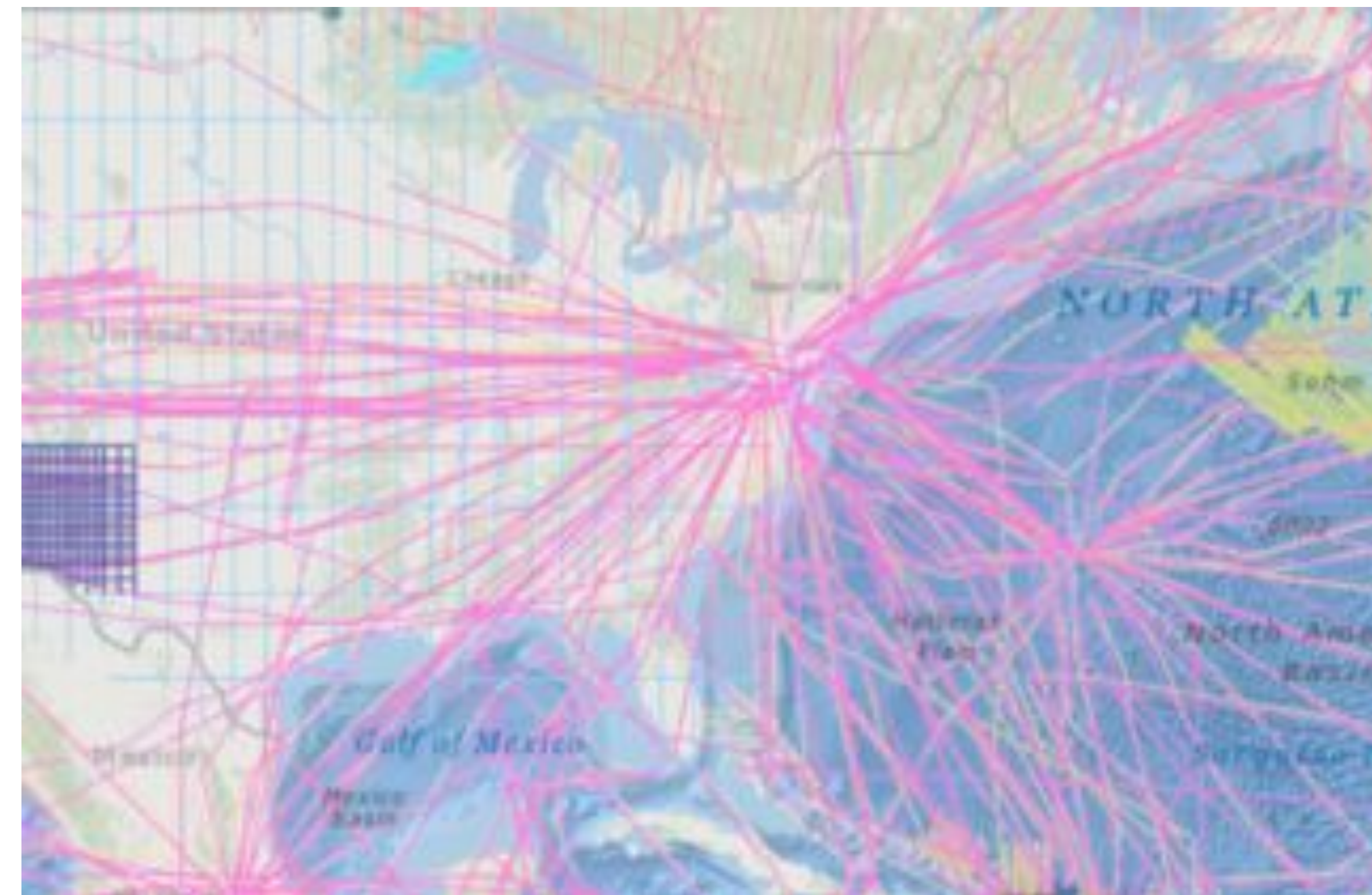
Lab 1b – Exploring Modalities

Atypical Data

CT Scan



Spectro
gram



Magnet
ic Data



Let's Get Started!

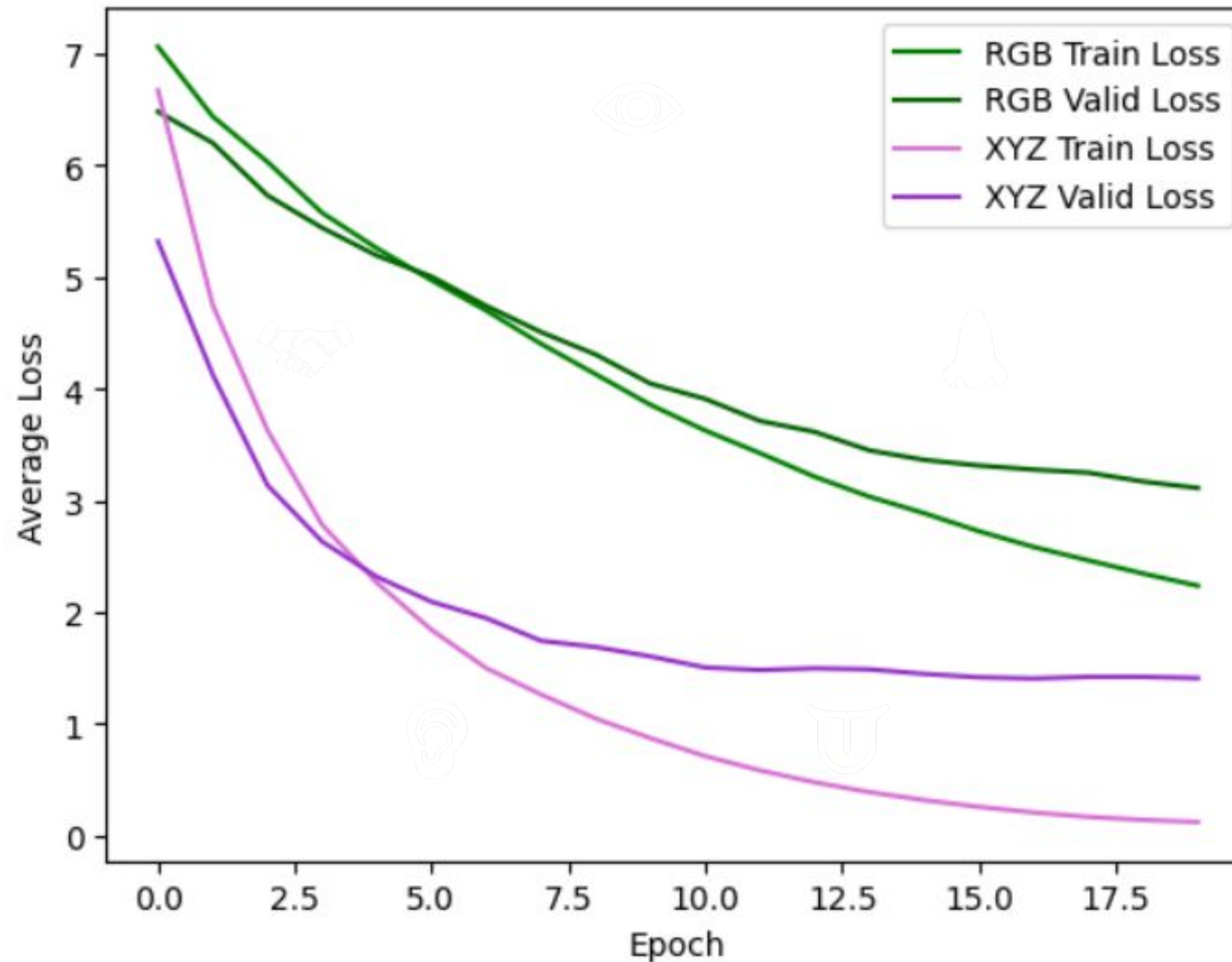
Agenda

- Part 1: Multimodal Data
- Part 2: Intermediate Fusion
- Part 3: Cross-modal Projection and OCR Pipelines
- Part 4: VSS and GraphRAG
- Part 5: Wrap-up & Assessment

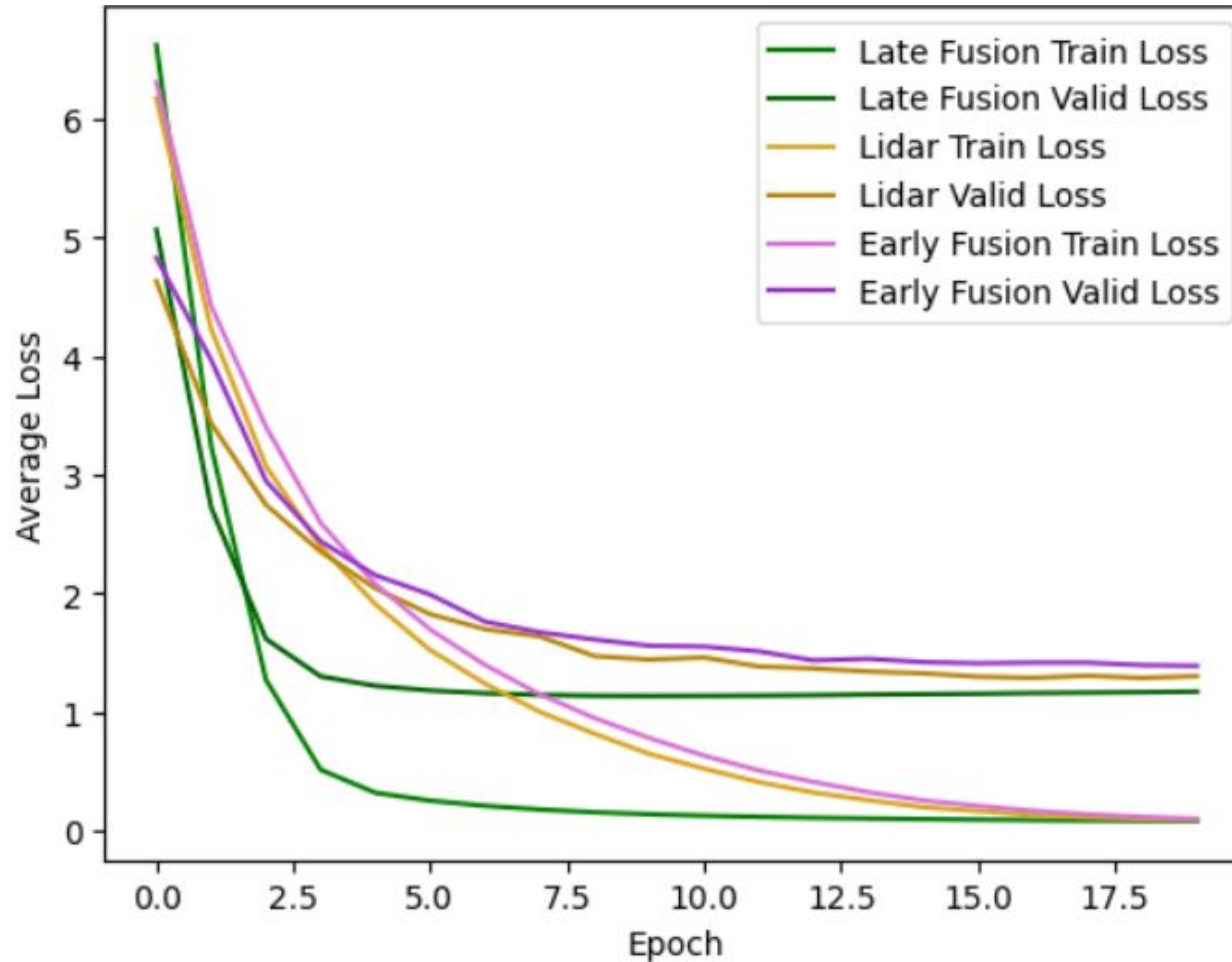


Experiment Results

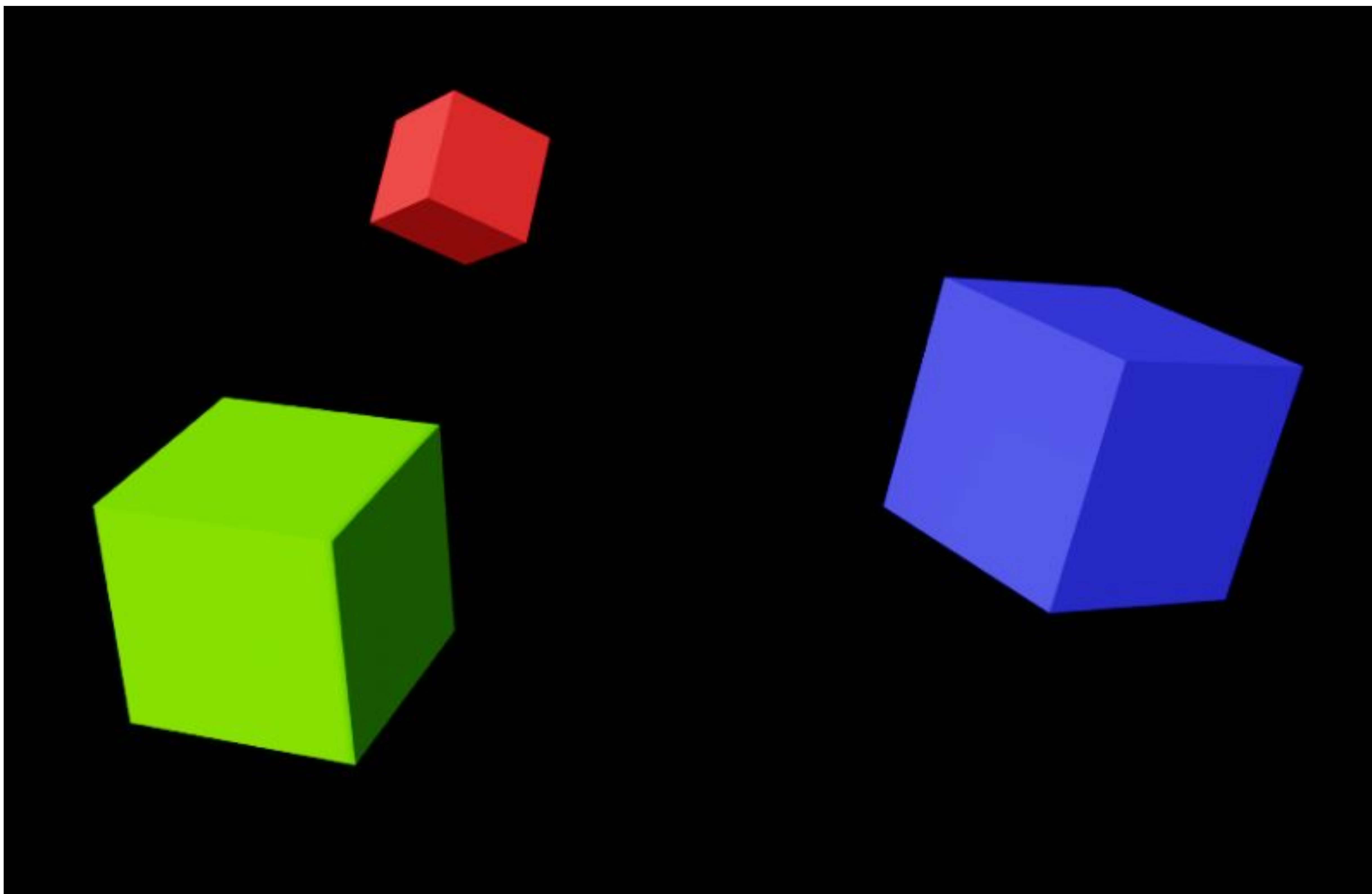
Single Modal Data - Results

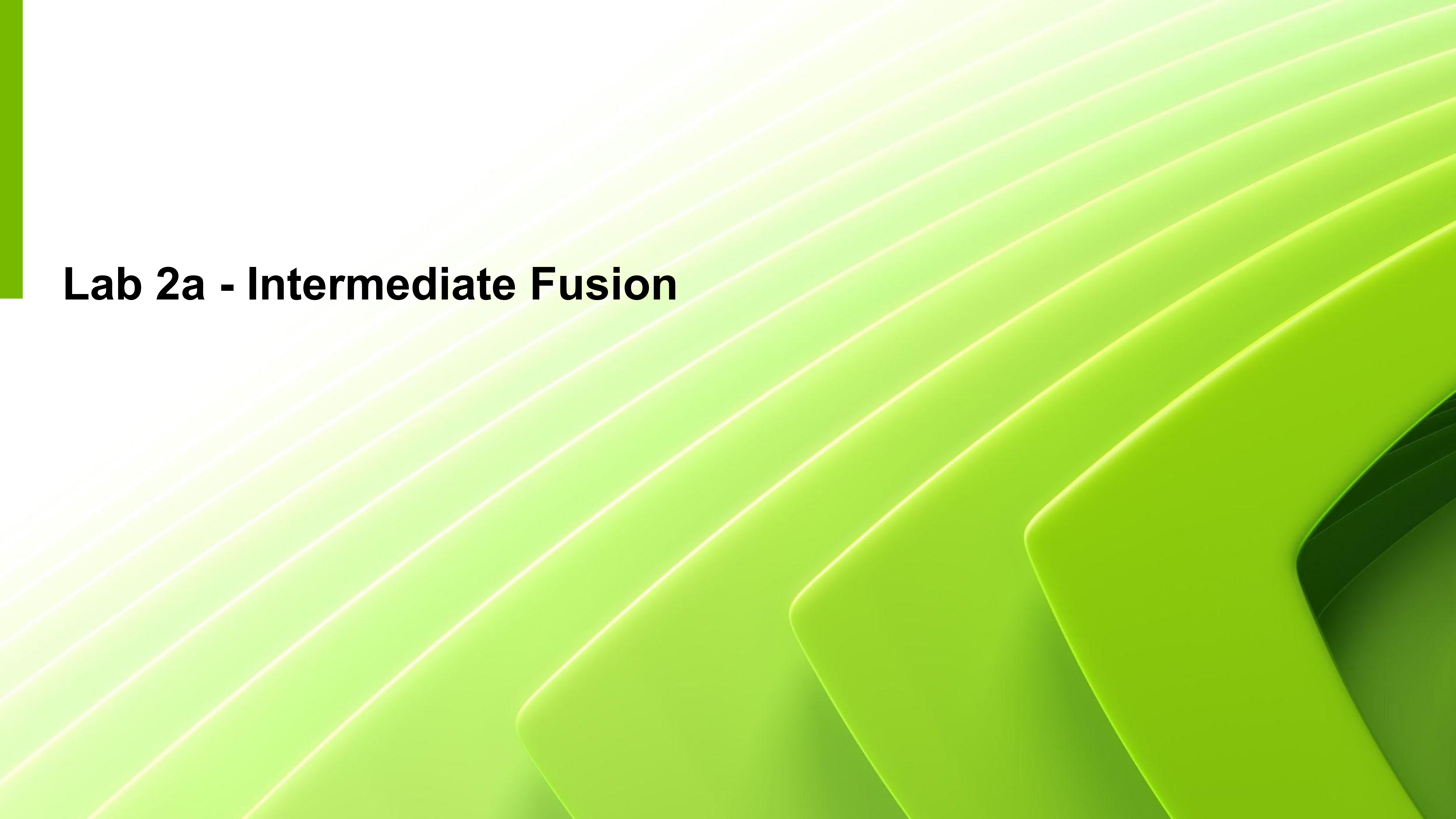


Multimodal Data - Results



Next Experiment

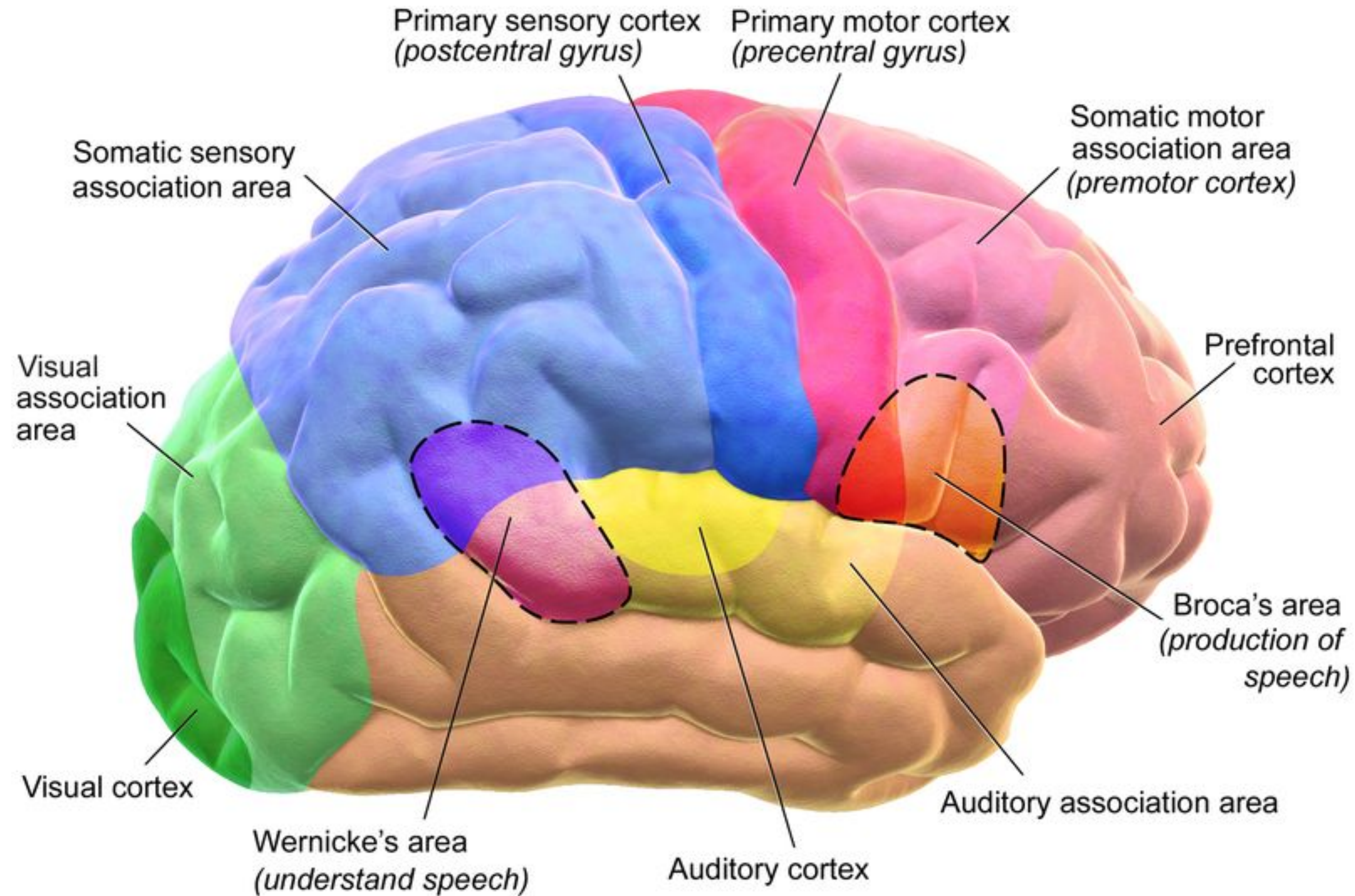




Lab 2a - Intermediate Fusion

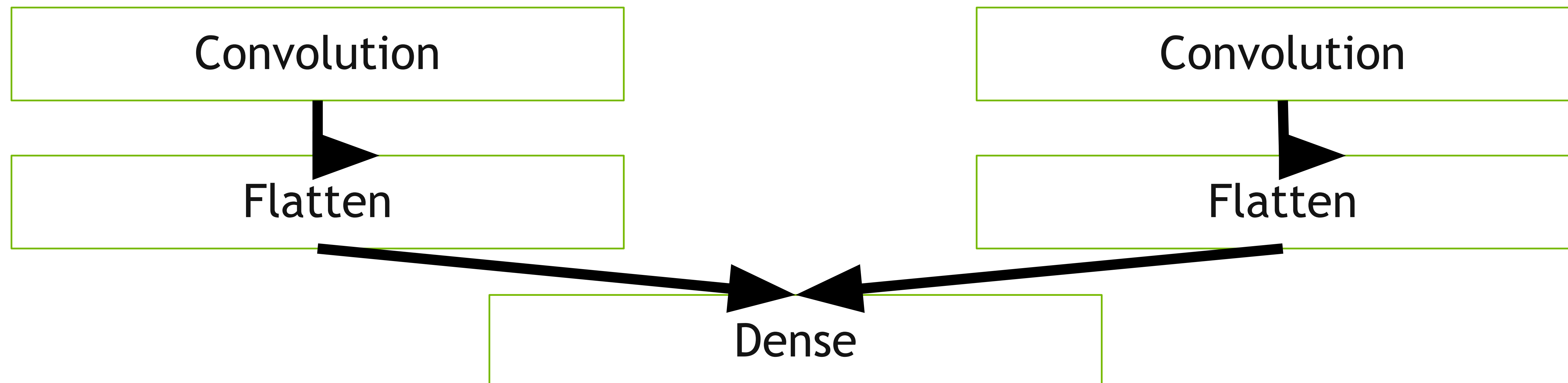
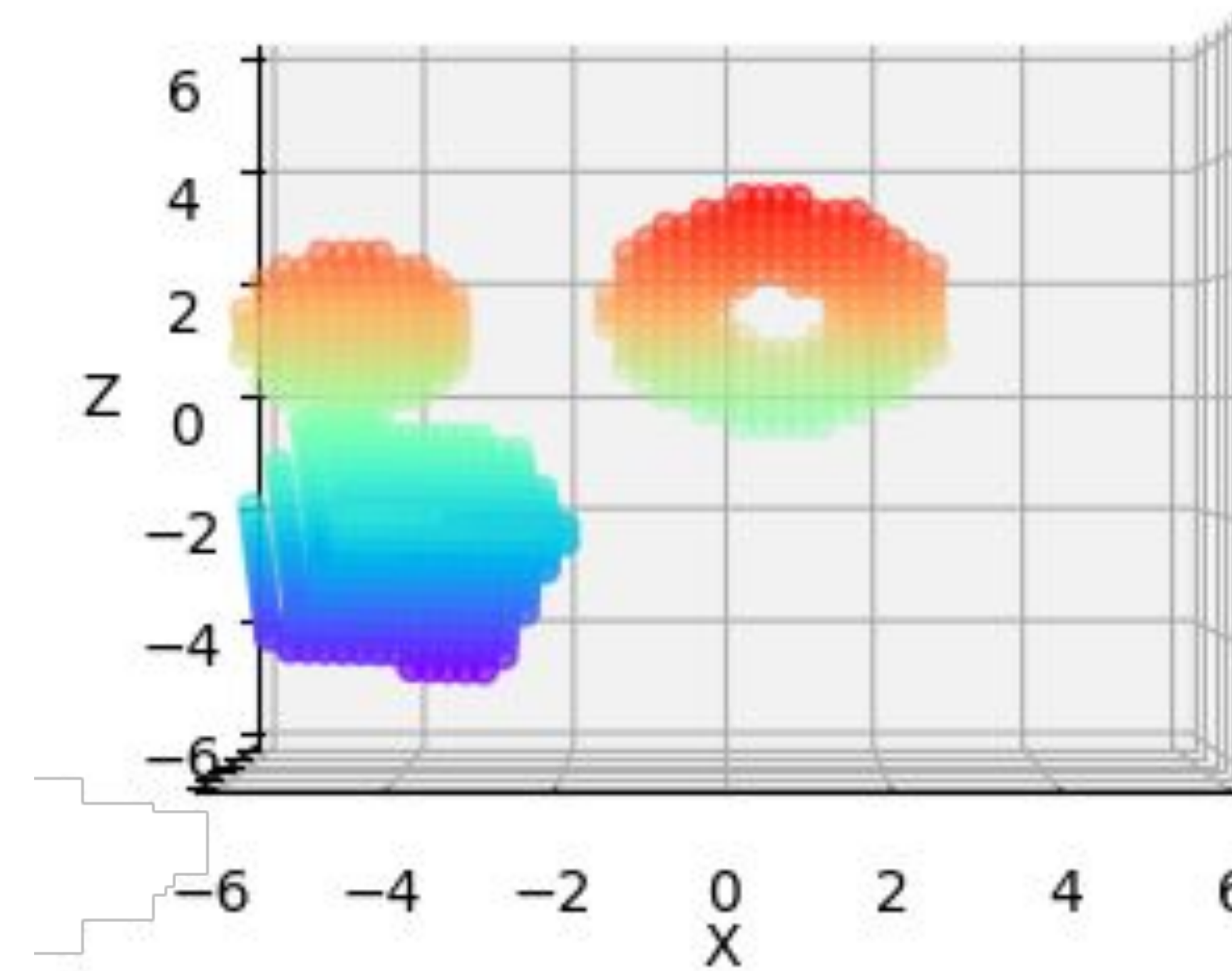
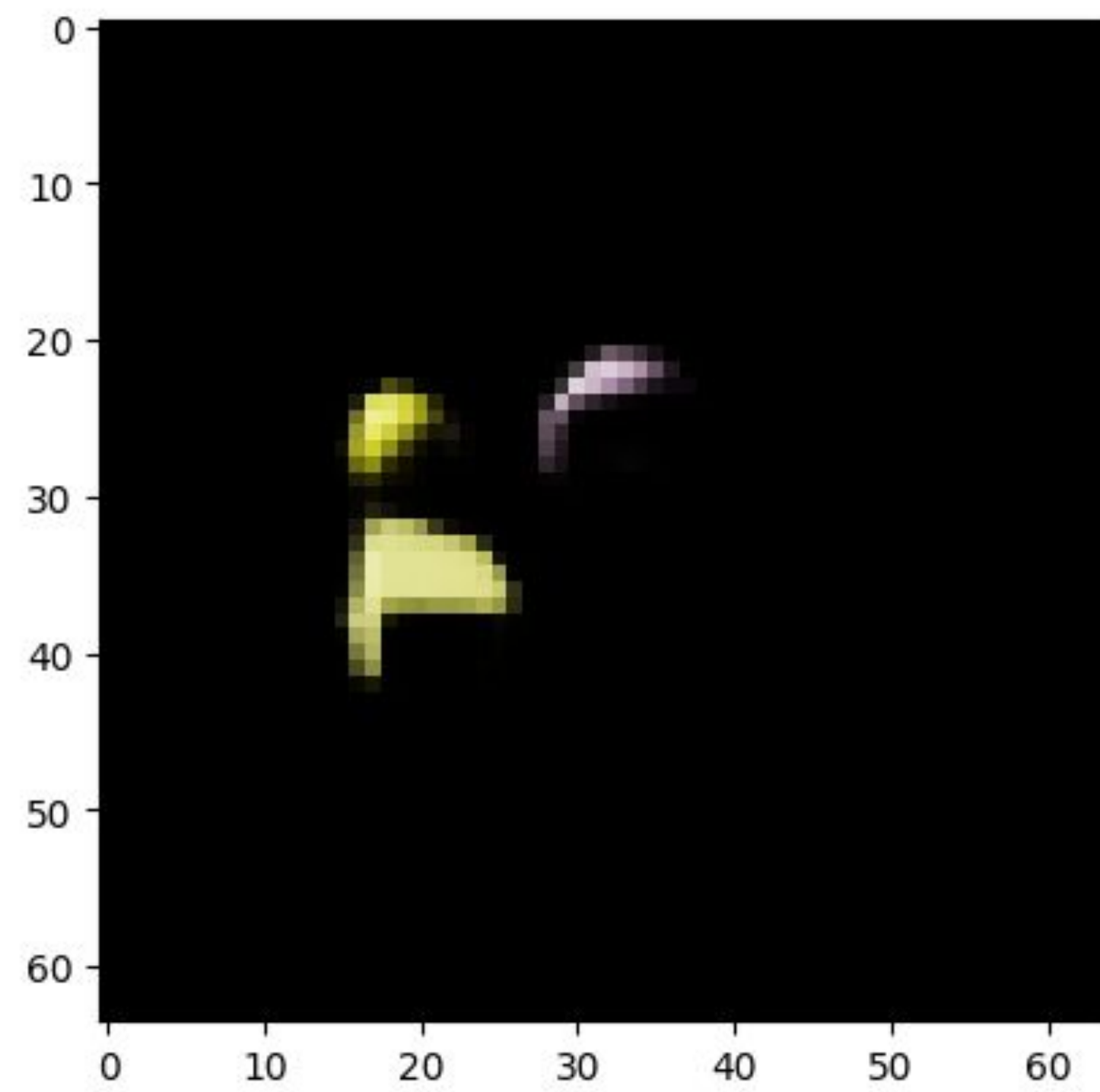
Human Inputs and Outputs

The Human Brain as a Multimodal Model

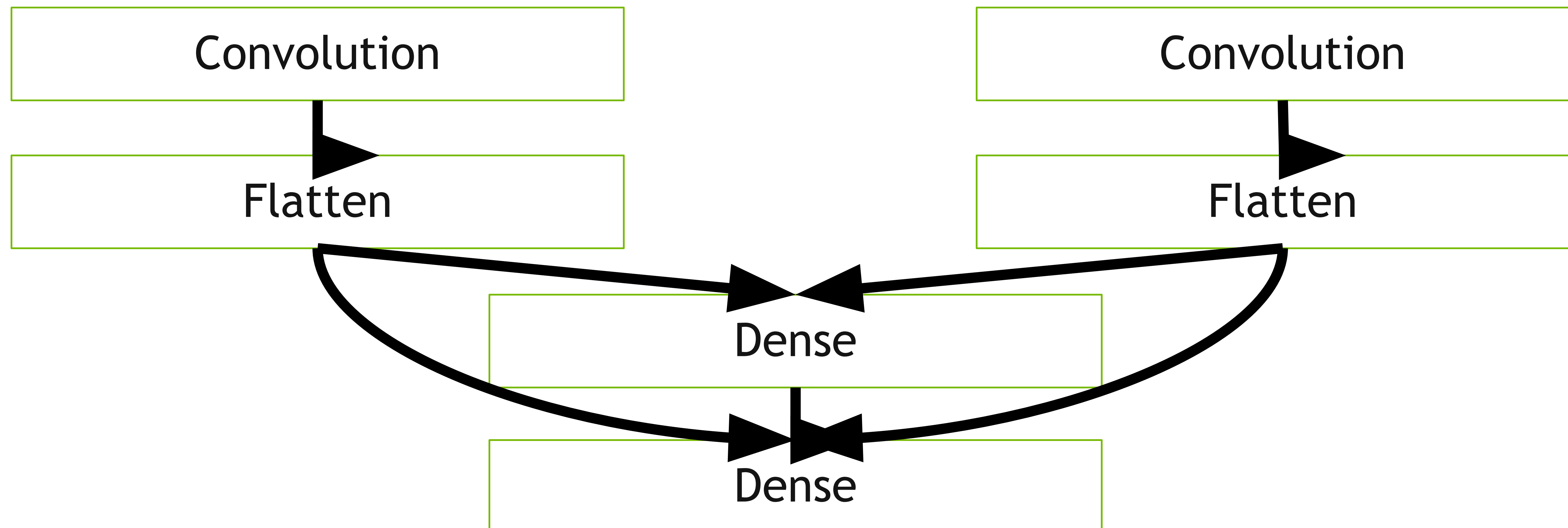
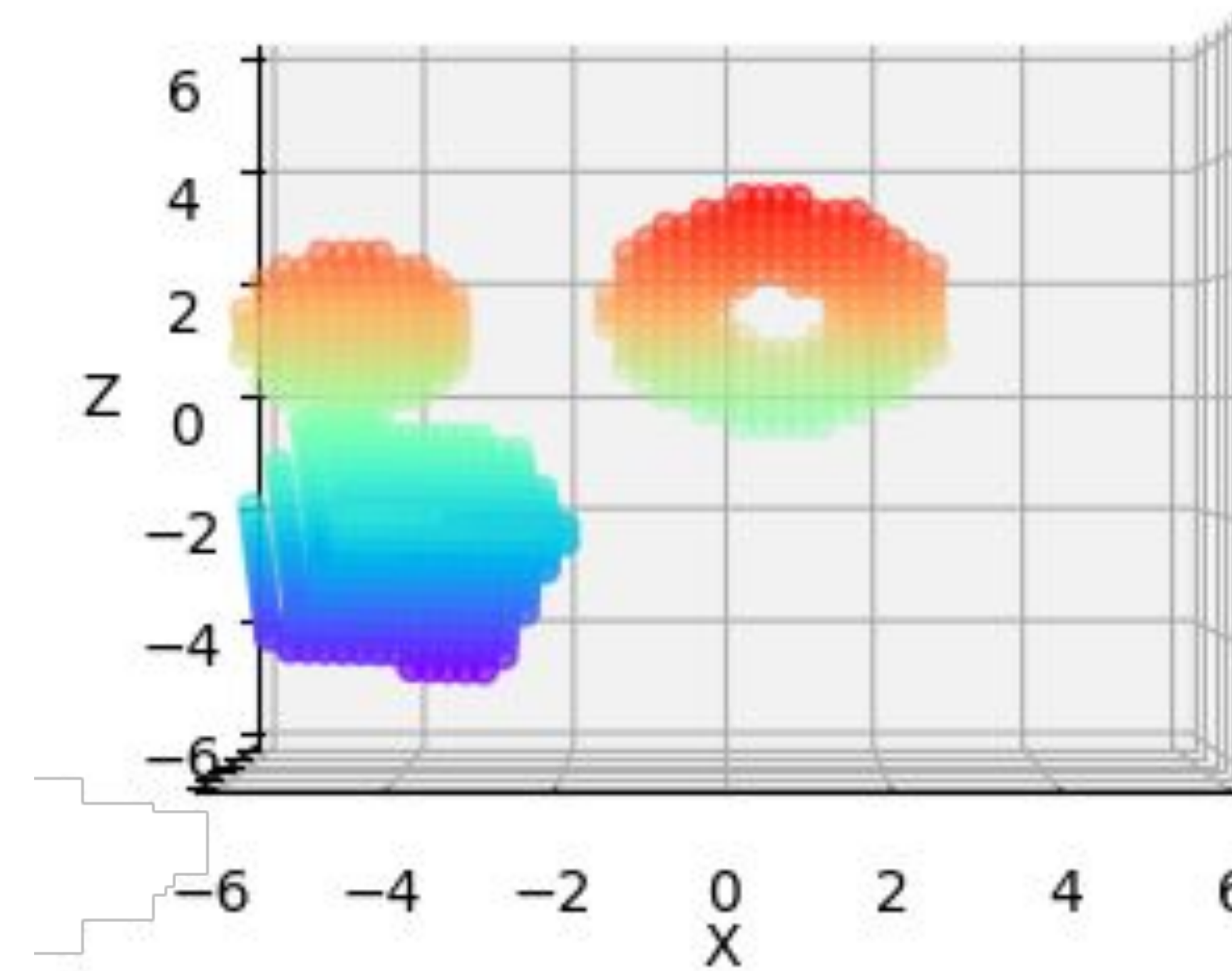
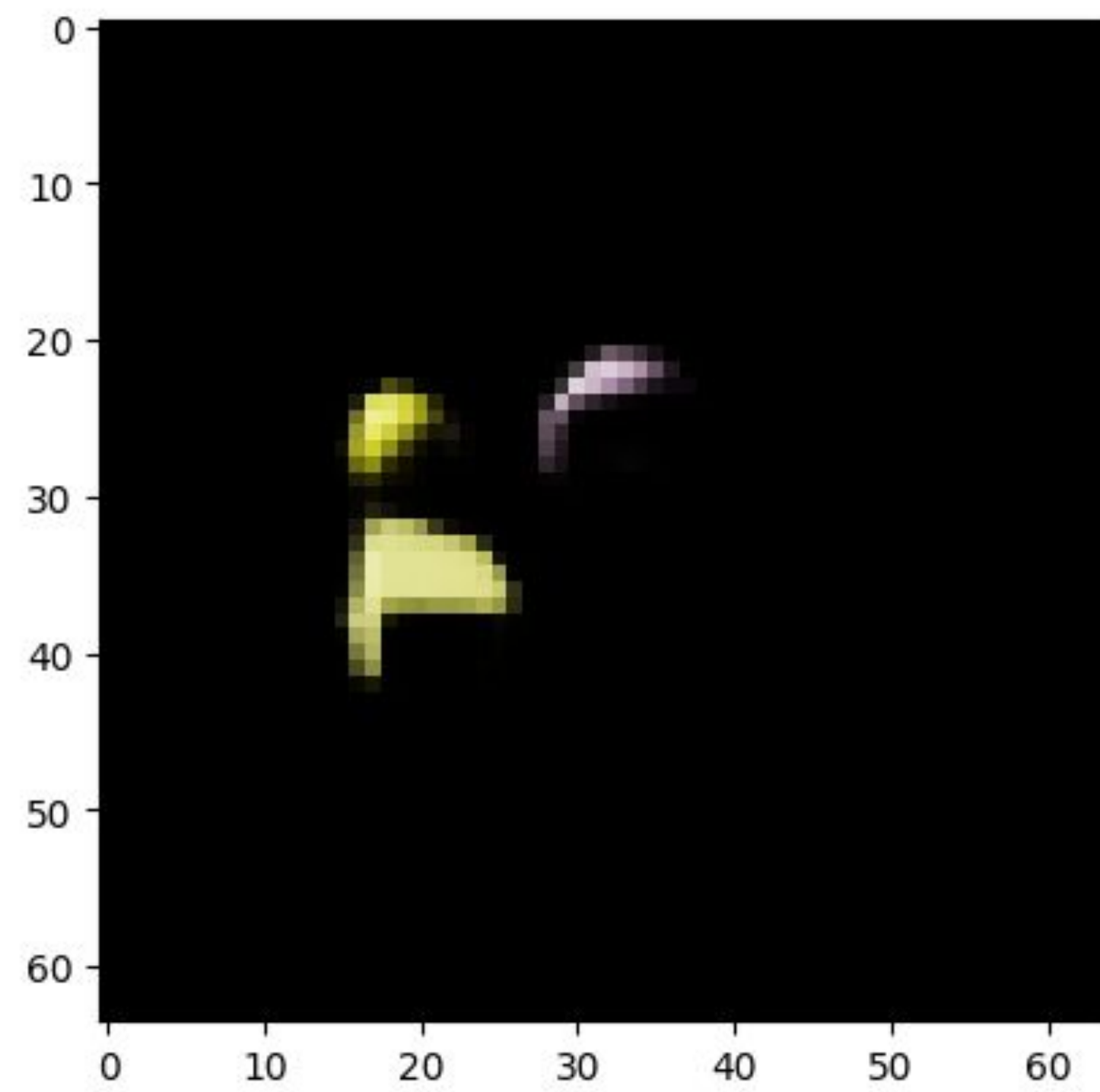


Blausen.com staff (2014). "[Medical gallery of Blausen Medical 2014](#)". *WikiJournal of Medicine* 1 (2). DOI:[10.15347/wjm/2014.010](#). ISSN [2002-4436](#)

Intermediate Fusion

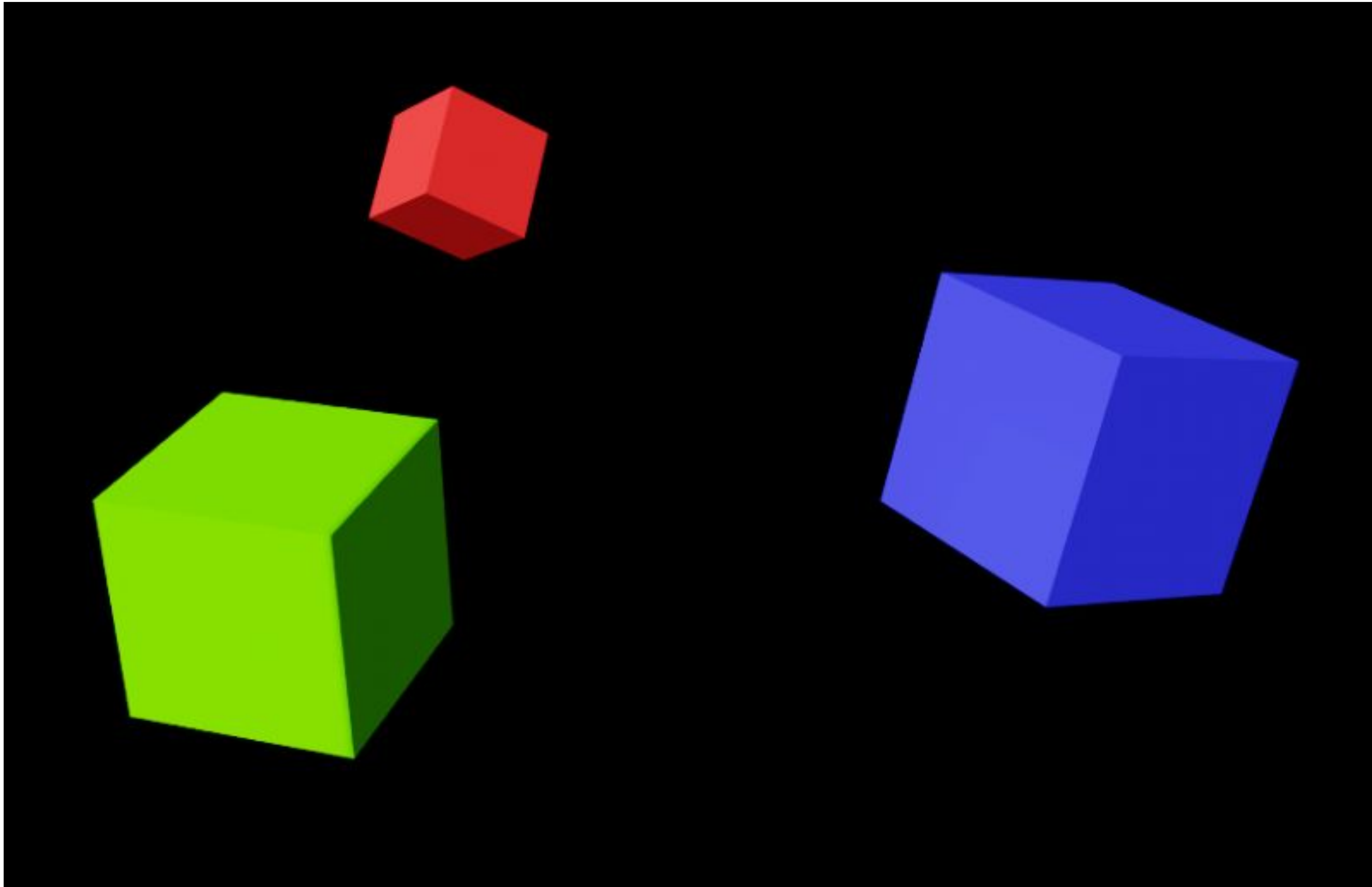


Intermediate Fusion



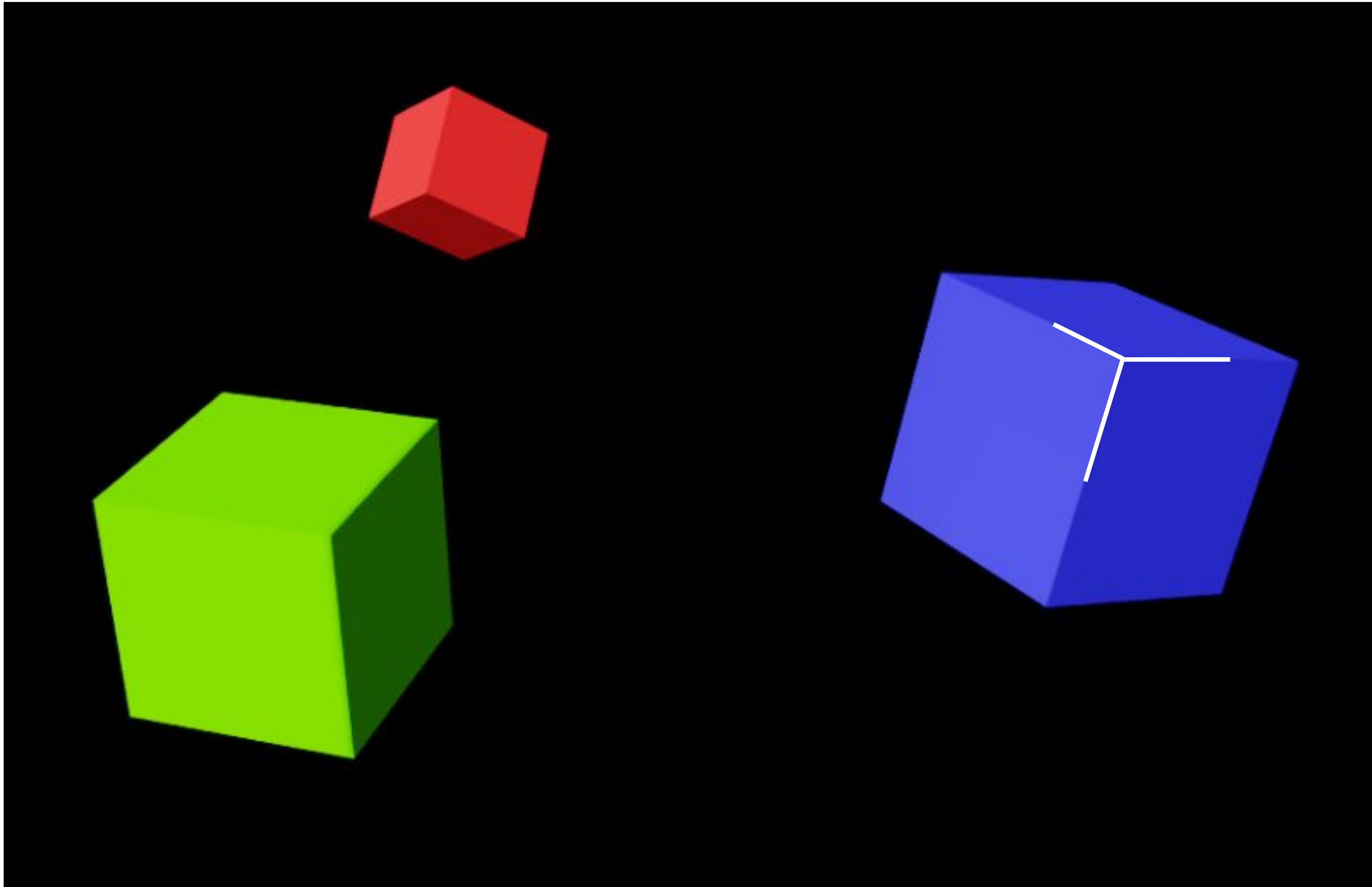
Intermediate Fusion

Finding the Center



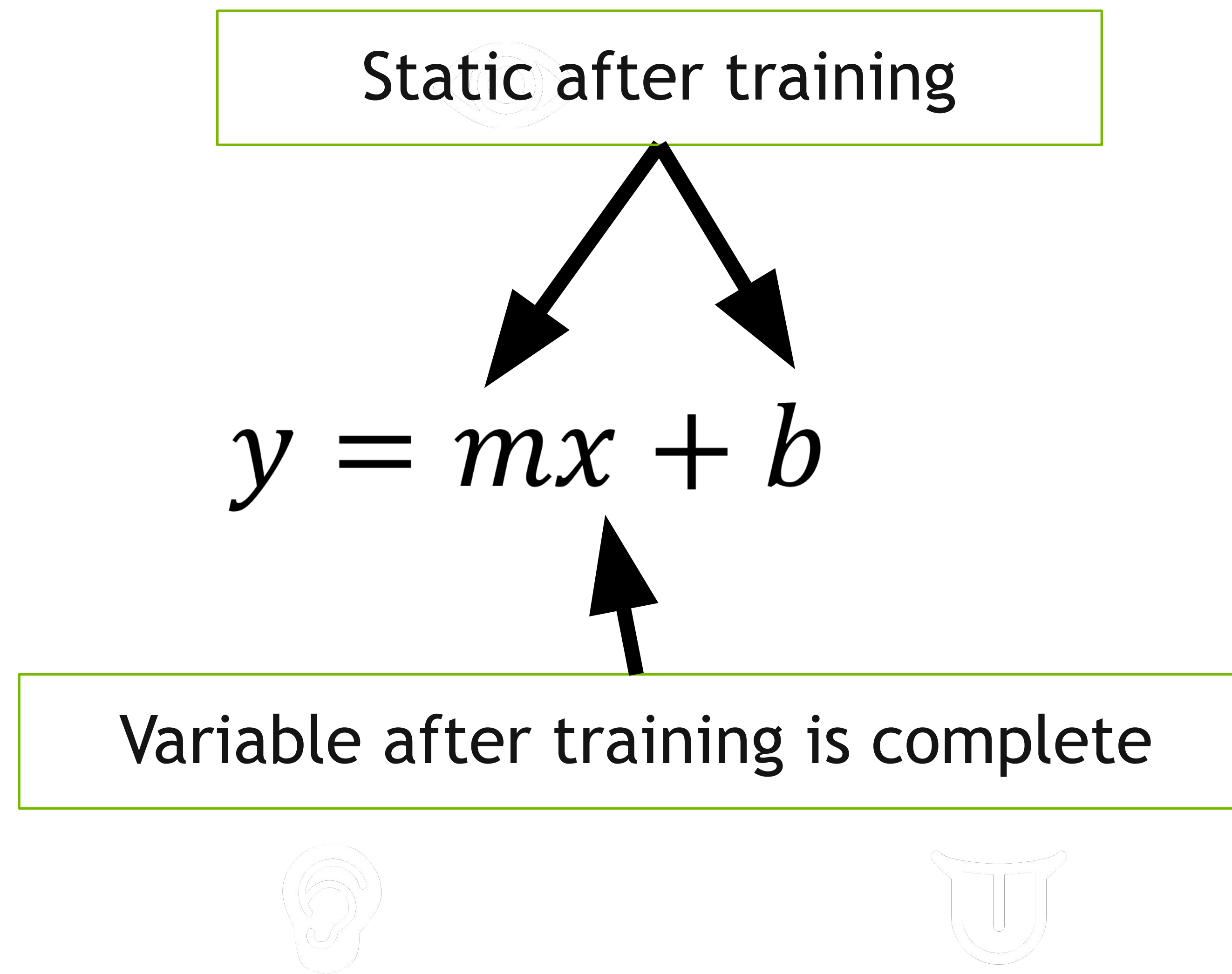
Intermediate Fusion

Finding the Center

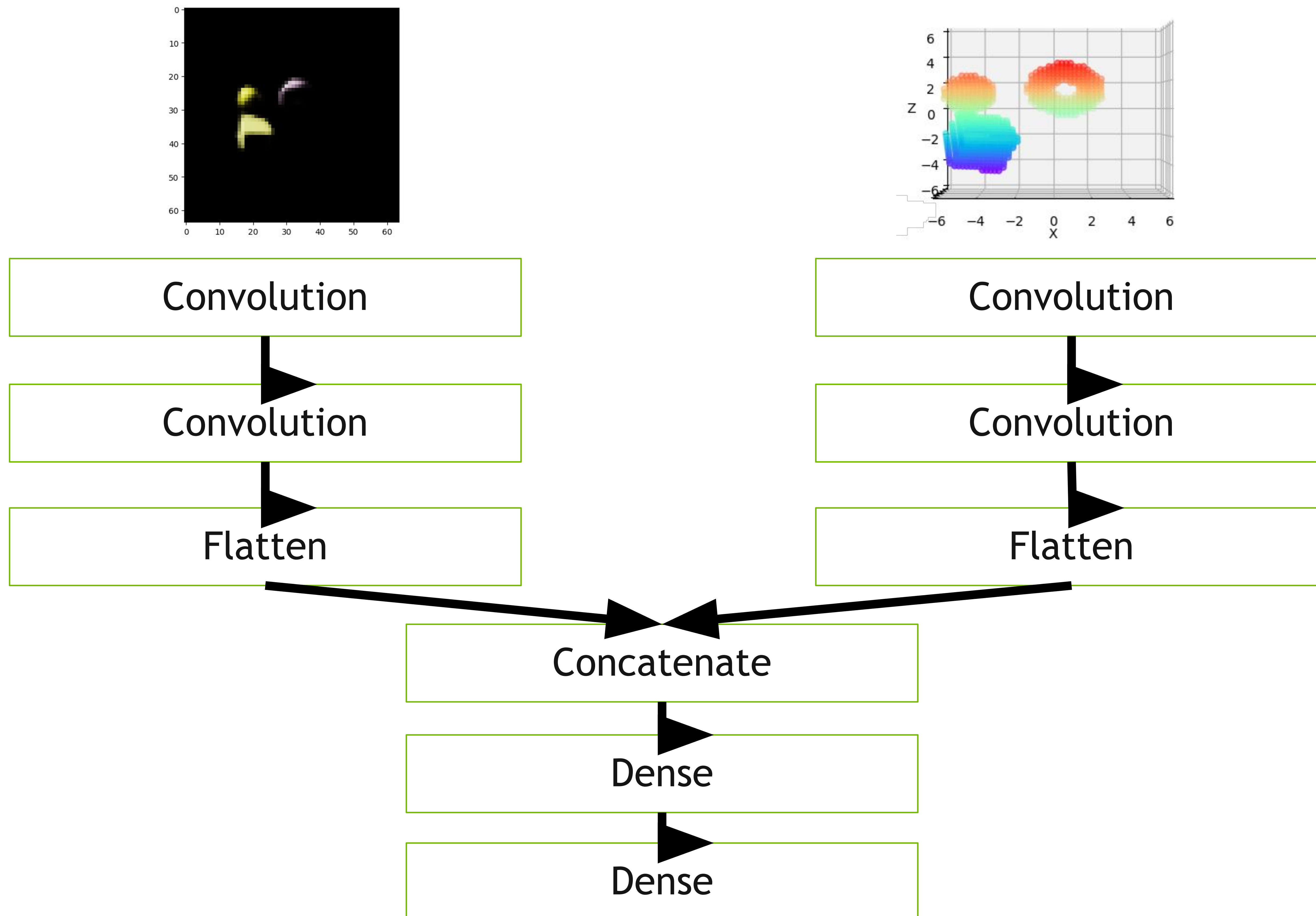


Intermediate Fusion

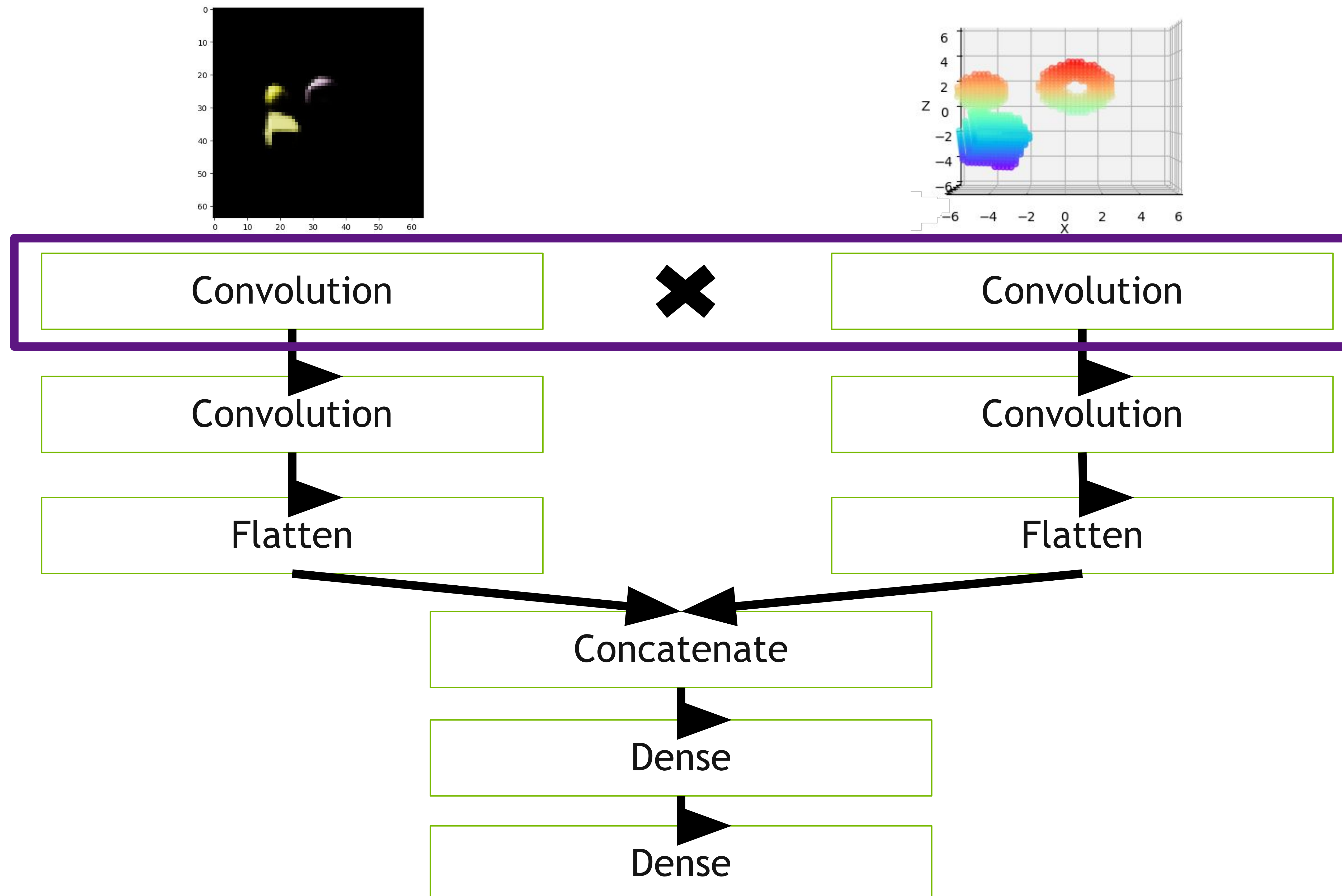
Finding the Center



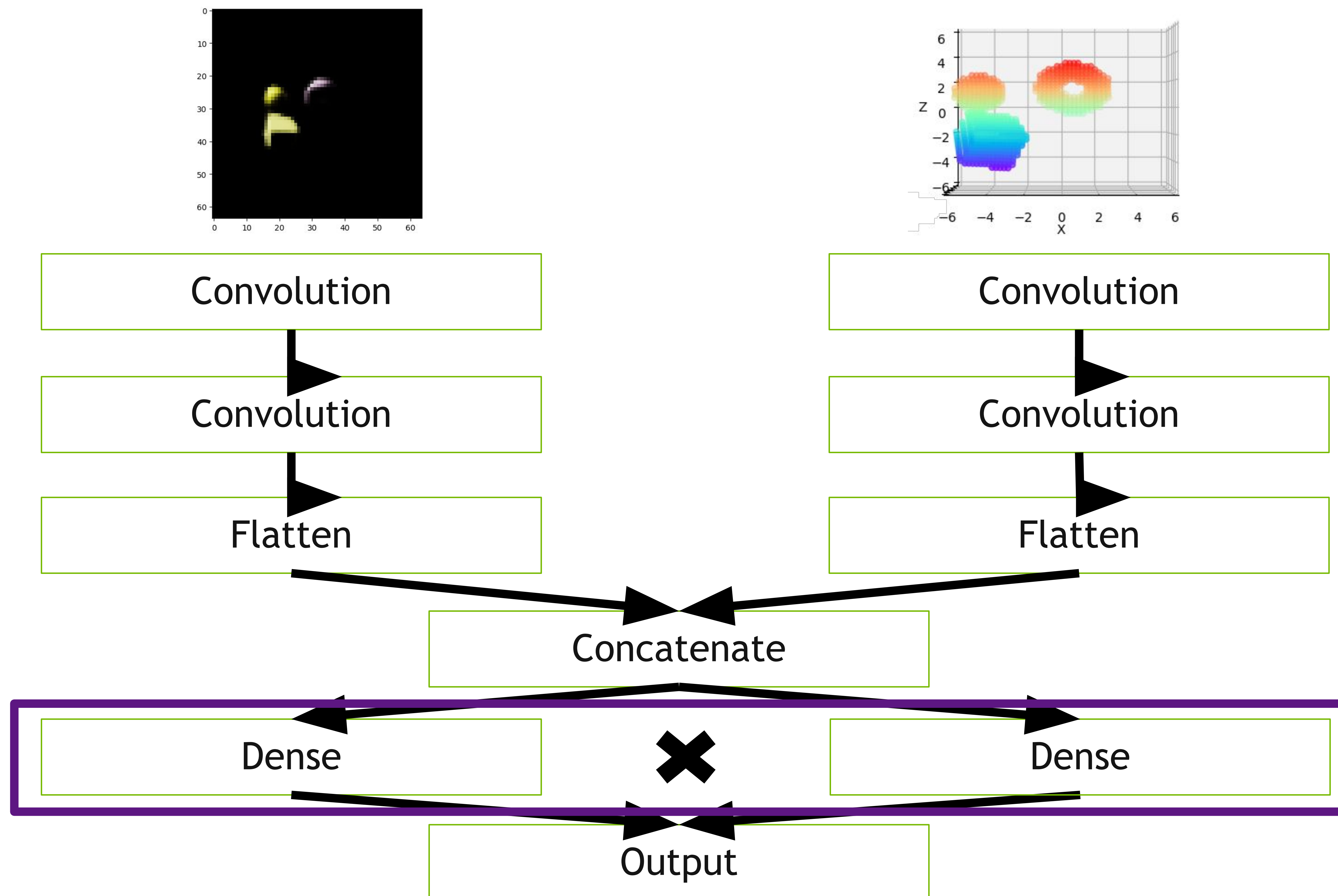
Intermediate Fusion



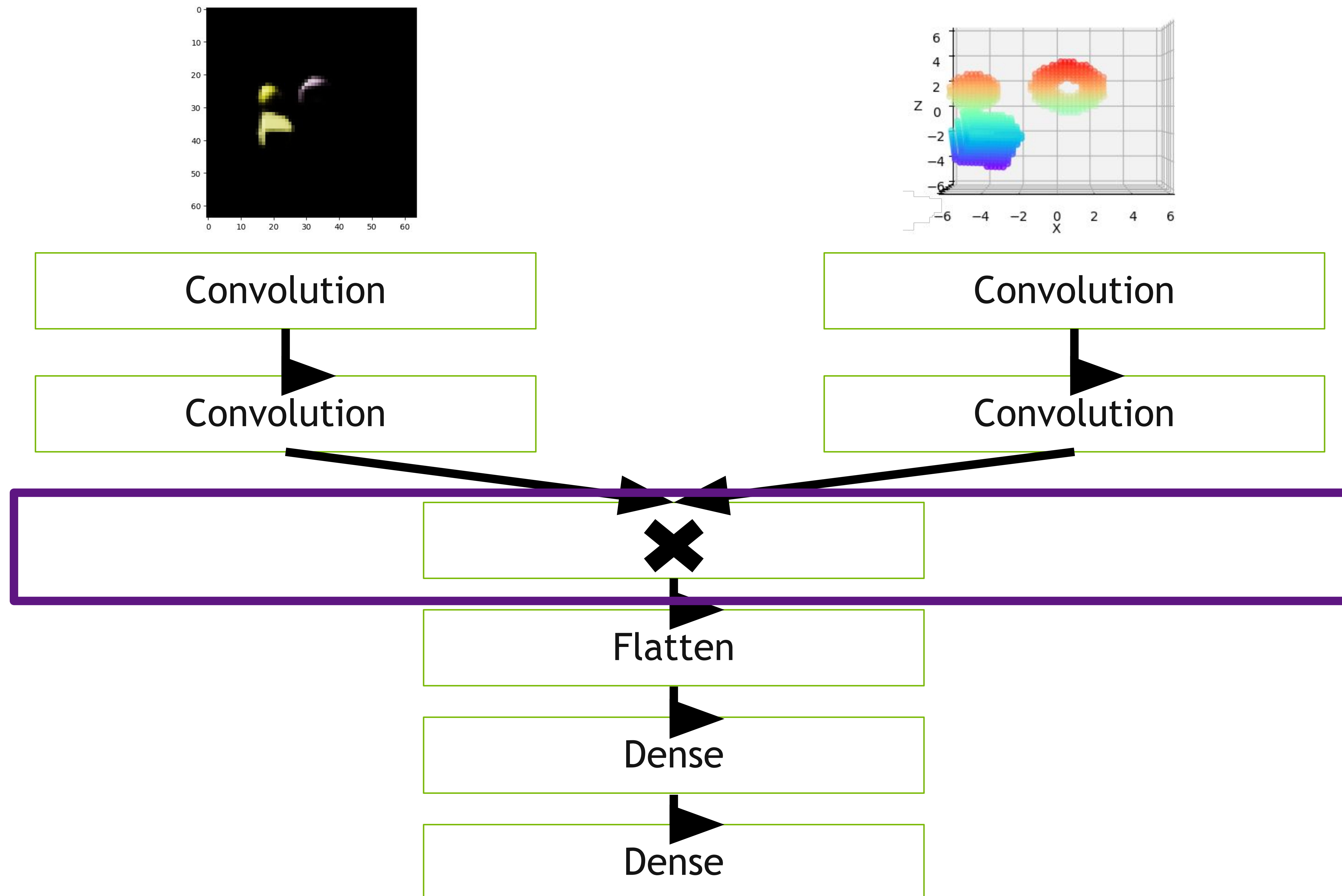
Intermediate Fusion



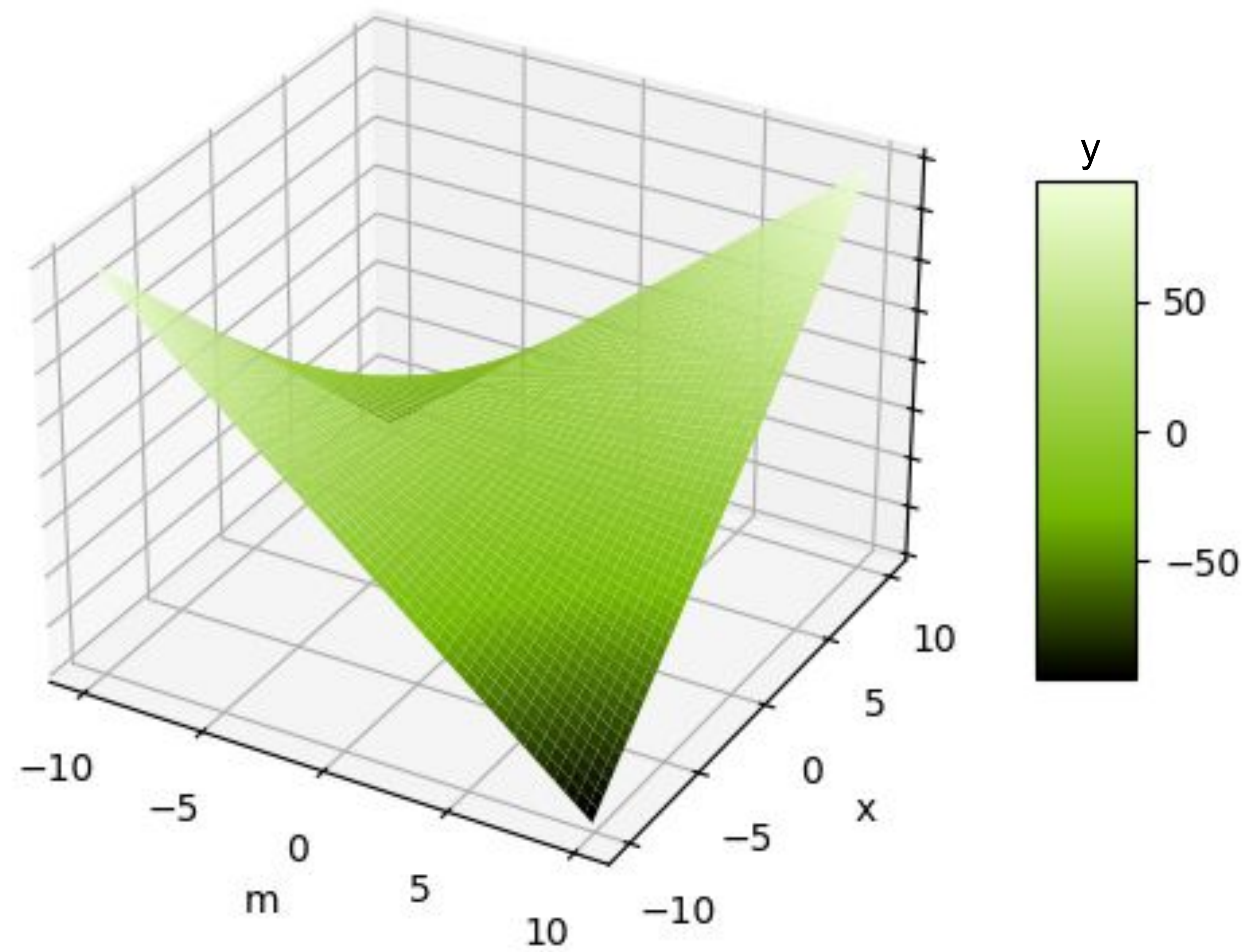
Intermediate Fusion



Intermediate Fusion



Multiplication Overfitting



Lab Experiment



Early Fusion



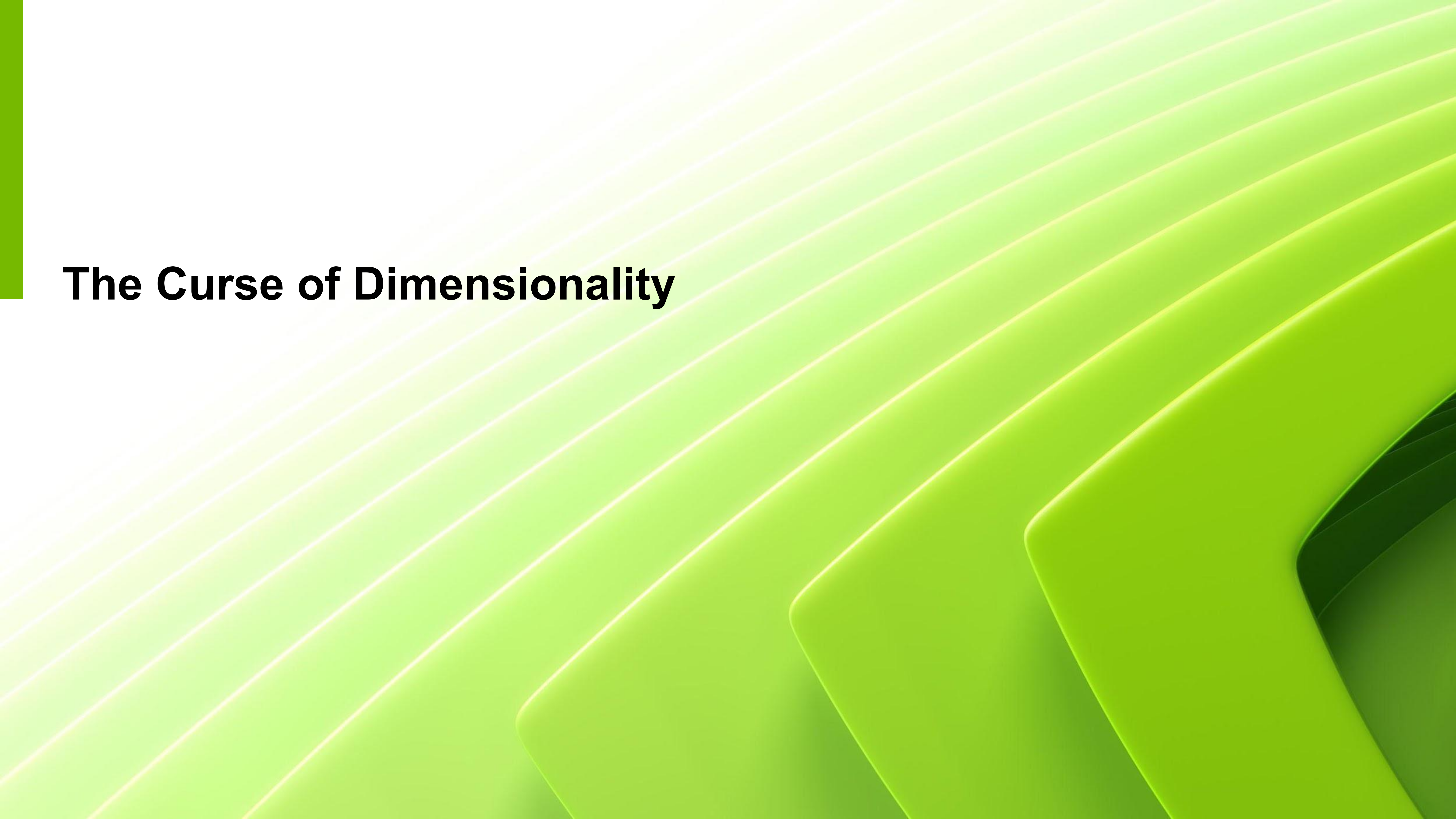
Late Fusion



Intermediate Fusion
Concatenate

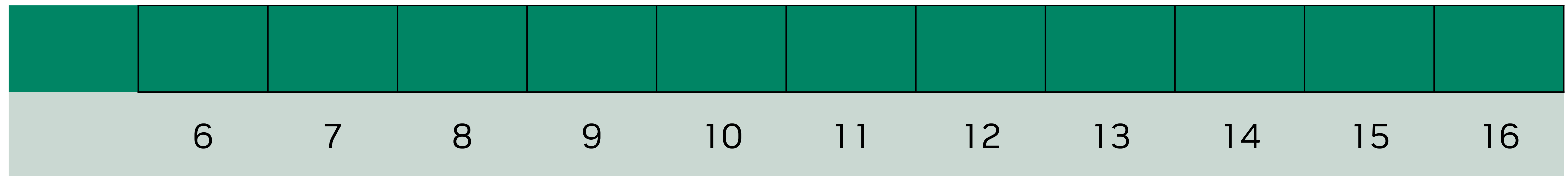


Intermediate Fusion
Multiplication



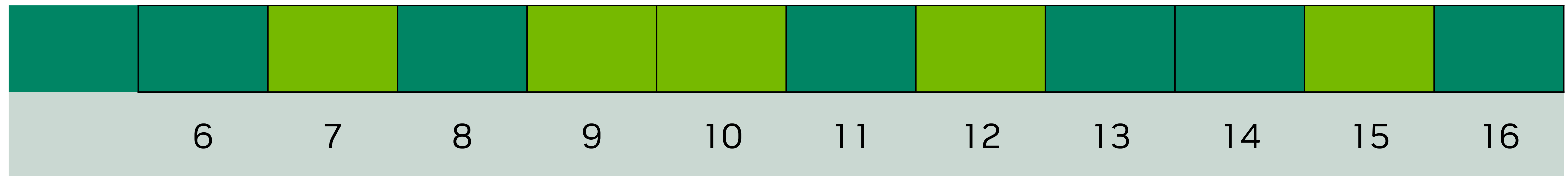
The Curse of Dimensionality

The Curse of Dimensionality



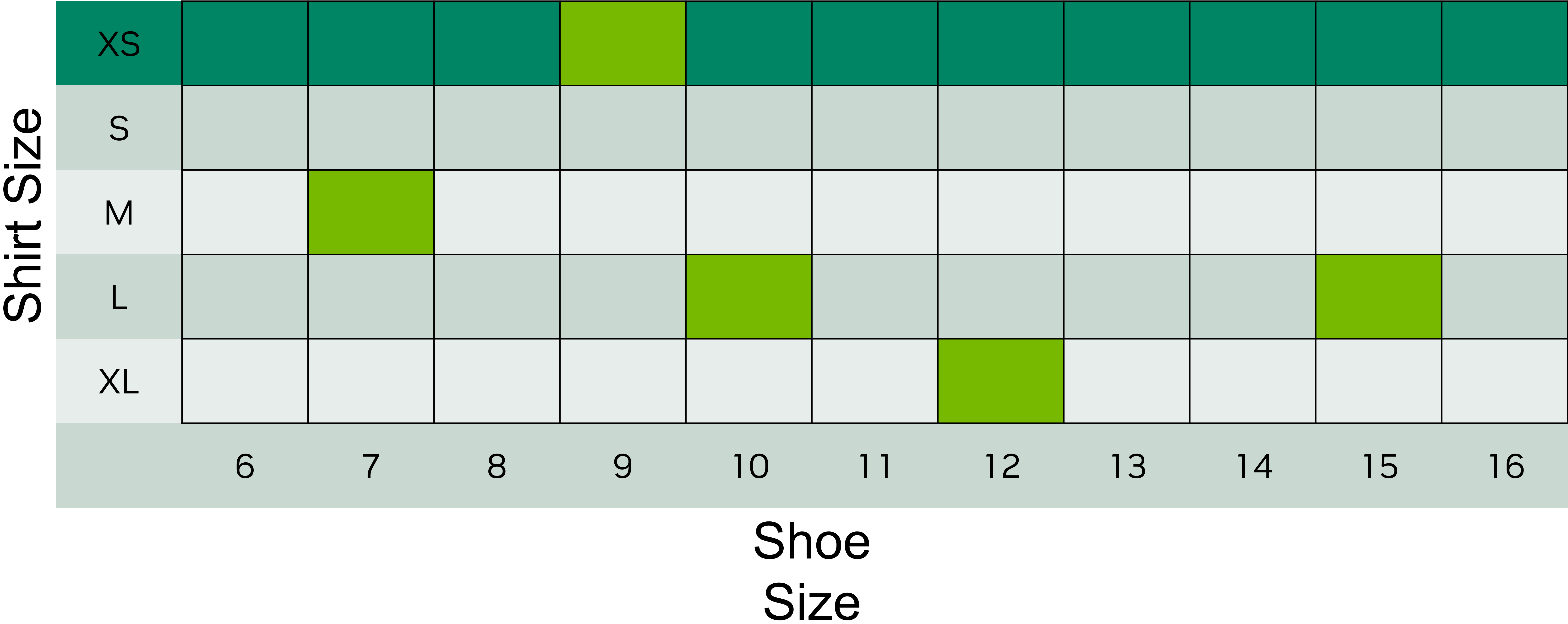
Shoe
Size

The Curse of Dimensionality



Shoe
Size

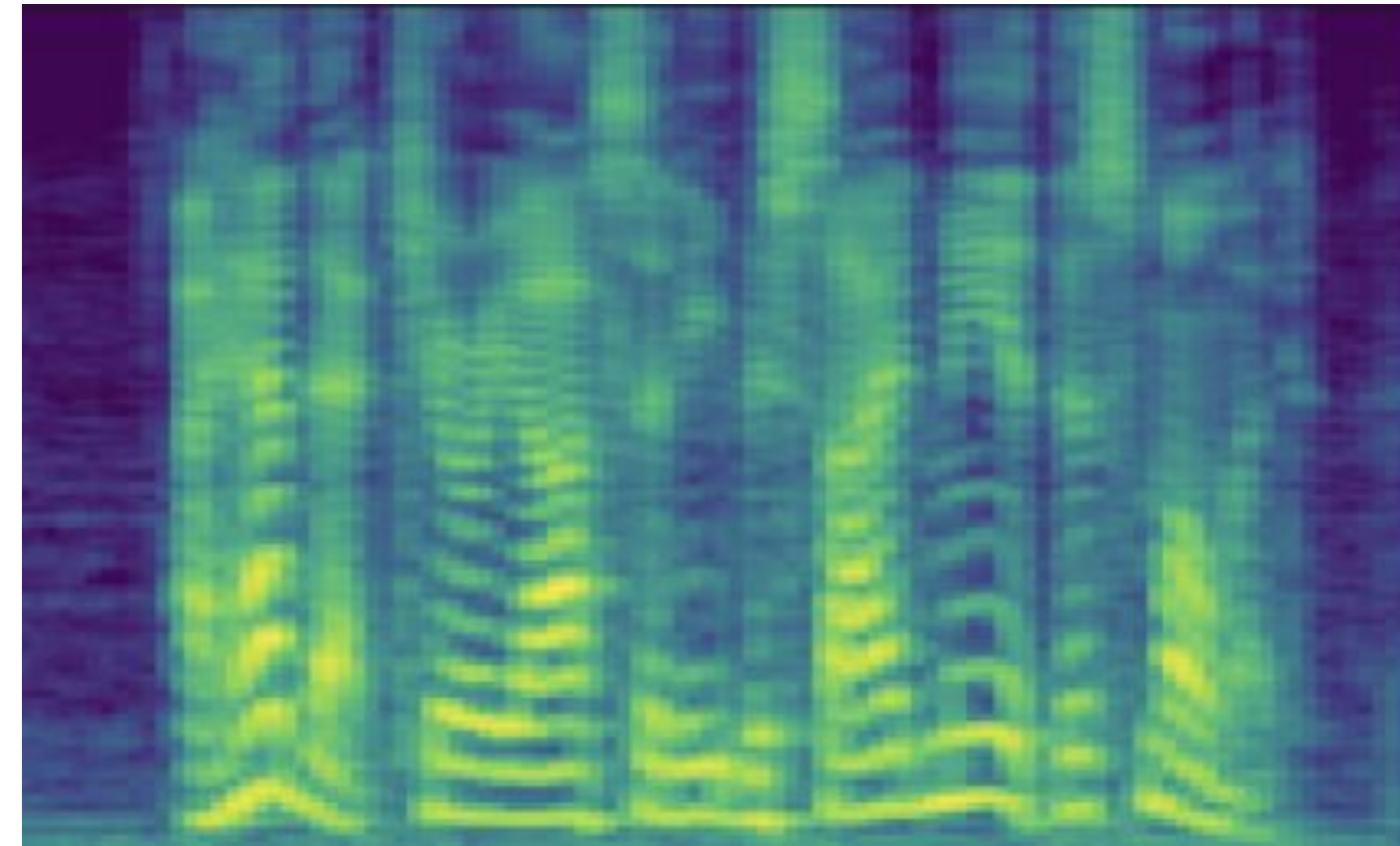
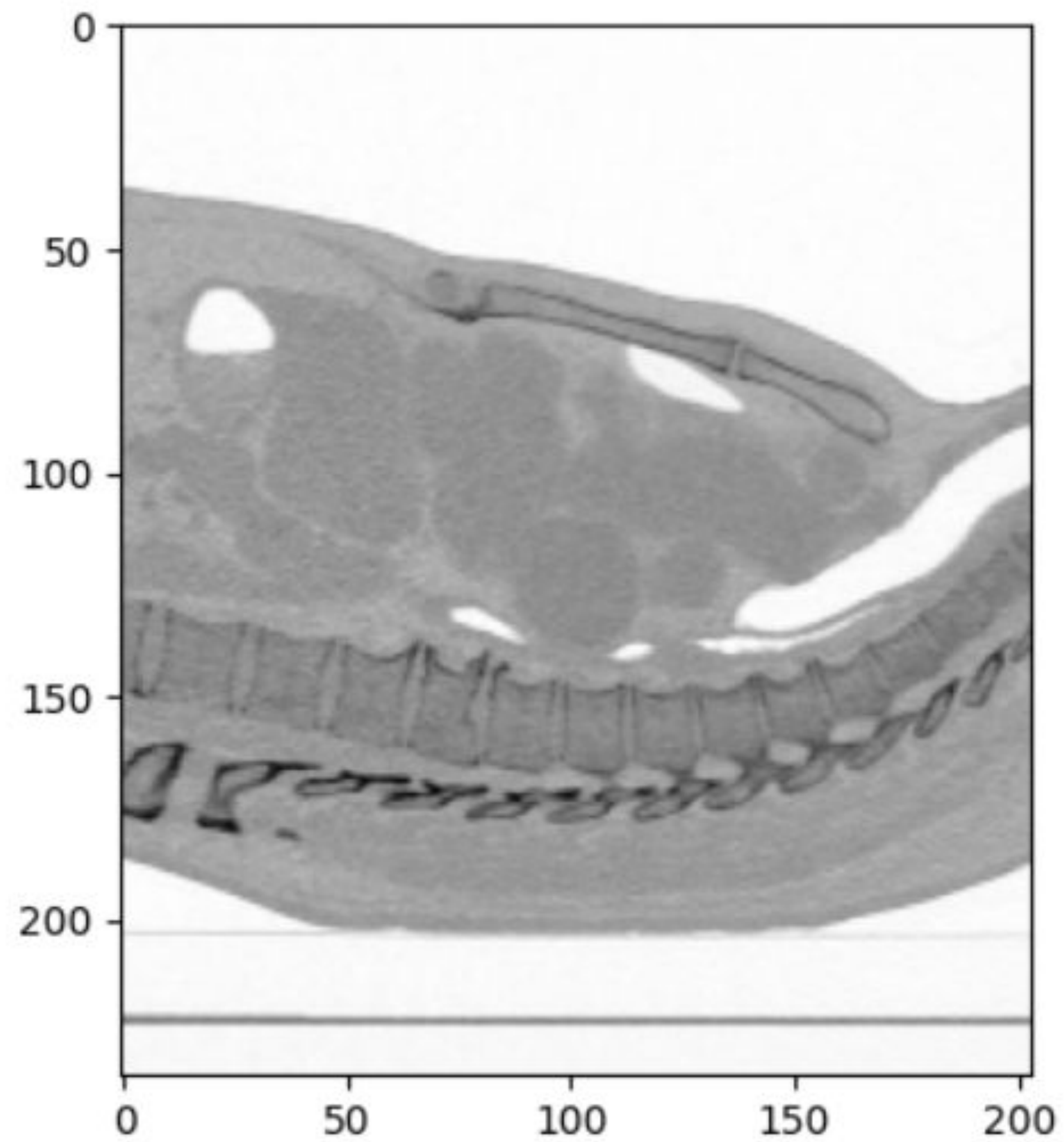
The Curse of Dimensionality



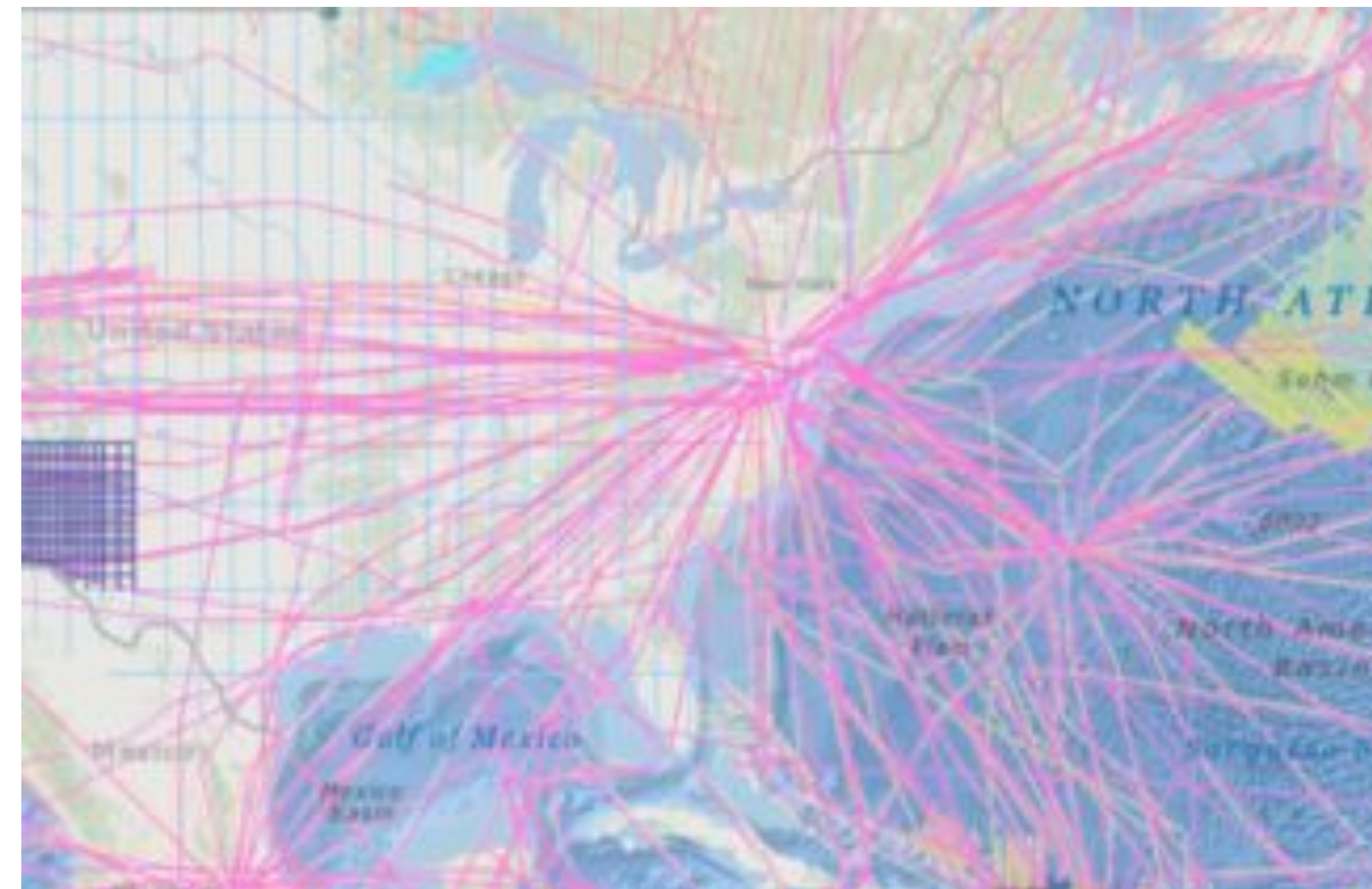
Choosing Data

Analysis Paralysis

CT Scan



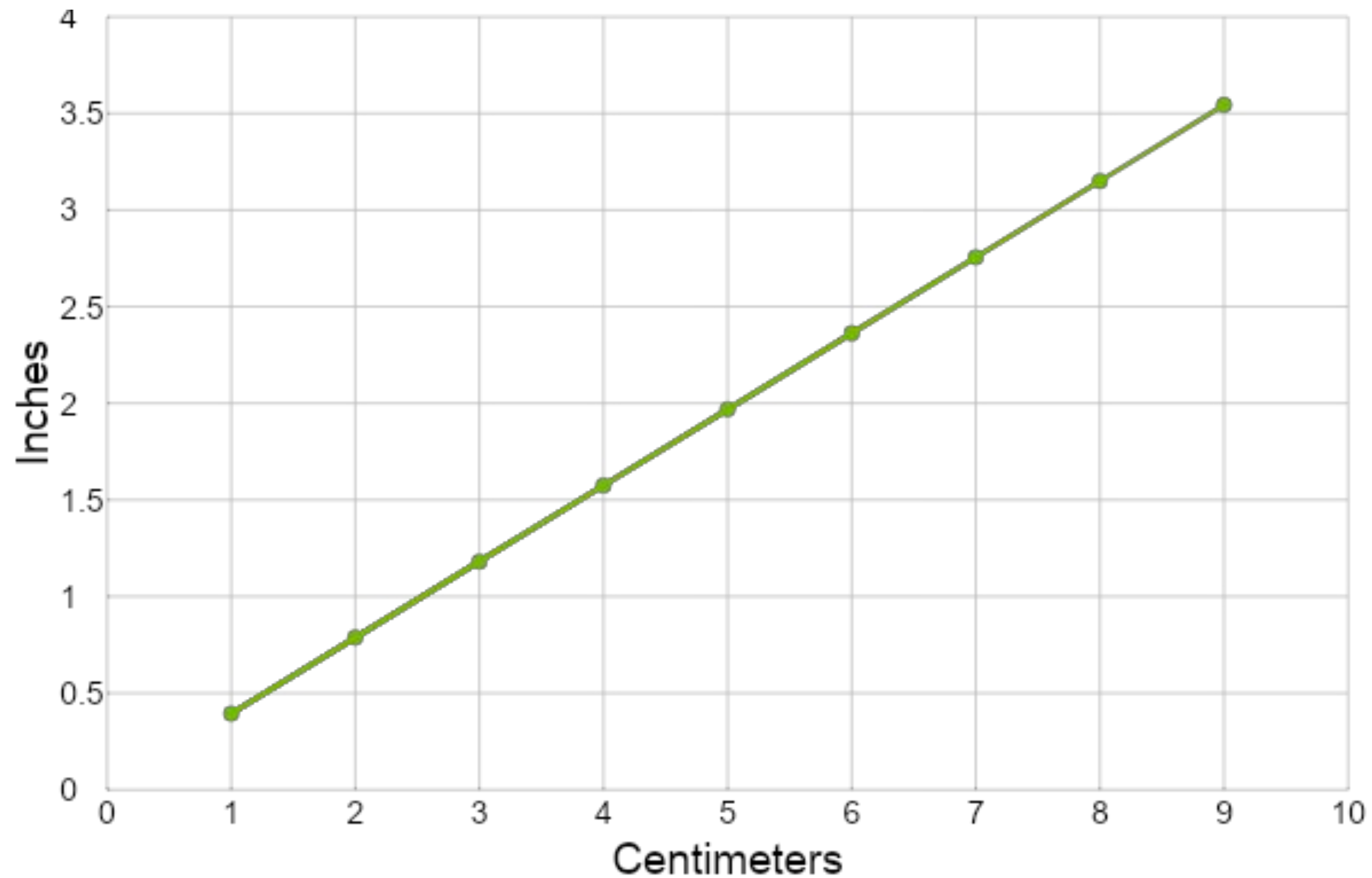
Spectro
gram



Magnet
ic Data

Choosing Data

Highly Correlated Data





Lab 2b – Contrastive Pretraining

Matching Text to Image

Is it Possible?



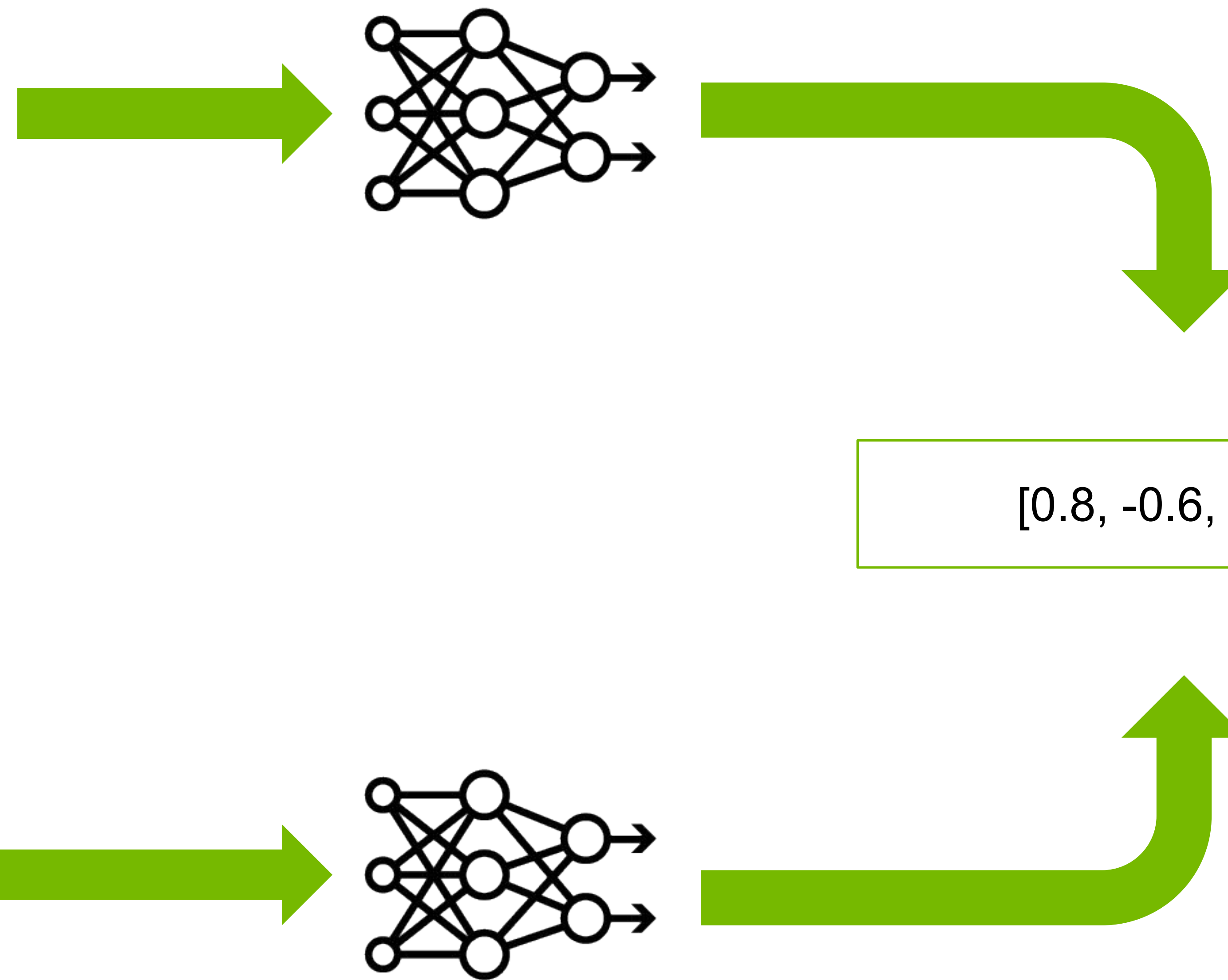
“A bunch of different
marbles”

Matching Text to Image

Is it Possible?

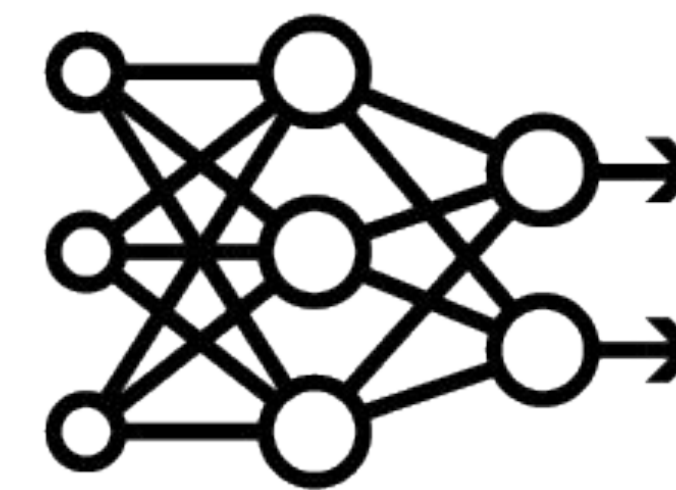


“A bunch of different
marbles”



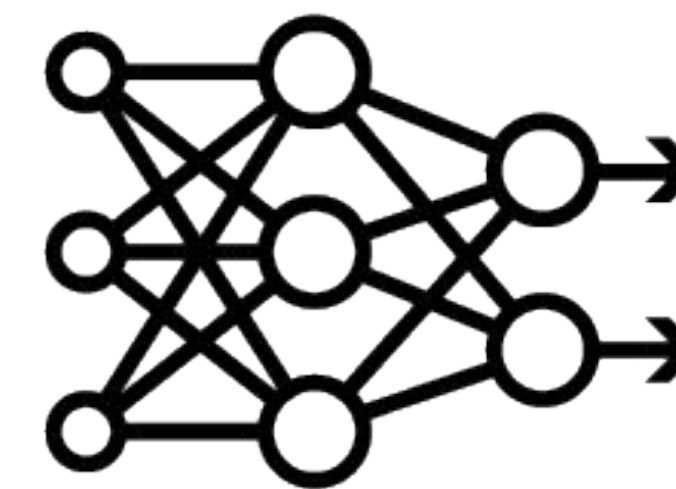
$[0.8, -0.6, 0.7]$

Cosine Similarity



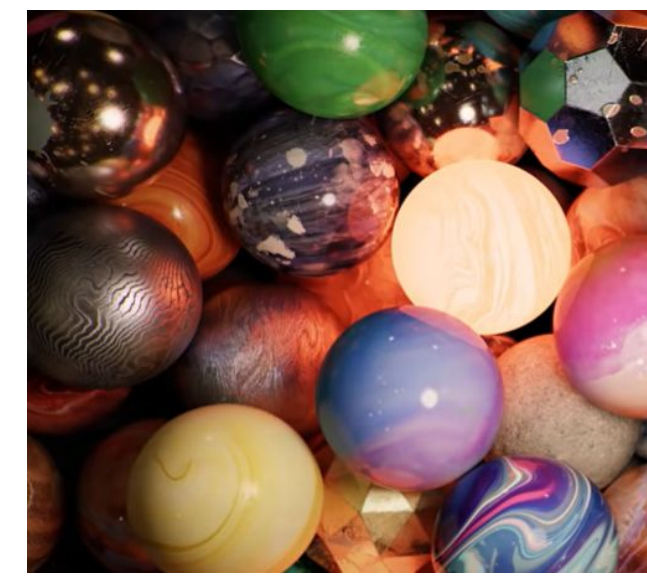
[0.0, 1.0]

“A bunch of different
marbles”



[1.0, 1.0]

Cosine Similarity

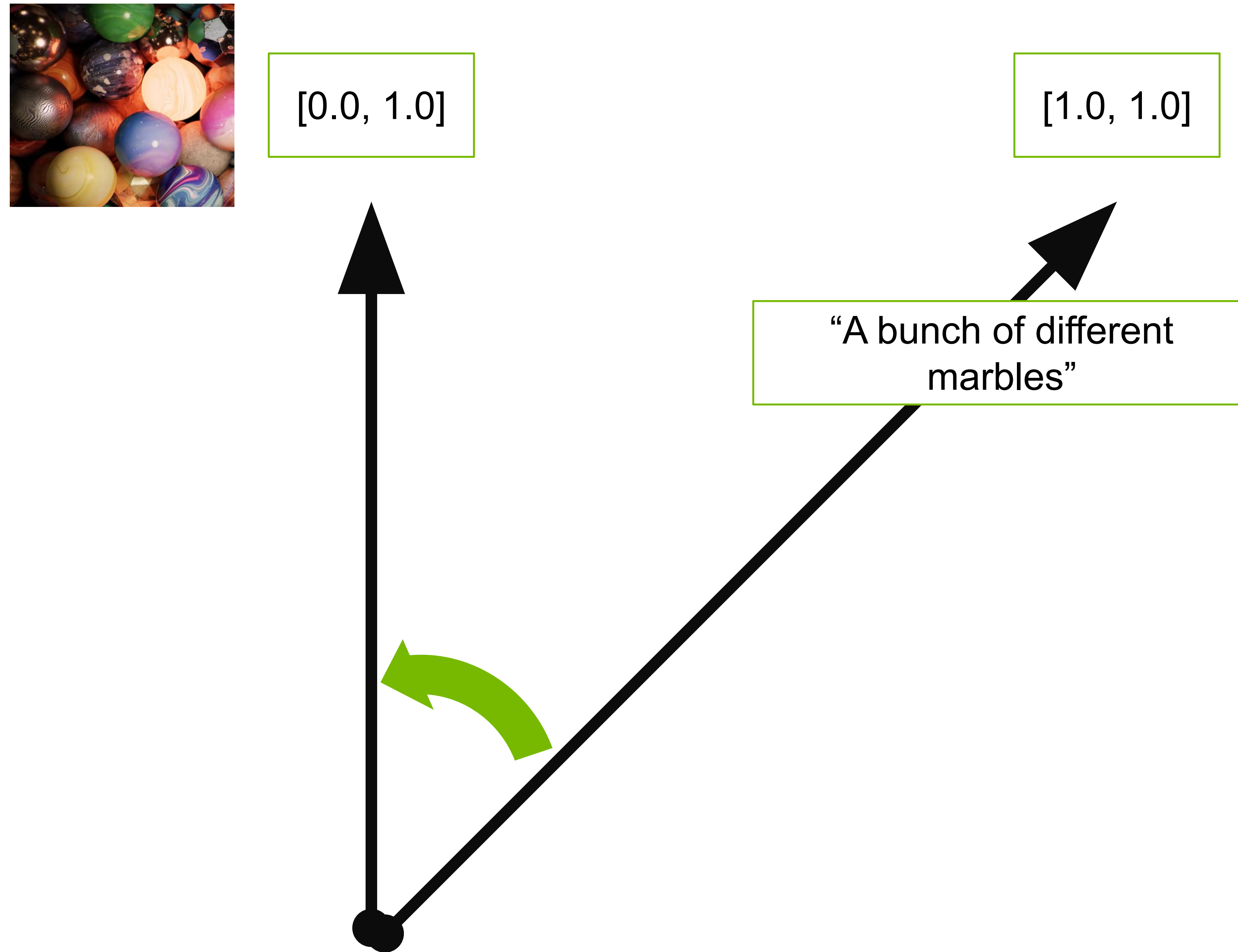


[0.0, 1.0]

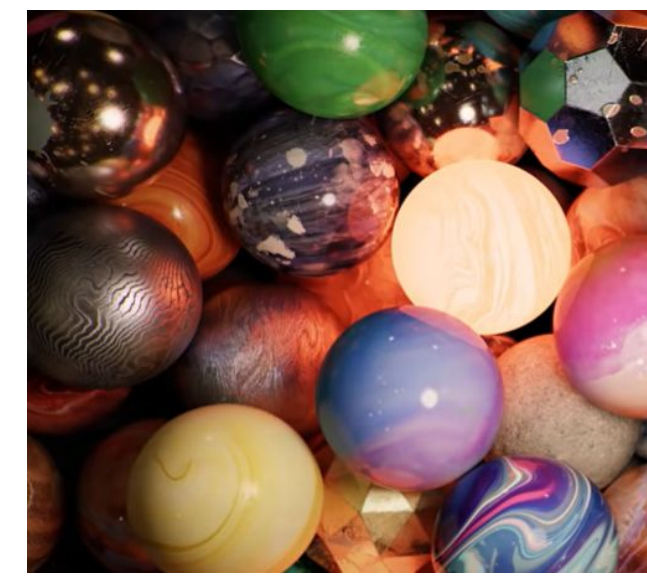
[1.0, 1.0]

“A bunch of different
marbles”

Cosine Similarity



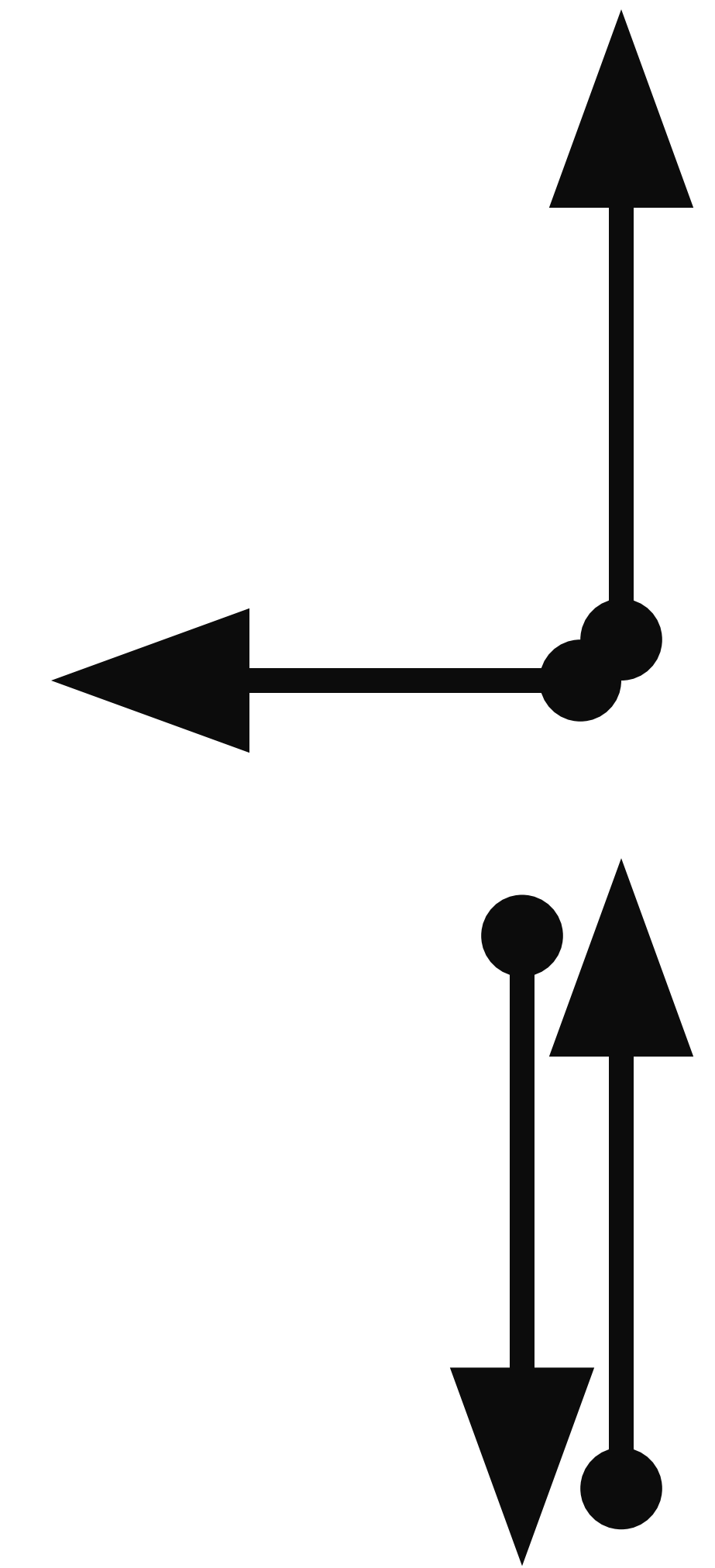
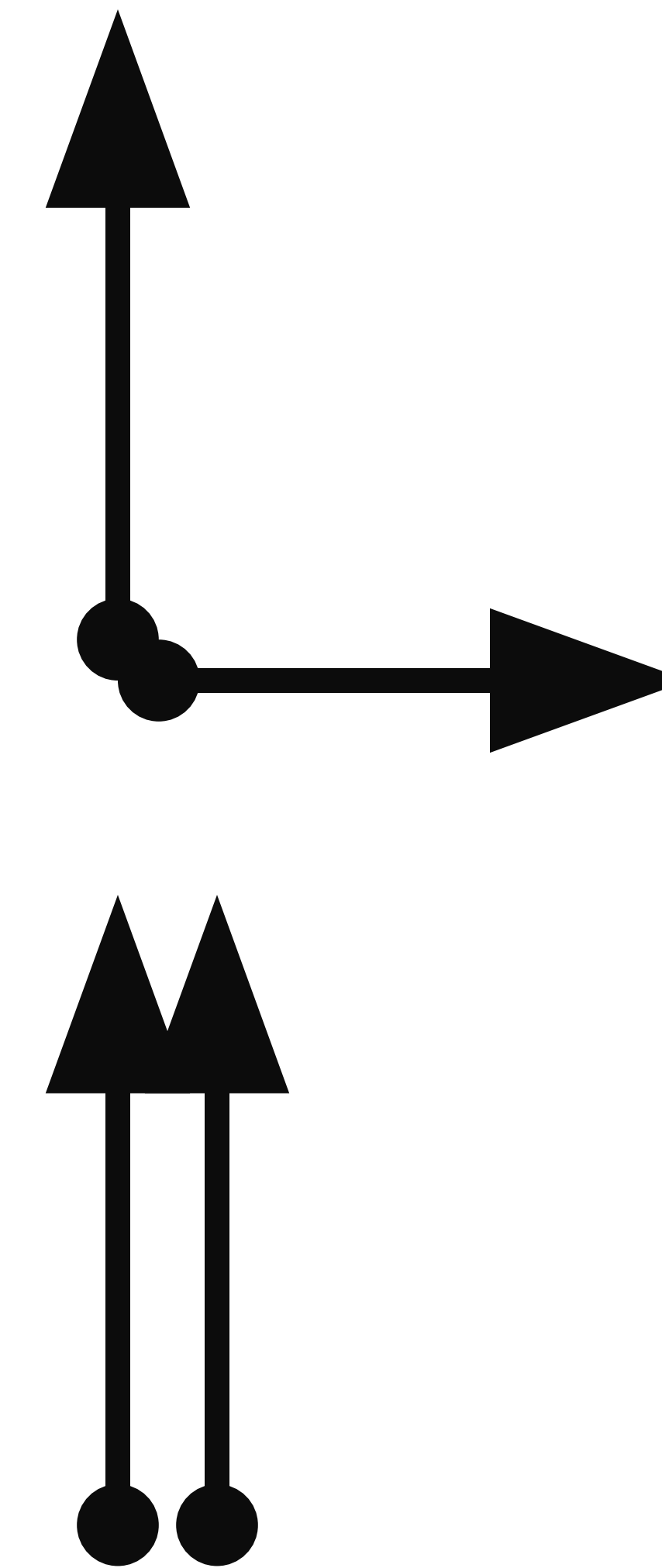
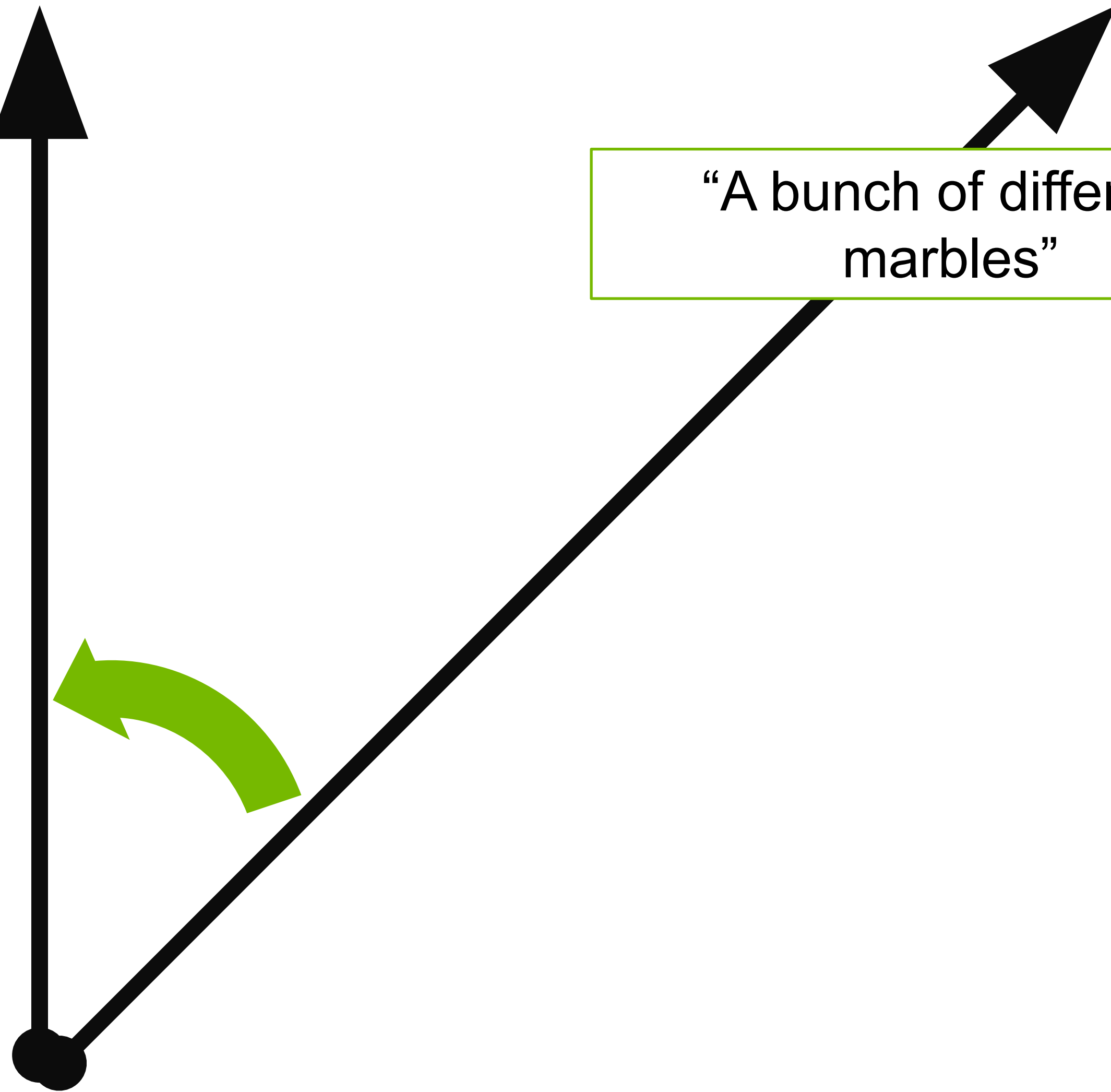
Cosine Similarity



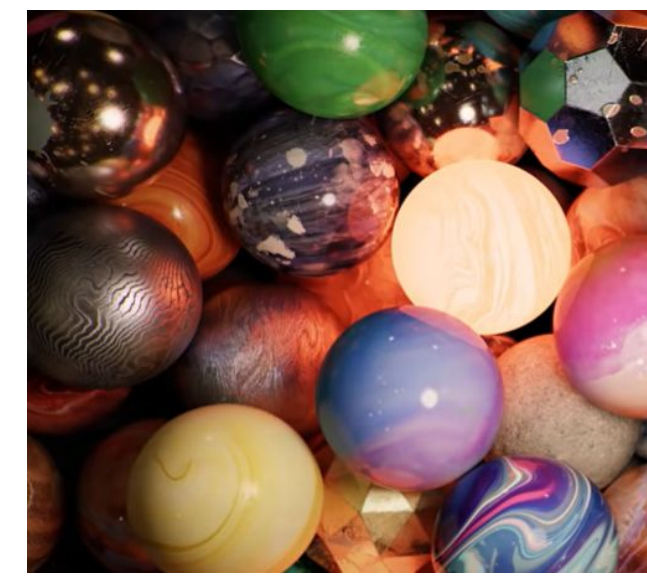
[0.0, 1.0]

[1.0, 1.0]

“A bunch of different
marbles”



Dot Product



$[0.0, 1.0]$

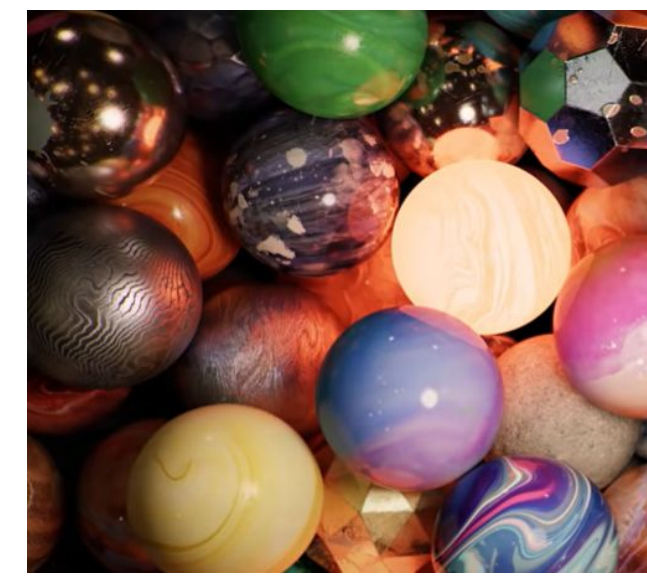
$[1.0, 1.0]$

“A bunch of different
marbles”

$[0.0, 1.0]$

$[1.0, 1.0]$

Dot Product



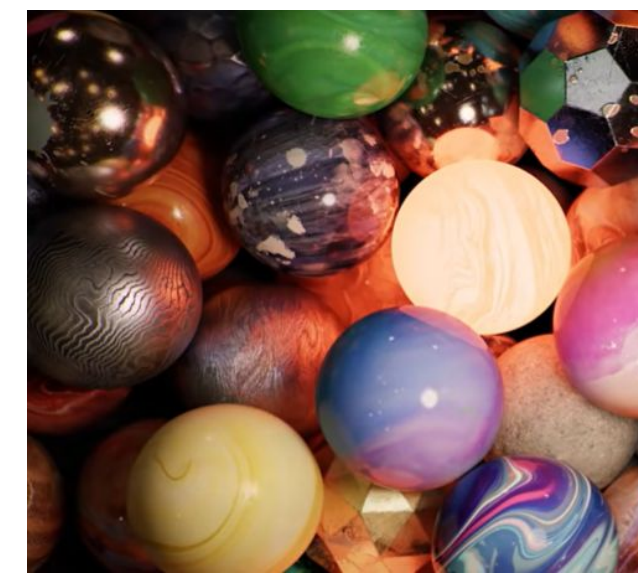
$[0.0, 1.0]$

$[1.0, 1.0]$

“A bunch of different marbles”

$[0.0, 1.0]$

Dot Product



$[0.0, 1.0]$

$[1.0, 1.0]$

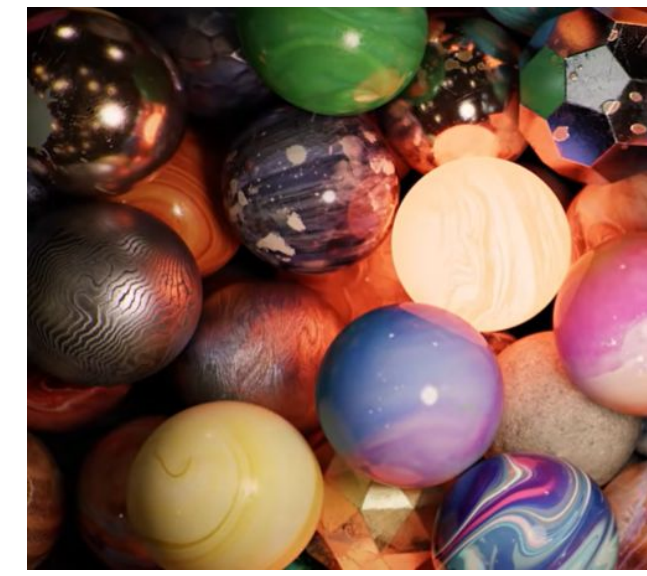
“A bunch of different marbles”

A

$[0.0, 1.0]$

B

Dot Product



[0.0, 1.0]

[1.0, 1.0]

“A bunch of different
marbles”

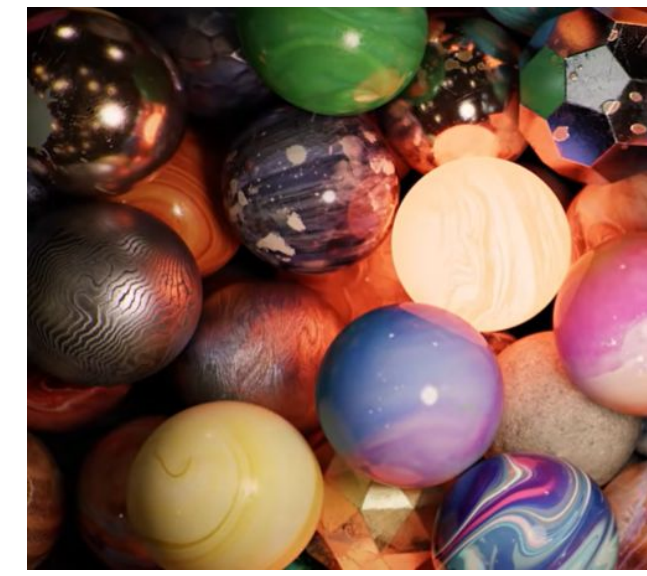
	A	B	A x B
x	0		0
y	1		

A

[0.0, 1.0]

B

Dot Product



[0.0, 1.0]

[1.0, 1.0]

“A bunch of different
marbles”

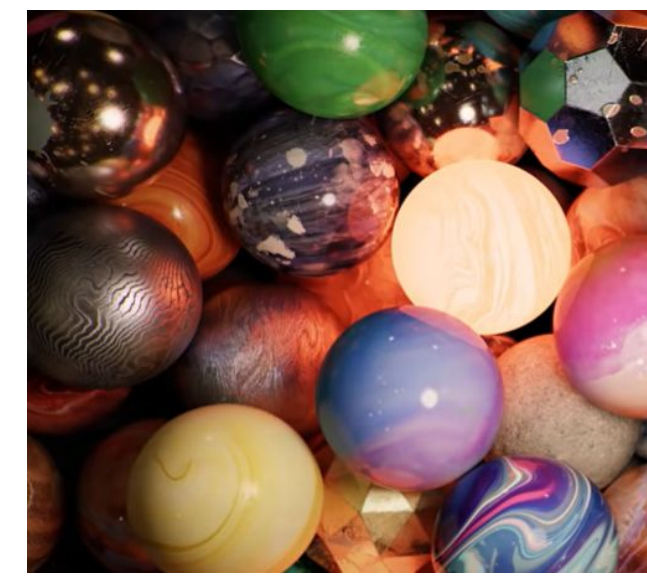
	A	B	A x B
x	0		0
y	1		

A

[0.0, 1.0]

B

Dot Product



[0.0, 1.0]

[1.0, 1.0]

“A bunch of different
marbles”

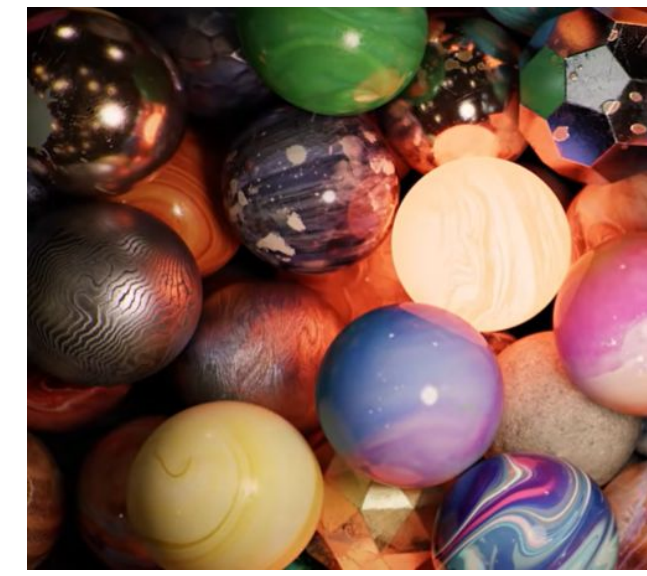
	A	B	A x B
x	0		0
y	1		

A

[0.0, 1.0]

B

Dot Product



[0.0, 1.0]

[1.0, 1.0]

“A bunch of different
marbles”

	A	B	A x B
x	0		0
y	1		

A

[0.0, 1.0]

B

CLIP Training



A happy corgi

A leather jacket

A green spiral

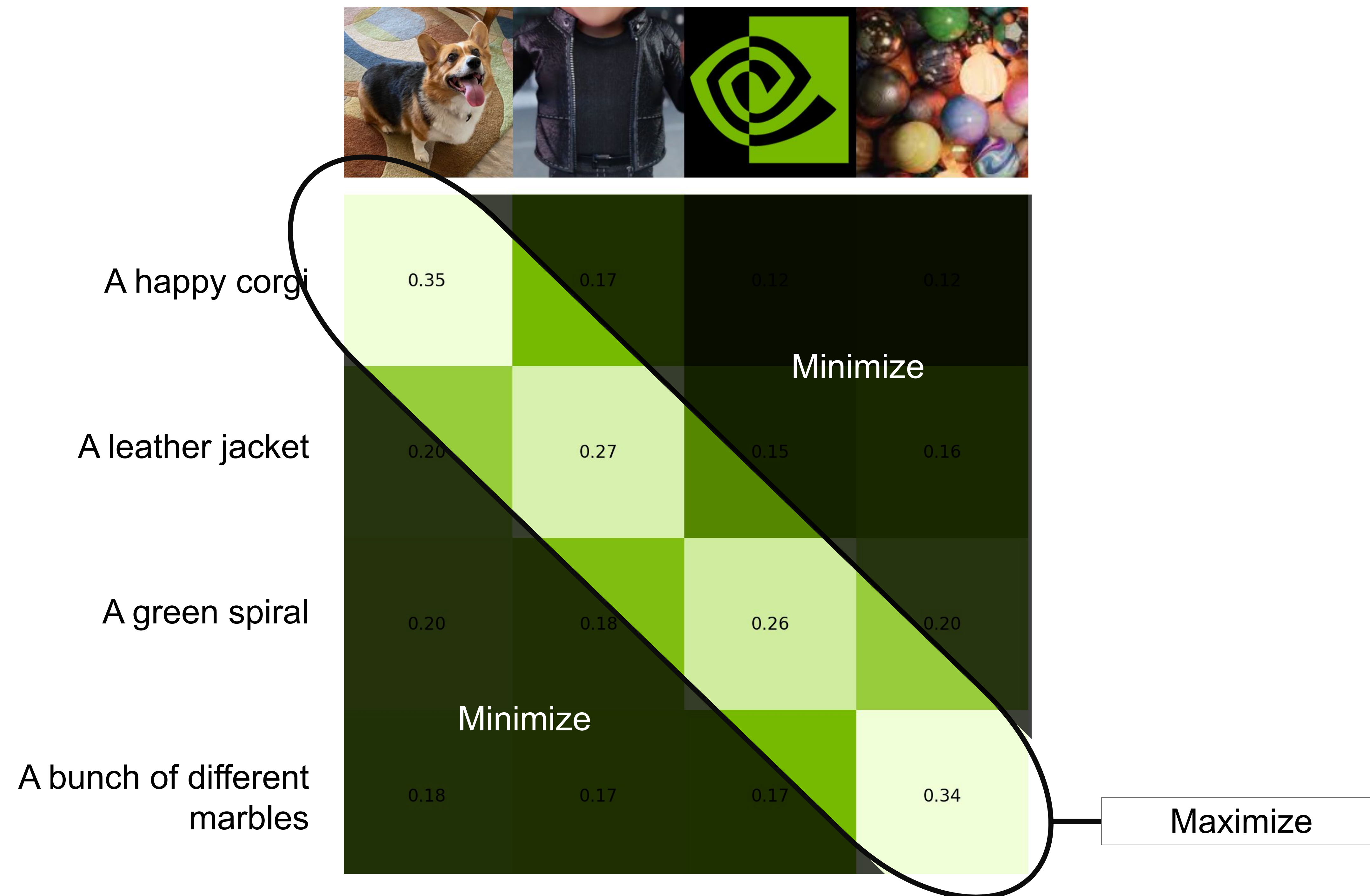
A bunch of different
marbles

0.35	0.17	0.12	0.12
0.20	0.27	0.15	0.16
0.20	0.18	0.26	0.20
0.18	0.17	0.17	0.34

Cosine similarity between
encoding for the NVIDIA
logo and encoding for
each text

Cosine similarity between encoding
for "A happy corgi" and encoding for
each image

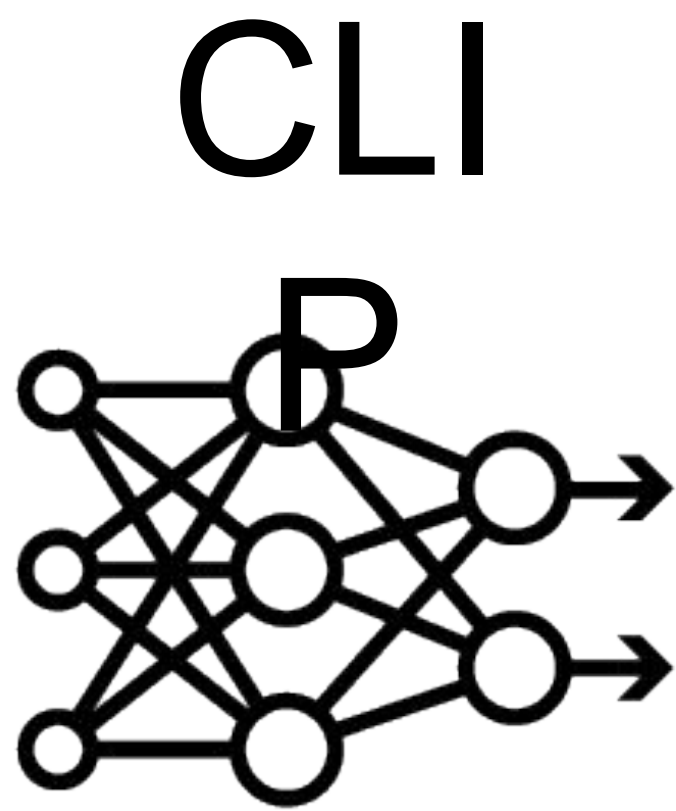
CLIP Training





RAG and Vector Databases

Vector Databases



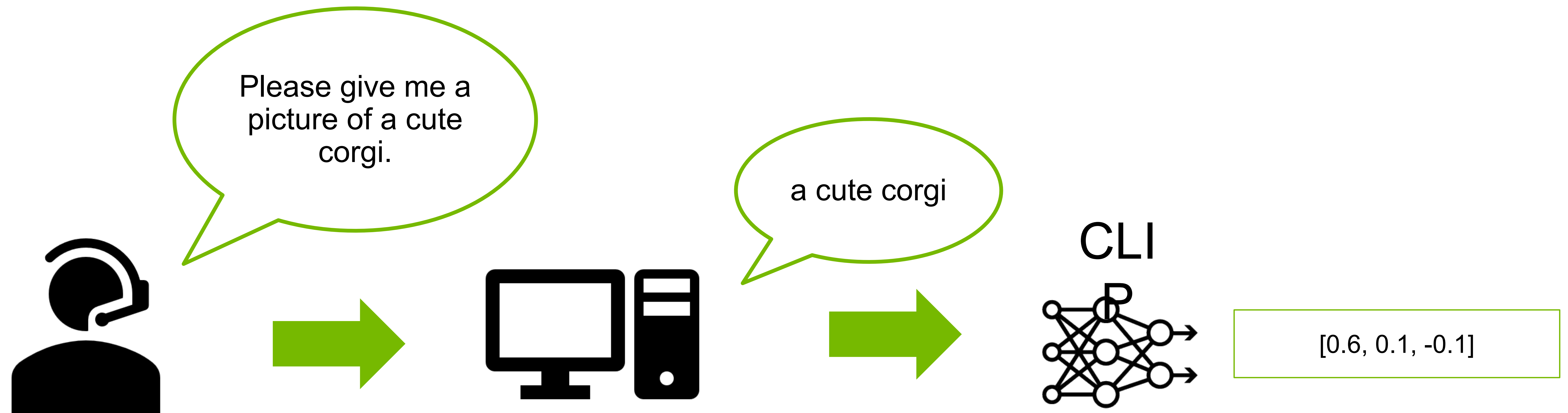
[0.7, 0.2, -0.1]

[-0.1, 0.1, -0.8]

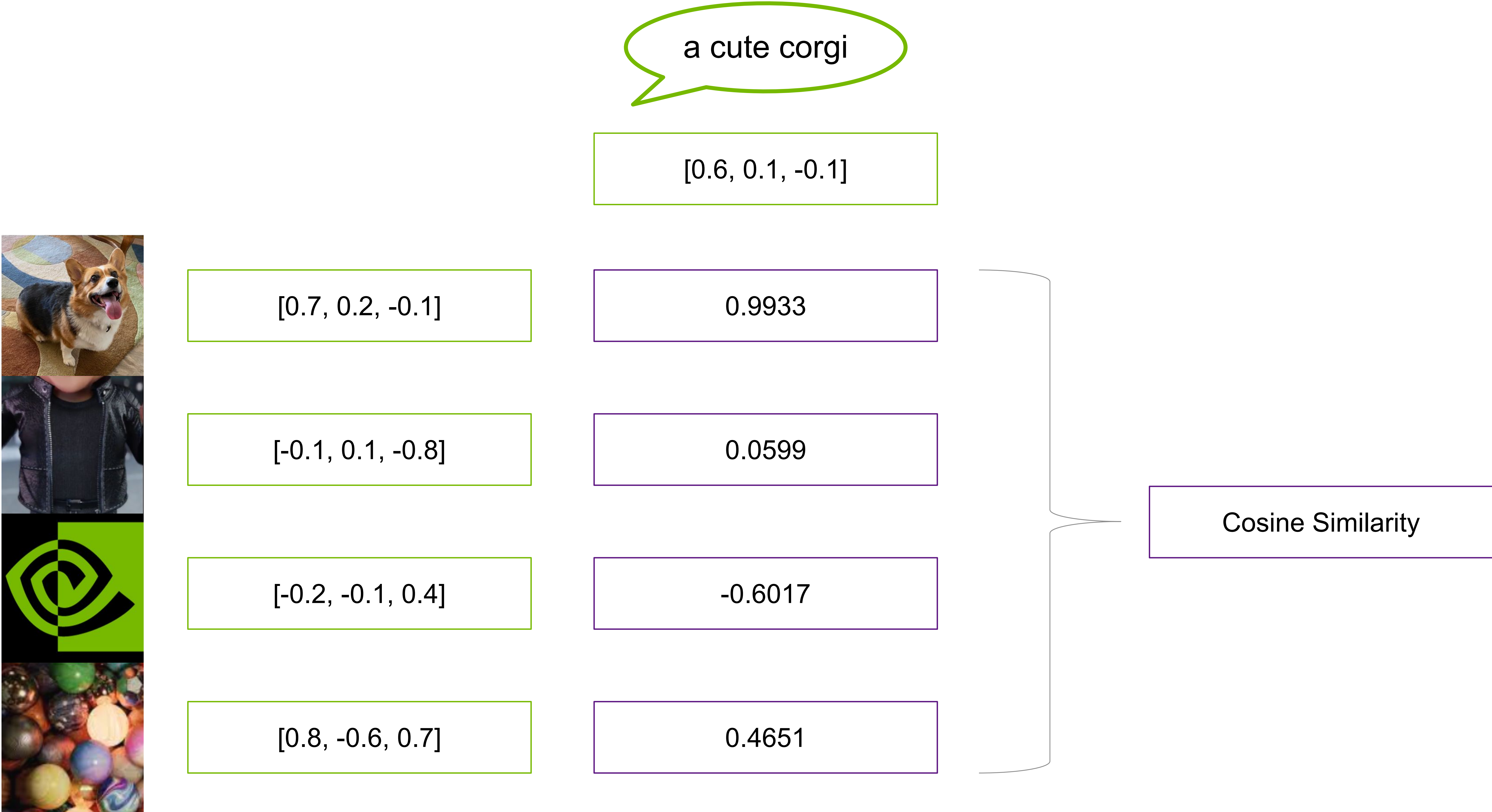
[-0.2, -0.1, 0.4]

[0.8, -0.6, 0.7]

Retrieval Augmented Generation (RAG)



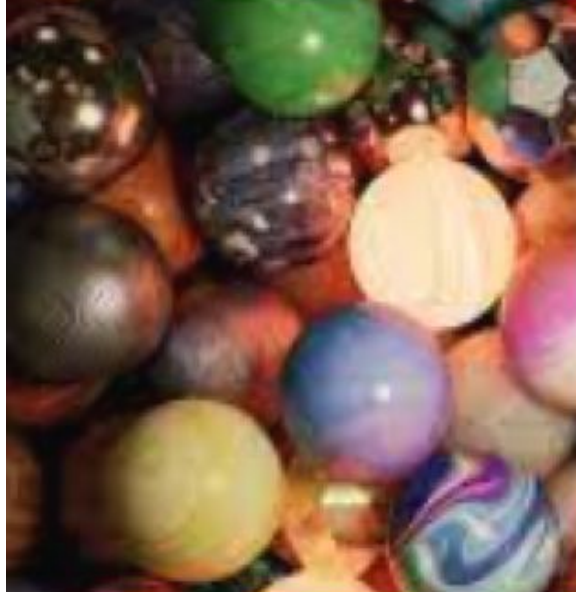
Retrieval Augmented Generation (RAG)



Retrieval Augmented Generation (RAG)

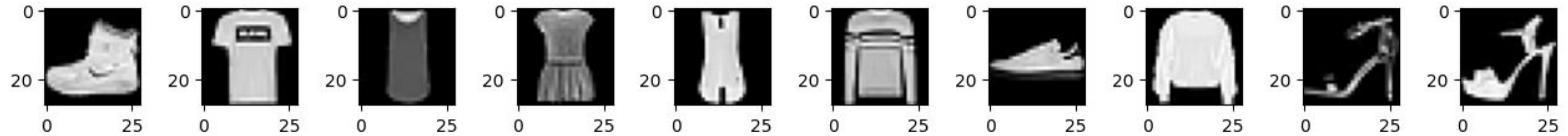
a cute corgi

[0.6, 0.1, -0.1]

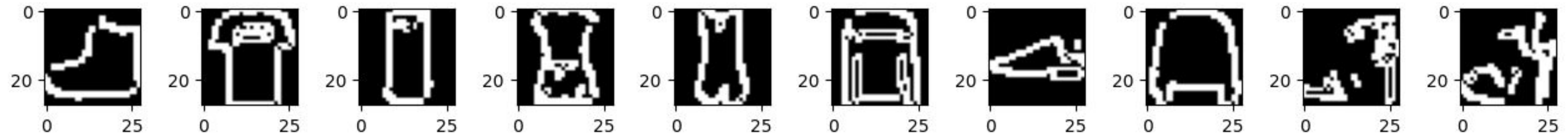
	[0.7, 0.2, -0.1]	0.9933	Best Match
	[-0.1, 0.1, -0.8]	0.0599	
	[-0.2, -0.1, 0.4]	-0.6017	
	[0.8, -0.6, 0.7]	0.4651	

Contrastive Outline-Image Pretraining

Image



Outline



BUILDING AI AGENTS WITH MULTIMODAL MODELS



Building AI Agents with Multimodal Models: NVIDIA DLI Workshop for Academia

Von Berlin Computer Vision Group

📍 Online event

Dez

20

10:30

NVIDIA DLI Certification Course

- **Free** for academic participants
 - Staff, students
- Email info@kineto.ai if you are in industry